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UCOS-MASTER-ARCH · CANONICAL

UCOS Ω∞ Master Architecture & Build Specification

The single context document for building the Universal Capability Operating System: a platform factory that composes marketplaces, commerce, mobility, media, fintech, ERP, CRM and future platforms from certified capabilities, context, policies and metadata.

VERSION	DATE	STATUS	KEYWORDS
1.0.0	2 Oct 2026	Canonical — supersedes all earlier fragments	MUST / SHOULD / MAY per RFC 2119

CONTENTS
34 sections · 26 diagrams · 16 wireframes

L0 Constitution	L1 Meta-model	L2 Data	L3 Knowledge	L4 Foundation	L5 Business	L6 Intelligence
L7 Composition	L8 Runtime	L9 Integration	L10 Experience	L11 Operations	L12 Certify & evolve	

The 13 canonical layers, foundation first. Select a layer to jump to its section.

PART I

FOUNDATIONS

1. Executive Summary

1.1 What UCOS is

UCOS Ω∞ is a **platform factory**. It does not hard-code any single application. It composes running platforms (marketplaces, commerce, mobility, media, fintech, ERP,

composes running platforms (marketplaces, commerce, mobility, media, fintech, ERP, CRM, ITSM, healthcare, government, education and future types) from:

```
PLATFORM  =  CAPABILITIES  +  CONTEXT  +  POLICIES  +  WORKFLOWS
           +  KNOWLEDGE    +  AI       +  EXPERIENCE +  GOVERNANCE
           —  composed, validated, certified, deployed, observed and evolved from M
```

Core equation: **Application = Capability Composition, NOT custom code.**

```
Amazon-like      = Identity + Product + Catalog + Inventory + Pricing + Offer + Car
                   + Payment + Logistics + Review + Recommendation + Marketplace +
Uber-like        = Identity + Geo + Temporal + Dispatch + Routing + Matching + Dyna
                   + Payment + Rating + Tracking
Netflix-like     = Identity + Subscription + Entitlement + Media + Streaming + Reco
                   + Billing + Analytics
Salesforce-like  = Identity + CRM + Workflow + Decision + Analytics + AI + Communic
SAP-like         = Identity + ERP + Finance/Ledger + Procurement + Inventory + Manu
                   + Workforce + Analytics
```

1.2 What "no code changes" means precisely

The source vision says "without writing new business code". A literal reading can't be built, so this specification defines it precisely:

Change type	Requires platform source-code change?	Mechanism
New tenant, company, marketplace, brand, store	**No**	Platform Definition (metadata)
New country, language, currency, tax regime, holiday calendar	**No**	Reference-data registries + jurisdiction packs
New product/service type, attribute, taxonomy node, variant	**No**	Ontology/Taxonomy/Attribute registries
New workflow, approval chain, SLA, trigger, offer rule, pricing rule	**No**	Workflow DSL, Decision tables, Temporal policies
New UI screen, portal, layout, theme, journey	**No**	UI Meta-Model + renderers

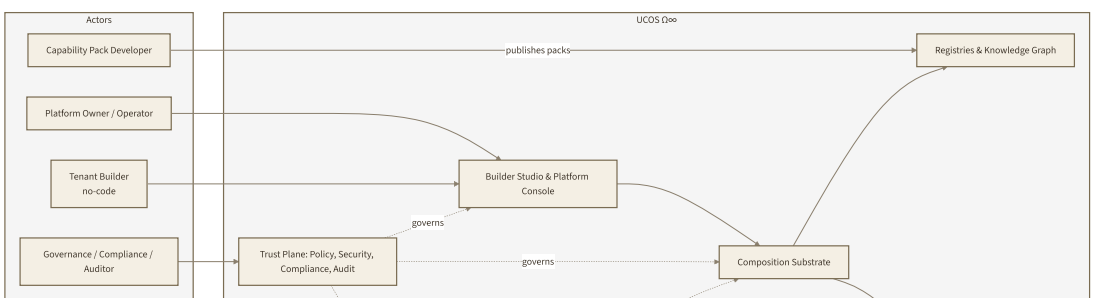
New compliance control mapping, policy, role	**No**	Compliance-as-code + Policy registry
New integration with a system that has a supported protocol	**No**	Connector manifest + mapping
Switch AI model/provider	**No**	Model registry + AI gateway routing
New capability primitive never modelled before (e.g. a new settlement algorithm)	**Yes — once**, as a certified **Capability Pack**	Built once, registered once, certified once, then reused by every tenant
New engine type (e.g. a new runtime kind)	Yes — platform release	Governed platform evolution

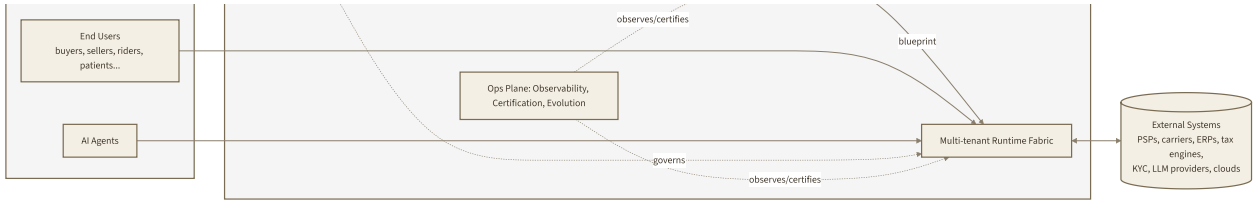
Invariant: there is never *tenant-specific* or *application-specific* code. Code exists only in (a) the platform kernel and engines, and (b) reusable, certified Capability Packs.

1.3 The five things that make UCOS different from a normal SaaS

1. **One meta-model:** every artifact is a Canonical Object with identity, ownership, context, lifecycle, policy, audit and evidence (§7).
2. **Capability Genome:** every capability ships as a self-describing manifest with data, API, events, workflows, UI, policies, security, compliance, tests and docs (§8).
3. **Composition Substrate:** a deterministic resolver turns a business intent into a full blueprint and **finds what is missing** (§11).
4. **Completeness gate:** no platform activates while `mandatory_gap_count > 0` or uncertified artifacts exist (§11.6, §28).
5. **Knowledge Once:** global registries (time, geography, currency, products, services, regulations) are authored once and referenced everywhere (§9, §10).

1.4 System context diagram





2. How to Use This Document

2.1 Reading order

Role	Read first	Then
Architect / Tech lead	§1, §3, §5, §6, §7, §8, §11	everything
Backend engineer	§7, §8, §13–§17	§18–§20
Frontend / designer	§22–§25	§7.3, §8, §15
SRE / Platform	§13, §14, §26, §27	§19, §28
Security / Compliance	§18–§21	§28
AI coding agent	§34 **first**, then §3, §7, §8	the section relevant to the task

2.2 Conventions

- IDs: `ucos:<kind>:<ulid>` (§7.2). Capability IDs: `<universe>.<capability>` e.g. `commerce.cart`.
- Event names: `<universe>.<object>.<pastTenseVerb>` e.g. `order.order.placed` (§15.5).
- All schemas are expressed in YAML for readability; the runtime stores them as JSON Schema 2020-12.
- Diagrams use Mermaid (renders in GitHub, GitLab, VS Code, most doc tools); wireframes use ASCII.
- "Universe" = a bounded context that owns a family of canonical objects. "Capability" = an installable unit of behaviour inside a universe.

2.3 Document authority

This document is **Constitution-level** (§18.1). Changes require a Constitutional Change Proposal (CCP) approved by the Architecture Authority (§18.4). Sub-specifications

(one per universe, generated per §12.5) MUST NOT contradict it.

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3. Constitutional Principles & Enforceable Laws

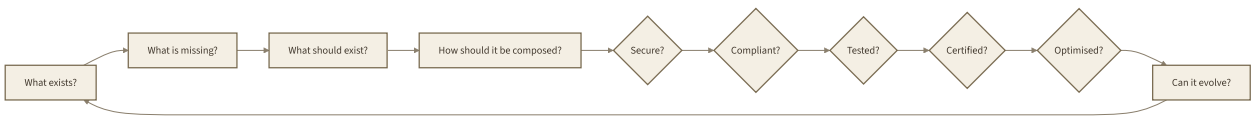
The chat stated more than 40 principles. They are consolidated here into **20 laws**.
Each law has an **enforcement mechanism**: a principle with no enforcement mechanism is not a law.

#	Law	Statement	Enforcement (where / how)
L1	**Knowledge Once**	Every fact, definition, rule or reference datum has exactly one canonical home. Author once → register once → certify once → reuse everywhere.	Registry uniqueness constraints; Composition linter rejects duplicate concept definitions (semantic dedupe via ontology `sameAs`); CI check `ucos lint --duplicates`.
L2	**Zero Hard Coding**	No tenant, industry, country, language, currency, gateway, model, workflow, UI or topology is hard-coded in platform or pack code.	Static analysis rules (banned literals: ISO country codes, currency codes, provider names outside adapters); pack certification check; code review checklist.
L3	**Canonical Ownership**	Every object type, capability and registry belongs to exactly one universe and has a named steward.	`owner` and `steward` are required fields in the meta-model; registry rejects orphan entries.
L4	**Zero Duplication / Zero Overlap**	Two capabilities MUST NOT provide the same object type or the same command.	Composition Substrate conflict detector (§11.5); pack publish fails on `provides` collision.
L5	**Zero Orphans**	Every artifact traces up to Constitution → Law → Policy → Capability → Requirement, and down to Test → Evidence → Certification.	Traceability graph check; certification gate `TRACE-001`.
L6	**Composition over Construction**	Applications are compositions of certified capabilities, never bespoke code.	No `apps/<tenant>` code directories permitted; repo policy.
L7	**Context Awareness**	Every execution carries an explicit Context (identity, tenant, jurisdiction, time, locale, channel, risk).	Runtime rejects calls without a resolved `ContextEnvelope` (§7.6).
L8	**Policy before Execution**	No command executes without a policy decision (allow/deny +	Policy Enforcement Point embedded in the gateway and the command bus; deny-by-

		obligations).	default.
L9	**Governed Change**	No change to definitions bypasses governance (proposal → validation → approval → certification → activation).	Definition lifecycle state machine (§7.5); activation requires a `Certification` reference.
L10	**Completeness before Activation**	A platform instance cannot be activated while `mandatory_gap_count > 0` (unless there is an explicit, authorised, time-boxed waiver).	Certification gate `GAP-000` (§11.6).
L11	**Security by Construction**	Zero trust, least privilege, encryption at rest/in transit, tenant isolation are defaults, never options.	Security baseline policies installed at Layer 0; pack certification `SEC-*` checks.
L12	**Compliance by Construction**	Regulatory obligations are resolved from context (jurisdiction × industry × data class × capability) and enforced as controls.	Compliance resolver (§20); evidence collectors; gate `CMP-*` .
L13	**Audit Everything**	Every command, decision, policy evaluation, AI output, approval, deployment and access is recorded immutably.	Append-only audit log with hash chaining (§19.9).
L14	**Observable Everything**	Every capability emits standard traces, metrics, logs and business signals.	Pack certification requires an OTel instrumentation manifest.
L15	**Vendor & Model Agnosticism**	External providers (cloud, PSP, LLM, carrier, tax) are reached only through adapters behind canonical ports.	Port/adapter architecture; dependency rules in build (`packs/*` may not import vendor SDKs except `adapters/*`).
L16	**Versioned & Evolvable**	Every definition is versioned (SemVer), backward-compatibility rules are enforced, and deprecations follow a published timeline.	Schema compatibility checker (§29.4).
L17	**Human Authority over Autonomy**	AI/automation can propose and act only within delegated, bounded authority. Constitutional, security-critical and irreversible financial changes require human approval.	Autonomy levels (§29.2); Authority objects for agents (§21.5).
L18	**Deterministic**	The same Platform Definition +	Composer is a pure function; snapshot IDs

	Composition	registry snapshot always yields the same blueprint (content-hash identical).	pinned; reproducibility test in CI.
L19	**Isolation & Sovereignty**	Tenant data, keys and (where required) compute are isolated. Data residency is honoured per jurisdiction.	Cell architecture + RLS + per-tenant keys (§14.3, §19.6).
L20	**Discover the Unknown**	The platform continuously asks *what exists, what is missing, what overlaps, what conflicts, what violates, what should evolve*.	Completeness Engine scheduled scans + on-change scans (§11, §29).

3.1 The Ultimate System Loop



3.2 Capability quality attributes (every capability MUST be)

Constitutional · Discoverable · Composable · Reusable · Extensible · Evolvable · Governed · Versioned · Certified · Observable · Secure · Context-aware · Auditable · Localisable · Tenant-isolated.

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4. Glossary

Term	Definition
Constitution	Top-level, human-authored rules (this document plus the per-universe constitutions).
Law	An enforceable invariant derived from the Constitution (§3).
Universe	A bounded context owning a coherent family of canonical object types, e.g. Temporal Universe, Offer Universe.
Domain	A grouping of universes for navigation and team ownership, e.g. Foundation, Commerce.
Canonical Object	An instance of any type that conforms to the root object envelope (§7.1).
Object Type	A registered definition (schema + behaviour bindings) of a kind of canonical object.

Capability	An installable unit of behaviour that provides object types, commands, queries, events, workflows, UI models and policies.
Capability Pack	The signed, versioned distribution artifact of one or more capabilities (§8).
Registry	An authoritative, versioned store of definitions or reference data, e.g. Capability Registry, Product Registry, Temporal Registry.
Knowledge Graph (UKG)	The graph of all definitions, instances (selected) and relationships, used for reasoning, discovery and traceability.
Context	The resolved situation of an execution: tenant, identity, jurisdiction, time, locale, channel, currency, risk, device, observer (§7.6).
Platform Definition (PD)	A tenant's declared intent: industry, business model, countries, channels, capabilities, compliance and security profile (§11.2).
Blueprint	The fully resolved, deterministic output of composition: every capability, object type, policy, workflow, UI, integration, infra and control needed (§11.4).
Platform Instance	A deployed, certified blueprint serving a tenant in one or more cells.
Gap	A required element that is missing, unresolved, conflicting or uncertified (§11.6).
Waiver	A time-boxed, authorised acceptance of a non-mandatory gap, or an exceptional acceptance of a mandatory one.
Certification	A signed attestation that an artifact passed defined gates with linked evidence.
Evidence	An immutable record proving that a control or test is satisfied.
Cell	An isolated deployment unit (region + stack) hosting a set of tenants.
Pack Developer	An engineer who builds Capability Packs, the only place where new business code is written.
Steward	The accountable owner of a universe, registry or object type.
Agent	An AI actor with an identity, delegated authority, tools, memory and guardrails.
Digital Twin	A live model of a real thing (platform, organisation, product, supply chain), built from definitions and telemetry, used for simulation.

5. Canonical Layer Architecture

5.1 Consolidation decision

The source material contains at least **eight incompatible layer numberings** (Layer 000–021, Level 0–14, Layer 0–27, Universe 001–047, Layer A–J, Layer 00–21, Level 0–

20, Layer 01–28). They are merged into **one 13-layer model with 4 cross-cutting planes**. Every concept from every numbering maps to exactly one layer (§33).

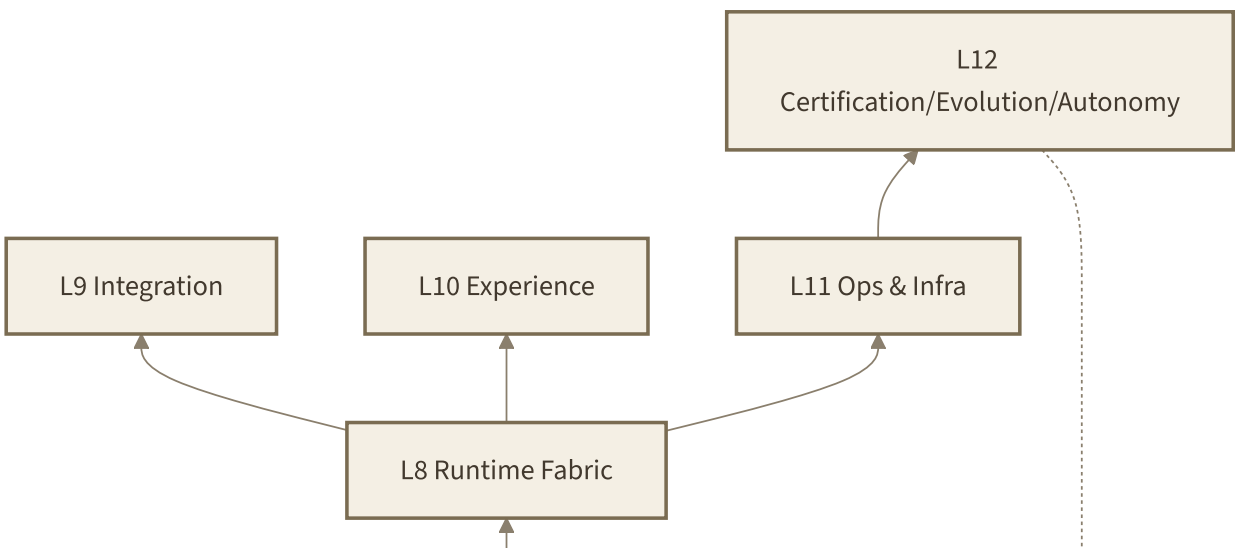
5.2 The canonical stack

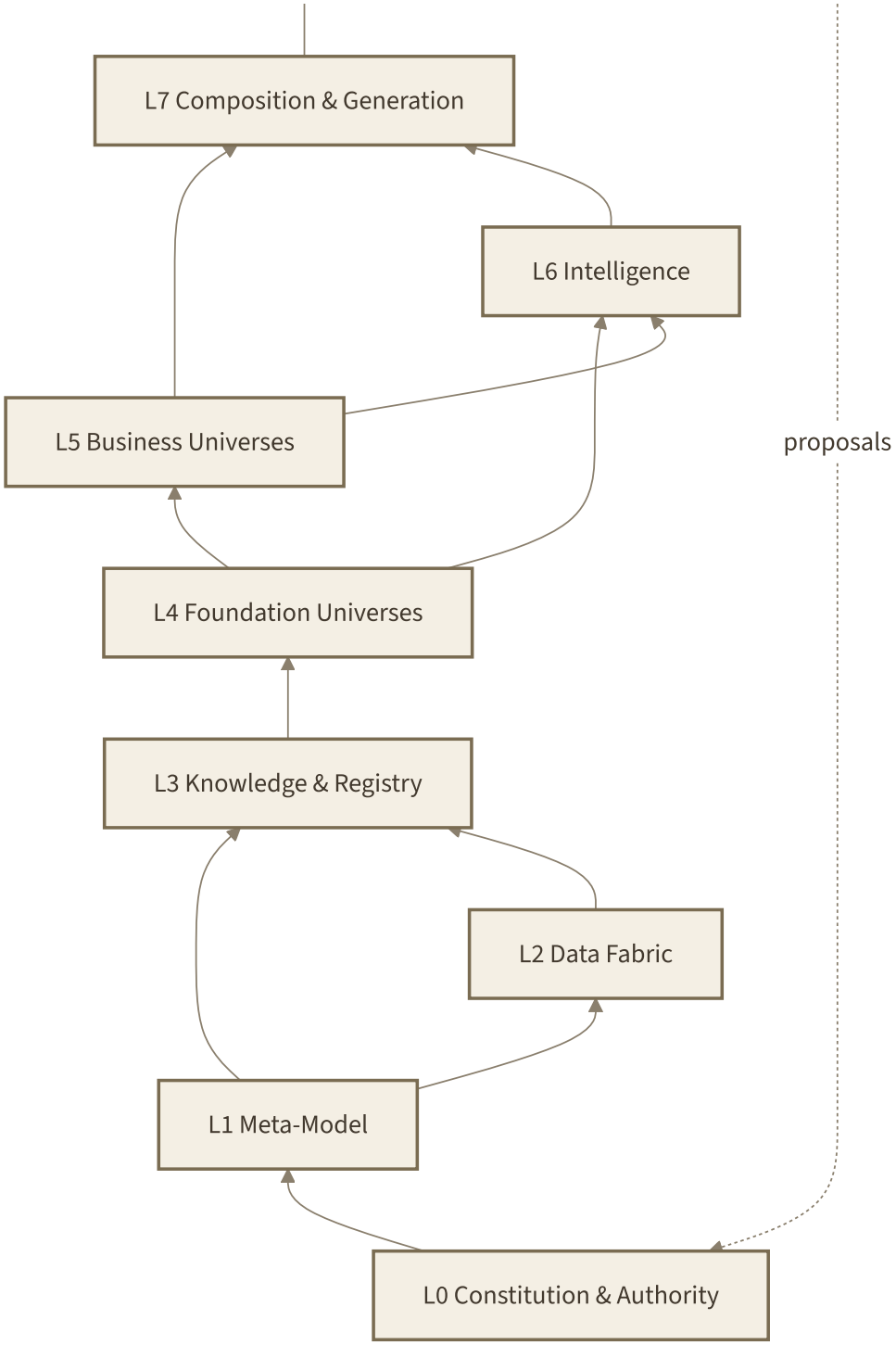
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┌──────────────────────────────────────────────────────────────────────────┐
│  L12  CERTIFICATION · EVOLUTION · AUTONOMY                               │
│       Certification Engine · Gap/Completeness Engine · Evolution Engine ·  │
│       Self-healing · Self-optimisation · Learning · Digital Twins · Simulation │
├──────────────────────────────────────────────────────────────────────────┤
│  L11  OPERATIONS · OBSERVABILITY · RESILIENCE · INFRASTRUCTURE           │
│       OTel · SLO · AIOps · DR/BCP · Chaos · FinOps · Cells · K8s · IaC · GitOps │
├──────────────────────────────────────────────────────────────────────────┤
│  L10  EXPERIENCE                                                         │
│       UI Meta-Model · Renderers (web, mobile, desktop, kiosk, POS, TV, voice,  │
│       chat, agent, AR/VR adapters) · Portals · Design System · i18n · a11y │
├──────────────────────────────────────────────────────────────────────────┤
│  L9   INTEGRATION                                                        │
│       API Gateway · GraphQL Federation · gRPC · Webhooks · Connectors · EDI ·  │
│       ETL/CDC · MCP Server · Protocol translation · Federation           │
├──────────────────────────────────────────────────────────────────────────┤
│  L8   RUNTIME FABRIC                                                     │
│       Command Bus · Object Service · Durable Workflow · Decision/Rules · Event Bus ·│
│       Temporal Scheduler · Query · Search · Graph · Vector · AI Runtime · Agents │
├──────────────────────────────────────────────────────────────────────────┤
│  L7   COMPOSITION & GENERATION                                           │
│       Builder Studio · Composition Substrate (resolvers) · Blueprint Store ·  │
│       Generators (SDK, docs, tests, IaC, UI bundles) · Platform Factory  │
├──────────────────────────────────────────────────────────────────────────┤
│  L6   INTELLIGENCE                                                       │
│       Model Registry · Prompt Registry · Agent Registry · Recommendation · Search │
│       AI · Forecasting · Fraud/Risk · Optimisation · Planning · Reasoning │
├──────────────────────────────────────────────────────────────────────────┤
│  L5   BUSINESS CAPABILITY UNIVERSES                                      │
│       Product · Service · Catalog · Offer · Pricing · Commerce · Marketplace · │
│       Order · Inventory · Fulfillment · Logistics · Payment · Ledger/Finance · │
│       Tax · Subscription · Entitlement · CRM · Support · Communication · │
│       Content/Media · Document · Asset · ERP/HRMS/Procurement · Industry Packs │
├──────────────────────────────────────────────────────────────────────────┤
```

```
┌──────────────────────────────────────────────────────────────────────────┐
│  L4   FOUNDATION UNIVERSES                                               │
│       Identity · Organization/Tenant · Geography · Temporal · Currency/Unit ·  │
│       Language/Localization · Context · Resource                         │
├──────────────────────────────────────────────────────────────────────────┤
│  L3   KNOWLEDGE & REGISTRY FABRIC                                        │
│       Ontology · Taxonomy · Universal Knowledge Graph · Registry Fabric ·      │
│       Global Product & Service Intelligence Network · Reference Data · Evidence  │
├──────────────────────────────────────────────────────────────────────────┤
│  L2   DATA FABRIC                                                        │
│       Relational · Document · Graph · Vector · Object · Stream · Time-series ·  │
│       Lakehouse · Data Contracts · Lineage · Quality · Residency · Retention     │
├──────────────────────────────────────────────────────────────────────────┤
│  L1   META-MODEL                                                         │
│       Canonical Object · Field Types · Cardinality · Relationships · Lifecycle ·  │
│       Versioning · Context Envelope · Traceability                       │
├──────────────────────────────────────────────────────────────────────────┤
│  L0   CONSTITUTION & AUTHORITY                                           │
│       Constitution · Laws · Authority Model · Governance · Policy Registry ·     │
│       Decision Records · Delegation · Ratification                       │
└──────────────────────────────────────────────────────────────────────────┘

  CROSS-CUTTING PLANES (apply to every layer):
  ▌TRUST PLANE      Security · Privacy · Compliance · Audit · Risk · Trust/Reputation
  ▌CONTEXT PLANE    Context resolution and propagation (identity, tenant, jurisdiction, t
  ▌OBSERVABILITY    Traces · Metrics · Logs · Business, security, AI and governance signa
  ▌GOVERNANCE PLANE Policy decision & enforcement, change control, certification status
```

5.3 Layer dependency rules

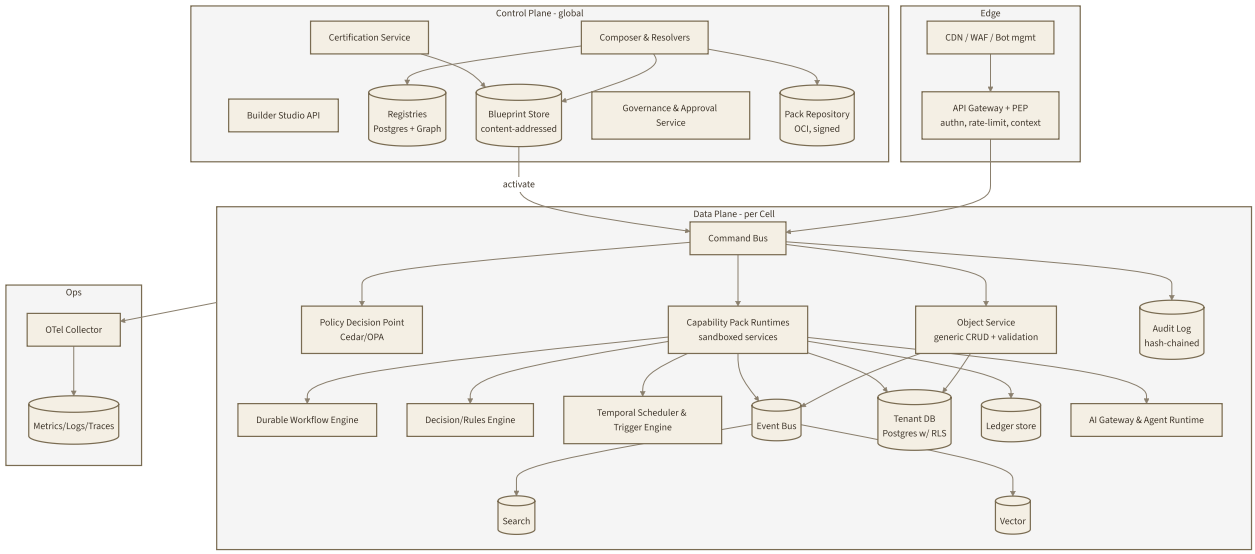




Rules:

1. A layer MAY depend only on layers below it (arrows point upward to dependents). The **only upward flow** is L12 → L0 *proposals*, which still pass through governance.
2. Business universes (L5) MUST NOT depend on each other directly for data ownership. They reference other universes' objects by `ref` (ID + version) and communicate through events and published queries.
3. Foundation universes (L4) MUST NOT depend on any L5 universe.
4. L8 runtime engines are generic. They MUST NOT contain business-universe-specific code; all business behaviour comes from packs and metadata.

5.4 Runtime-view architecture (what actually runs)



Control plane vs data plane

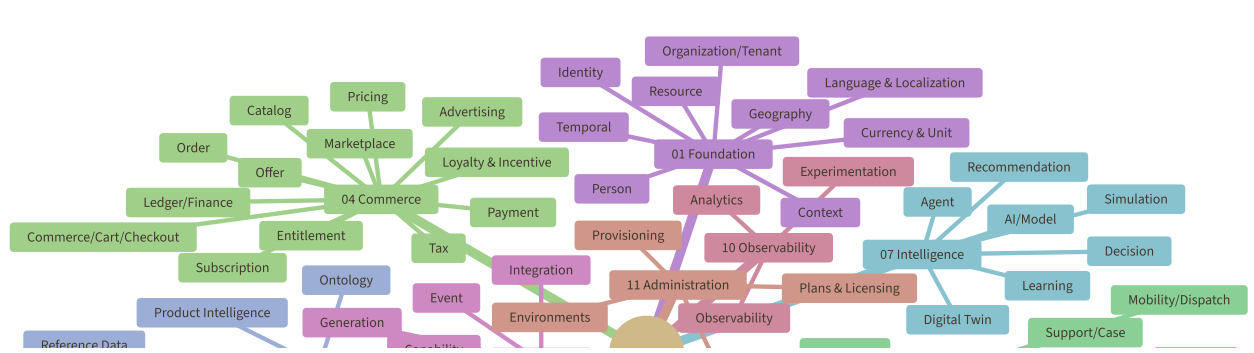
- The **control plane** (global, highly governed) owns definitions: registries, packs, compositions, blueprints, certifications, approvals.
- The **data plane** (one per cell or region, horizontally scaled) executes certified blueprints for tenants. It reads blueprints from a local, signed cache and never mutates definitions.

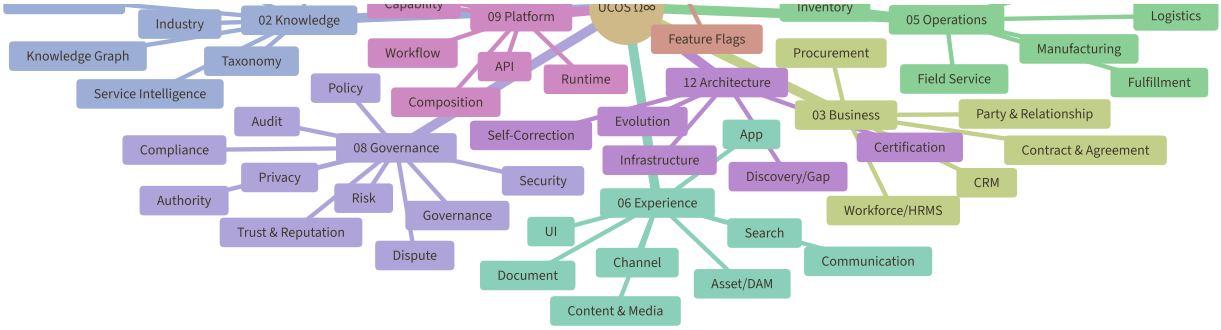
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6. Universe Catalog & Canonical Ownership

6.1 Domains → Universes

The 72 universes named in the Genesis Directive, plus those added in later turns of the chat, are grouped into 12 domains. These domains also form the Platform Console navigation (§24.1).





6.2 Universe catalog (canonical homes)

Legend: **Phase** = delivery phase from §32. **Owns** = the canonical object types for which this universe is the *only* writer.

#	Universe	Layer	Owns (canonical objects)	Depends on	Phase
U01	Authority	L0	Authority, Delegation, Ratification, DecisionRecord	—	0
U02	Governance	L0	ChangeProposal, Approval, Review, Waiver	Authority	0
U03	Policy	L0	Policy, PolicySet, Obligation	Authority	0
U04	Meta	L1	ObjectType, FieldType, RelationshipType, LifecycleModel	—	0
U05	Ontology	L3	Concept, ConceptRelation, Mapping (sameAs/broader/narrower)	Meta	0
U06	Taxonomy	L3	Taxonomy, TaxonomyNode, Classification	Ontology	0
U07	Knowledge Graph	L3	KnowledgeObject, Assertion, Provenance, TrustScore	Ontology	1
U08	Reference Data	L3	CodeList, CodeValue (ISO 3166/4217/639, UN/LOCODE, units)	Meta	0
U09	Identity	L4	Identity, Credential, Session, Role, Permission, Group	Authority, Policy	0
U10	Organization & Tenant	L4	Tenant, Organization, OrgUnit, Workspace, Brand, Store	Identity	0
U11	Person	L4	Person, ContactPoint, Preference	Identity	0
U12	Geography	L4	Place, Address, Jurisdiction, Zone, Geofence. ServiceArea	Reference Data	0

U13	**Temporal**	L4	TimePoint, TimeRange, Duration, Recurrence, Calendar, AvailabilityWindow, Slot, Reservation, SLAClock, Trigger, TemporalPolicy	Geography (timezones), Reference Data	0
U14	Currency & Unit	L4	Currency, ExchangeRate, Money (value type), Unit, Quantity (value type)	Reference Data, Temporal	0
U15	Language & Localization	L4	Locale, Translation, MessageBundle, FormatProfile	Reference Data	0
U16	Context	L4	ContextProfile, ContextRule (resolution rules)	all L4	0
U17	Resource	L4	Resource (people, vehicles, rooms, machines, compute), ResourcePool, Capacity	Temporal, Geography	2
U18	Product (Intelligence)	L3/L5	GlobalProduct (GPID), ProductModel, Variant, Attribute, AttributeValue, Brand, Manufacturer	Taxonomy, Unit	1
U19	Service (Intelligence)	L3/L5	GlobalService (GSID), ServicePlan, ServiceLevel, DeliveryModel	Taxonomy, Temporal	1
U20	Industry	L3	Industry, IndustryTemplate, IndustryStandard	Taxonomy	1
U21	Catalog	L5	Catalog, CatalogEntry/Listing, Collection, Assortment	Product, Service, Organization	1
U22	Offer	L5	Offer, Promotion, Coupon, Voucher, Bundle, Campaign, Incentive	Catalog, Pricing, **Temporal**, Policy	1
U23	Pricing	L5	PriceList, PriceRule, PriceComponent, Quote	Currency, Decision, Temporal	1
U24	Commerce (Cart/Checkout)	L5	Cart, CartLine, Checkout	Catalog, Offer, Pricing, Tax	1
U25	Marketplace	L5	Seller, SellerAgreement, CommissionPlan, MarketplacePolicy	Organization, Ledger	2
U26	Order	L5	Order, OrderLine, OrderAdjustment, ReturnRequest	Commerce, Payment, Fulfillment	1

U27	Subscription	L5	Subscription, SubscriptionPlan, BillingSchedule, UsageRecord	Temporal, Pricing, Payment	2
U28	Entitlement	L5	Entitlement, Feature, License, Right, Quota	Subscription, Identity	2
U29	Payment	L5	PaymentIntent, PaymentMethod (tokenised), Refund, Payout, Dispute (chargeback)	Ledger, Currency, adapters	1
U30	Ledger & Finance	L5	LedgerAccount, JournalEntry, Posting, Settlement, Invoice, CreditNote, RevenueSchedule	Currency, Temporal	2
U31	Tax	L5	TaxRegime, TaxRule, TaxCategory, TaxDetermination, TaxDocument	Geography, Product, Temporal	1
U32	Loyalty & Incentive	L5	LoyaltyProgram, PointsAccount, Reward, Referral	Ledger, Offer	3
U33	Advertising	L5	Campaign(Ad), Placement, Auction(Ad), Creative	Content, Decision	4
U34	Inventory	L5	StockItem, StockLevel, Reservation(Stock), Location(Stock)	Product, Geography	1
U35	Fulfillment	L5	FulfillmentOrder, Shipment, Package, ReturnShipment	Order, Inventory, Logistics	1
U36	Logistics	L5	Carrier, Route, Leg, DeliverySlot (ref Temporal Slot), TrackingEvent, Fleet, Vehicle	Geography, Temporal, Resource	2
U37	Mobility & Dispatch	L5	Trip, RideRequest, Match, DriverShift	Logistics, Geography, Temporal, Pricing	5
U38	CRM	L5	Account, Contact, Lead, Opportunity, Pipeline, Activity	Person, Organization	3
U39	Support & Case	L5	Case, Ticket, Incident, KnowledgeArticle, SLAInstance	Temporal (SLAClock), Communication	3
U40	Communication	L5	Message, Template, Channel, Conversation, Notification, Campaign(Msg)	Person, Localization, Consent	1
U41	Content & Media	L5	ContentItem, MediaAsset, Rendition, Playlist, Stream	Asset, Entitlement	4
U42	Document	L5	DocumentTemplate, Document,	Asset,	2

			Signature, DocumentVersion	Localization	
U43	Asset (DAM)	L5	BinaryAsset, Rendition, AssetRights	Data/Object store	1
U44	Search	L6	SearchIndexDefinition, Synonym, RankingProfile	Knowledge, Vector	1
U45	ERP / Procurement	L5	PurchaseRequisition, PurchaseOrder, Supplier, GoodsReceipt	Ledger, Inventory	4
U46	Workforce / HRMS	L5	Employee, Position, Shift (ref Temporal), Attendance, PayrollRun	Person, Temporal, Ledger	4
U47	Manufacturing	L5	BillOfMaterials, WorkOrder, Routing, ProductionSchedule	Product, Inventory, Temporal	6
U48	Healthcare / Education / Gov (Industry packs)	L5	Industry-specific objects as packs (e.g. Encounter, Course, Permit)	Foundation + relevant universes	6
U49	AI / Model	L6	Model, ModelVersion, Provider, Route, Evaluation, Guardrail	Policy, Security	3
U50	Prompt	L6	PromptTemplate, PromptVersion	AI	3
U51	Agent	L6	Agent, AgentAuthority, Tool, Memory, AgentRun	Identity, Authority, AI	3
U52	Recommendation	L6	RecommenderDefinition, Strategy, Placement(Rec)	AI, Knowledge	4
U53	Decision	L8	DecisionModel (DMN), DecisionTable, DecisionLog	Policy	1
U54	Simulation	L12	Scenario, SimulationRun, Assumption, Outcome	Digital Twin	5
U55	Digital Twin	L12	TwinDefinition, TwinInstance, StateSnapshot	Knowledge, Observability	5
U56	Learning	L12	LearningSignal, Insight, ModelFeedback	AI, Analytics	5
U57	Capability	L7	CapabilityDefinition, CapabilityPack, ExtensionPoint	Meta	0
U58	Composition	L7	PlatformDefinition, Blueprint, GapReport, ResolutionTrace	Capability, Context, Policy	0
U59	Generation	L7	GeneratorDefinition, GenerationRun, GeneratedArtifact	Composition	2

U60	Workflow	L8	WorkflowDefinition, WorkflowInstance, HumanTask	Temporal, Decision	1
U61	Event	L8	EventType, EventSchema, Subscription(Event)	Meta	0
U62	Integration	L9	Connector, ConnectorInstance, Mapping, SyncJob, Webhook	Event, Security	2
U63	API	L9	APIDefinition, APIProduct, APIKey, RateLimitPlan	Capability	1
U64	UI / App / Channel	L10	ViewDefinition, Layout, Component, Theme, App, Portal, Journey	Meta, Localization	1
U65	Security	Plane	SecurityControl, Key, Secret(ref), ThreatModel, Finding	Authority	0
U66	Privacy & Consent	Plane	Consent, Purpose, ProcessingActivity, DataSubjectRequest, RetentionRule	Identity, Compliance	1
U67	Compliance	Plane	Regulation, Clause, Control, ControlMapping, Obligation	Policy, Geography	1
U68	Audit & Evidence	Plane	AuditRecord, Evidence, EvidenceCollector	—	0
U69	Risk	Plane	RiskModel, RiskScore, RiskAssessment	Decision, AI	2
U70	Trust & Reputation	Plane	Rating, Review, TrustScore, Verification (KYC/KYB)	Identity, Risk	2
U71	Dispute & Resolution	L5	Claim, Dispute, Arbitration, Appeal	Order, Payment, Case	3
U72	Observability & Analytics	L11	SignalDefinition, SLO, Dashboard, Metric, Report, Experiment, FeatureFlag	Event	1
U73	Infrastructure & Deployment	L11	Cell, Environment, Deployment, Release	—	0
U74	Certification	L12	CertificationGate, Certification, CertificationRun	Audit & Evidence	0
U75	Discovery & Gap	L12	GapRule, Gap, GapScan	Composition	0
U76	Evolution	L12	EvolutionProposal, Deprecation, Migration	Governance	3

6.3 Ownership resolution of commonly duplicated concepts

These were the main duplication risks in the source material. The rulings below are binding.

Concept	Canonical home	Everyone else…
Any date, time, period, validity, schedule, recurrence, slot, SLA clock, expiry, trigger time	**Temporal (U13)**	stores a `TemporalRef` / `TemporalPolicyRef`, never `start_date` / `end_date` pairs
Money, currency, FX	**Currency & Unit (U14)** — `Money` value type; FX rates are temporal facts	MUST use `Money {amount (decimal string), currency}`
Address, country, region, jurisdiction, timezone	**Geography (U12)**	references a `Place` / `Jurisdiction`
Product master facts (brand, specs, GTIN)	**Product Intelligence (U18)**	tenants create **Listings** (U21) that reference GPID and add tenant-specific data
Price	**Pricing (U23)**	Offer applies adjustments; Catalog never stores price
Discounts/promotions/coupons	**Offer (U22)**	Pricing evaluates offers via the Decision engine
What a customer may use (features, plans, quotas)	**Entitlement (U28)**	Identity permissions are about *actions*; entitlements are about *commercial rights*
Financial truth	**Ledger (U30)**	Payment, Marketplace, Subscription post journal entries; nothing else stores balances
Review/rating	**Trust & Reputation (U70)**	Product pages query it
Notifications of any kind	**Communication (U40)**	other universes emit events + template IDs
Files, images, video	**Asset (U43)**	others hold `AssetRef`
Approval chains	**Workflow (U60)** + **Governance (U02)** for definition changes	—
Rules and thresholds	**Decision (U53)**	no `if` statements on business thresholds inside

		business thresholds inside pack code
Catalog vs Product	Product = global truth; Catalog = a tenant's curated assortment	—
Marketplace vs Commerce	Commerce = buying flow; Marketplace = multi-sided economics (sellers, commissions, payouts)	—

PART II

THE META-SYSTEM

7. Universal Meta-Model

Everything in UCOS is a **Canonical Object**: definitions (object types, capabilities, policies, workflows, UI views) and instances (orders, products, people) alike. One meta-model lets the platform reason about itself.

7.1 Root Canonical Object envelope

Every object, of any type, carries this envelope. Type-specific data lives in `attributes`.

```
# ucos:meta:CanonicalObject  (JSON Schema 2020-12 in the registry)
CanonicalObject:
  id:            UGID            # required, immutable — §7.2
  type:          TypeRef         # e.g. "commerce.order.Order@1.4"
  subtype:       string?         # taxonomy-driven specialisation
  tenant:        TenantRef       # "ucos:tenant:01J..." ; "global" for shared regis
  name:          LocalizedText?
  description:   LocalizedText?
  status:        LifecycleState  # §7.5
  version:       integer         # optimistic-concurrency counter (instances)
  defVersion:    SemVer?         # for definitions only
```

```
  owner:         PartyRef        # accountable party (org/person/agent)
  steward:       PartyRef?       # for definitions & registries — required there
  context:       ContextRef      # §7.6 — context in which it was created/valid
  attributes:    object          # validated against the ObjectType schema
  relationships: RelationshipRef[]  # §7.4 (materialised for hot paths; full set ir
  policies:      PolicyRef[]     # attached policies (beyond type-level ones)
  classification:                # §19.7 data classification
    confidentiality: public|internal|confidential|restricted
    dataClasses: [PII, PCI, PHI, FINANCIAL, ...]
  compliance:    { jurisdictions: [JurisdictionRef], controls: [ControlRef] }
  temporal:                      # bitemporal — §14.6
    validFrom: TimePoint ; validTo: TimePoint?
    recordedAt: TimePoint ; supersededAt: TimePoint?
  provenance:    { source: SourceRef, method: manual|import|generated|ai, confidenc
  audit:         { createdBy, createdAt, updatedBy, updatedAt, lastAuditRecord: Aud
  evidence:      EvidenceRef[]
  metadata:      { tags: string[], labels: map<string,string>, extensions: object }
```

7.2 Universal identity scheme (UGID)

The source listed many identifier families (UID, GID, PID, OID, CID, BID, SID, DID, AID, RID, EID, TID, KID, GPID, GSID). They are unified as **one format with a kind prefix**:

```
ucos:<kind>:<ULID>                          e.g. ucos:order:01J9Z3K8Q4V6X2...
ucos:<kind>:<namespace>/<natural-key>       e.g. ucos:currency:iso4217/INR   (refere
```

Legacy name	Kind prefix	Notes
UGID	(all)	the general form
UID / PID	`person` , `identity`	Person vs authenticating Identity are separate
OID / BID	`org` , `brand`	
TID	`tenant`	
CID	`customer` → modelled as `party-role`	Customer is a *role* a Party plays, not a separate identity
SID	`service` (GSID for global services)	
GPID	`gproduct`	Global Product ID (§10)
AID	`agent`	AI agent identity

DID	`did:` W3C DID (optional)	for verifiable credentials and cross-org trust
DeviceID	`device`	
EID / KID / RID	`event` , `knowledge` , `resource`	

Rules: IDs are opaque, immutable and never reused. Natural keys (GTIN, ISBN, VIN, LEI, tax IDs) are stored as `identifiers[]` attributes with a scheme, never as the primary ID. ULIDs give time-ordered, index-friendly keys.

7.3 Object Type definition (how new things are modelled without code)

```
# registry: ucos:objecttype:commerce.order.Order@1.4.0
kind: ObjectType
id: commerce.order.Order
version: 1.4.0
universe: U26-order
steward: ucos:org:platform-commerce-team
extends: ucos.meta.CanonicalObject
description: "A confirmed commitment by a buyer to purchase offers from one or more
attributes:
  orderNumber:   { type: String, pattern: "^[A-Z0-9-]{6,32}$", generated: sequence(
  buyer:         { type: Reference, target: party.PartyRole, cardinality: "1", cons
  channel:       { type: Reference, target: experience.Channel, cardinality: "1" }
  lines:         { type: Composition, target: commerce.order.OrderLine, cardinality
  totals:        { type: Object, schemaRef: commerce.order.OrderTotals }   # Money
  placedAt:      { type: TemporalRef, kind: TimePoint }                    # Tempora
  fulfillmentWindow: { type: TemporalRef, kind: TimeRange, optional: true }
  shippingAddress:   { type: Reference, target: geo.Address, optional: true }
  customAttributes:  { type: Extension, governedBy: tenant }               # tenant-
indexes:
  - [tenant, orderNumber]  # unique
  - [tenant, buyer, placedAt]
lifecycle: commerce.order.OrderLifecycle@1.0   # state machine (§16.1)
classification: { confidentiality: confidential, dataClasses: [PII, FINANCIAL] }
retention: ucos:retention:financial-records-default   # resolved per jurisdiction
events: [order.order.placed, order.order.confirmed, order.order.cancelled, order.or
commands: [PlaceOrder, ConfirmOrder, CancelOrder, AmendOrder]
uiModels: [order.list@1, order.detail@1, order.timeline@1]
```

```
search: { indexed: [orderNumber, buyer.name, lines.title], facets: [status, channel
```

7.4 Field types, cardinality and relationships

Canonical field types: `String, Text, LocalizedText, Number, Integer, Decimal, Boolean, Enumeration (taxonomy-backed), Money, Quantity (value+unit), Percentage, TemporalRef (TimePoint|TimeRange|Duration|Recurrence|TemporalPolicy), Coordinate, GeoShape, Reference, Composition, Array, Object, Map, Binary, AssetRef, Media, Json, Graph, PolicyRef, RuleRef, WorkflowRef, ModelRef, Extension, Computed, Encrypted<T>` .

Types the source called "Date / Time / DateTime / Duration / Currency / Location" are deliberately **not primitive**. They are value types owned by the Temporal, Currency and Geography universes, which enforces Knowledge Once.

Cardinality: `1` , `0..1` , `1..N` , `0..N` , plus modifiers `conditional(expr)` , `inherited` , `derived(expr)` , `computed(expr)` , `generated(strategy)` .

Relationship object:

```yaml
Relationship:
  id: UGID
  type: RelationshipTypeRef        # from the registry below
  source: ObjectRef                # id + type + version
  target: ObjectRef
  context: ContextRef
  validity: TemporalRef            # TimeRange
  constraints: ExpressionRef[]
  attributes: object               # e.g. quantity, share %, role
  evidence: EvidenceRef[]
  audit: AuditRef
```

Relationship type registry (seed set, extensible):

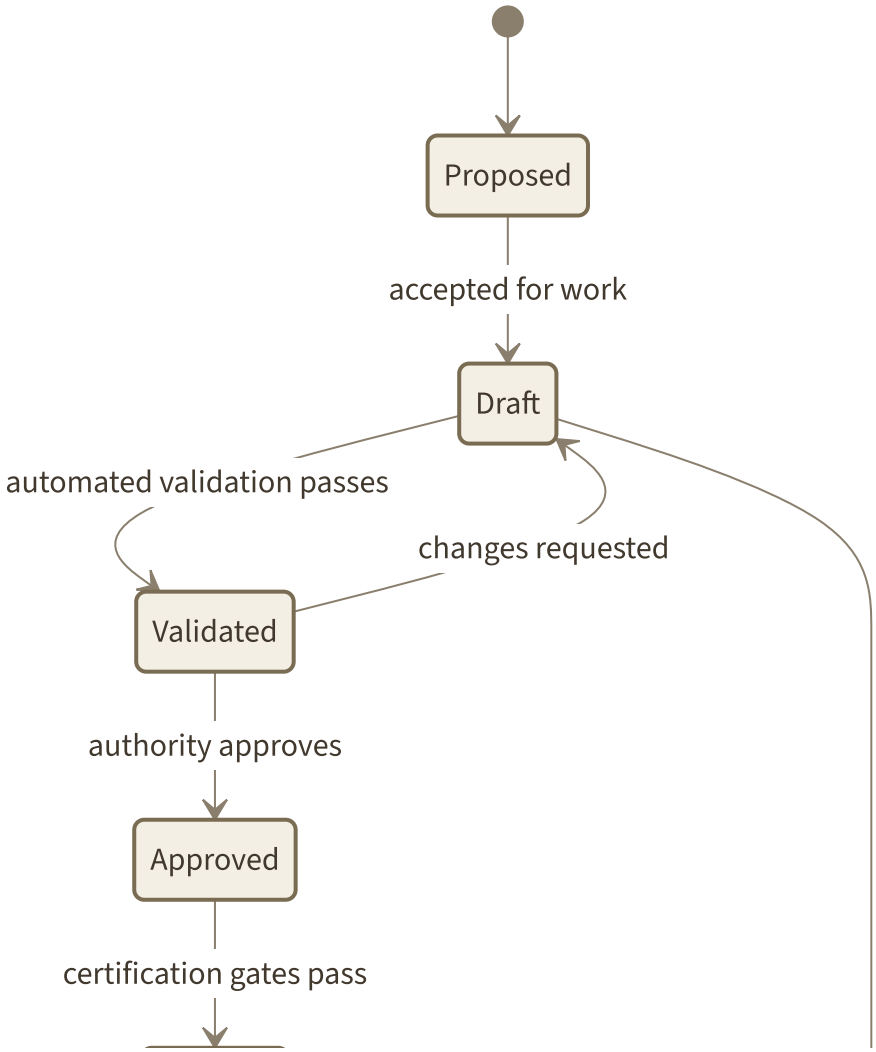
Category	Types
Structural	`contains` , `partOf` , `composedOf` , `aggregates` , `inheritsFrom` , `versionOf` , `supersedes` , `replaces` , `equivalentTo (sameAs)` , `similarTo`
Ownership & authority	`owns` , `belongsTo` , `memberOf` , `stewardOf` , `authorizes` , `delegatesTo` , `approves` , `certifies` , `governs` , `controls`

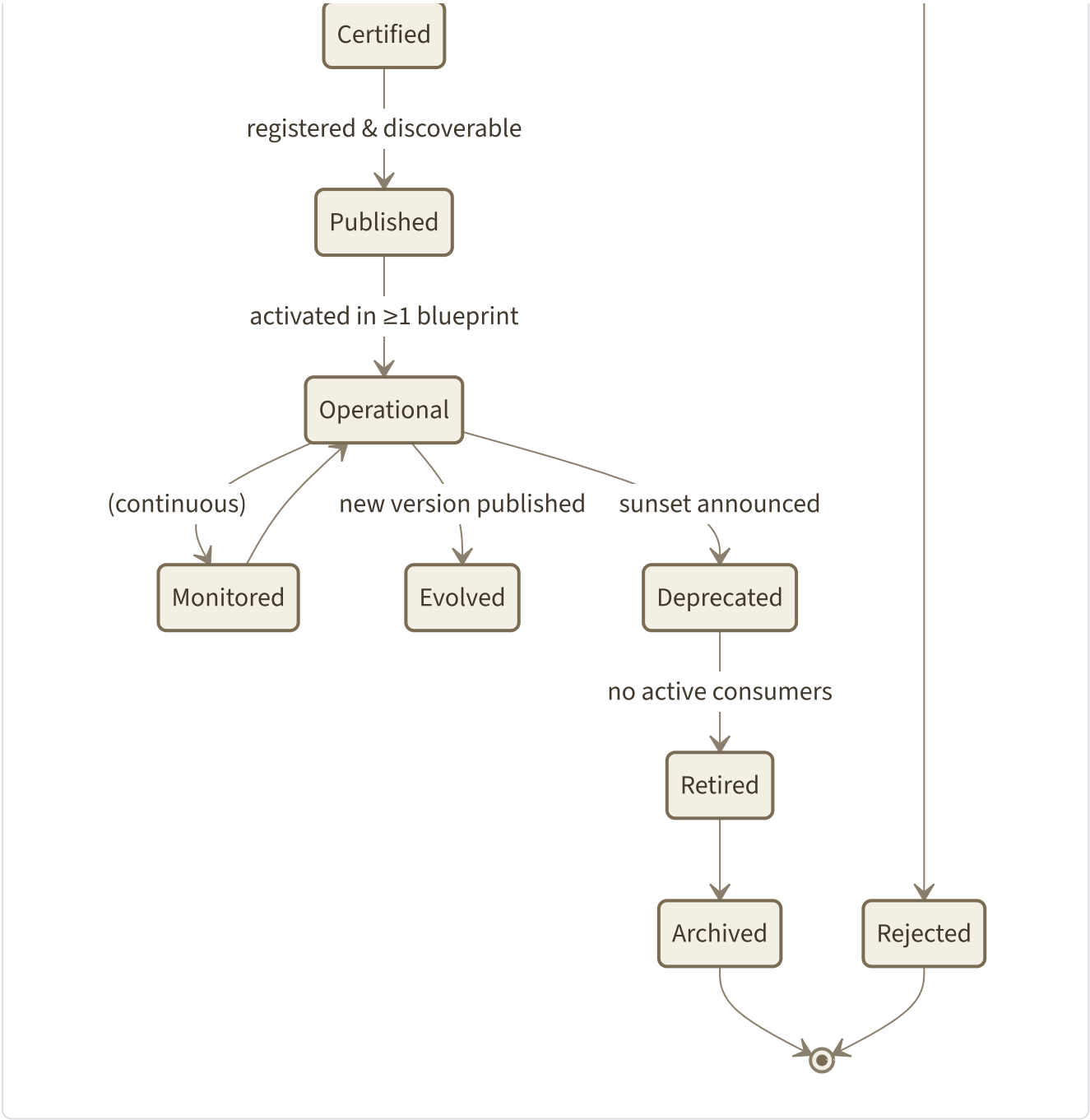
Dependency	`dependsOn` , `requires` , `provides` , `consumes` , `produces` , `extends` , `conflictsWith` , `references`
Operational	`executes` , `monitors` , `observes` , `evaluates` , `validates` , `verifies` , `audits` , `learnsFrom`
Commercial	`sells` , `purchases` , `offers` , `ships` , `settles` , `books` , `schedules` , `subscribesTo` , `reviews` , `rates` , `supports` , `fulfills`
Social	`communicatesWith` , `collaboratesWith` , `follows` , `refers`

Every relationship type declares `domain` , `range` , `inverse` , `cardinality` , `transitive?` , `symmetric?` , and whether it is **materialised** (stored with the object for fast reads) or **graph-only**.

7.5 Universal lifecycle

Two lifecycles exist. **Definitions** (types, capabilities, policies, workflows, views, packs) follow the governed lifecycle below. **Instances** (orders, products…) follow their own type-specific state machines (§16).





Transition guards (examples):

- `Validated → Approved` requires an `Approval` from an authority whose scope covers the definition's universe (§18).
- `Approved → Certified` requires every applicable `CertificationGate` (§28.2) to be green.
- `Deprecated → Retired` requires `activeConsumers == 0` (computed from the graph) or a forced migration (§29.4).

7.6 Context model

Every request, event, workflow step and AI call carries a **ContextEnvelope**. Context is resolved once at the edge and propagated (W3C Baggage + signed header `ucos-context`).

context).

```
ContextEnvelope:
  tenant: TenantRef
  platformInstance: BlueprintRef      # which certified blueprint is executing
  actor:  { identity: IdentityRef, kind: human|service|agent|device, onBehalfOf: Id
  roles: RoleRef[]
  organization: OrgRef?
  jurisdiction: JurisdictionRef[]     # resolved from buyer/seller/store/data locat
  locale: LocaleRef                   # language + region formats
  currency: CurrencyRef
  timezone: TZRef
  time: { now: TimePoint, asOf: TimePoint? }   # asOf enables time-travel queries &
  channel: ChannelRef                 # web, ios, android, pos, voice, agent, api..
  device: { id, trust: low|medium|high, fingerprintRisk }
  geo: { coordinate?, place? }
  risk: { score: 0..100, signals: [] }
  consent: ConsentSnapshotRef
  simulation: boolean                 # true → no external side effects (§29.6)
  trace: { traceId, spanId }
```

Context dimensions (consolidated from all versions): Identity, Organizational, Tenant, Temporal, Spatial/Geographic, Jurisdictional/Regulatory, Linguistic, Cultural, Economic/Financial, Commercial/Marketplace, Channel/Device, Security, Risk, Behavioral, Operational, Environmental, Knowledge, AI, Observer, Reality (physical/digital/simulated), Hypothetical/Simulation.

Dimensions that need no runtime fields (Cultural, Observer, Reality) are resolved through `ContextRule`s. For example, `if locale=ar-* then layoutDirection=rtl`, or `if simulation=true then reality=simulated`.

7.7 Traceability model

Every artifact MUST be reachable in the graph along this chain (the "Graph of Everything"):



`ucos trace <id>` returns the full chain. A missing link is reported as a Gap of kind

```
TRACEABILITY (§11.6).
```

---

8. Capability Genome & Capability Pack Specification

8.1 The genome

Every capability follows one structure. This solves the gap the source repeatedly called "the biggest missing artifact", the **Universal Capability Ontology**:

```
Capability
 ├── Identity & Purpose          id, version, universe, steward, purpose, outcomes
 ├── Constitution binding        laws/policies it implements
 ├── Ontology & Taxonomy         concepts it introduces/uses
 ├── Data Model                  object types provided / extended / referenced
 ├── Commands & Queries          API model (sync)
 ├── Events                      emitted / consumed
 ├── Workflows                   state machines, human tasks
 ├── Decisions                   decision tables / models it exposes for configuration
 ├── Temporal bindings           triggers, SLAs, schedules (by reference)
 ├── Policies                    default authz policies, obligations
 ├── Security model              data classes, scopes, threat model ref
 ├── Compliance model            control mappings
 ├── AI model                    agents/tools/models it offers or uses
 ├── UI model                    views, components, journeys
 ├── Integration                 ports (outbound) + adapters it supports
 ├── Configuration schema        what a tenant may configure
 ├── Dependencies                requires / optional / conflicts / extensionPoints
 ├── Analytics & Signals         metrics, SLOs, business KPIs
 ├── Tests                       contract, unit, scenario, security, perf
 ├── Documentation               user, admin, API, runbooks
 ├── Cost & capacity model       cost drivers, scaling hints
 ├── Risks                       known risks + mitigations
 ├── Certification               required gates
 └── Evolution path              deprecations, successors, migrations
```

8.2 Capability manifest (`capability.yaml`): full example

```yaml
apiVersion: ucos.io/v1
kind: Capability
metadata:
  id: commerce.cart
  version: 2.3.0
  universe: U24-commerce
  steward: ucos:org:team-commerce-core
  purpose: "Collect offers a buyer intends to purchase, price them in context, and
  outcomes: [cart-conversion, accurate-pricing]
  maturity: certified            # experimental | beta | certified | deprecated
implements:
  laws: [L1, L2, L7, L8, L13]
spec:
  provides:
    objectTypes:
      - commerce.cart.Cart@2
      - commerce.cart.CartLine@2
    commands: [CreateCart, AddLine, UpdateLineQuantity, RemoveLine, ApplyCoupon, Re
    queries:  [GetCart, GetCartByBuyer]
    events:
      emits:    [commerce.cart.created, commerce.cart.line-added, commerce.cart.lir
      consumes: [catalog.listing.unpublished, pricing.pricelist.changed, inventory.
    decisions:
      - id: commerce.cart.max-lines         # tenant-configurable decision table
      - id: commerce.cart.abandonment-rule
  requires:
    - capability: catalog.listing        version: "≥1.2 <2"
    - capability: pricing.engine         version: "≥3"
    - capability: identity.core          version: "≥1"
    - capability: temporal.core          version: "≥1"     # abandonment timer, co
  optional:
    - capability: offer.engine           enables: [ApplyCoupon]
    - capability: inventory.availability enables: [real-time-stock-check]
    - capability: tax.determination      enables: [tax-inclusive-estimates]
  conflicts: []
  extensionPoints:
    - id: commerce.cart.line-validators   # other packs may contribute validators
      contract: ucos.ext.Validator@1
    - id: commerce.cart.ui.line-widgets
      contract: ucos.ext.UIWidget@1
```

```
    contract: ucos.ext.uiwidget@1
temporal:
  triggers:
    - id: cart-abandonment
      kind: inactivity
      durationRef: decision:commerce.cart.abandonment-rule.idleDuration   # defau
      emits: commerce.cart.abandoned
configuration:
  schema:
    guestCheckoutAllowed: { type: Boolean, default: true }
    cartTTL: { type: TemporalRef, kind: Duration, default: "P30D" }
    mergeOnLogin: { type: Enumeration, values: [merge, replace, ask], default: me
policies:
  defaults:
    - policy: commerce.cart.owner-only     # buyer can read/write own cart
    - policy: commerce.cart.agent-scope    # agents need delegated 'cart:write' a
security:
  dataClasses: [PII]          # buyer link
  scopes: [cart:read, cart:write]
  threatModel: docs/threat-model.md
compliance:
  controls: [ucos.ctrl.data-minimisation, ucos.ctrl.audit-logging]
ai:
  tools:
    - id: cart.add_line          # exposed via MCP/agent tool registry
      command: AddLine
      authority: cart:write
ui:
  views: [cart.mini@1, cart.page@2, cart.drawer@1]
  journeys: [buy-flow@1]
signals:
  metrics: [cart_created_total, cart_conversion_ratio, cart_abandon_ratio, add_li
  slos:
    - name: AddLine latency
      objective: "p99 < 150ms"
      window: 28d
tests:
  contract: tests/contract/
  scenarios: tests/scenarios/*.feature
  security: tests/security/
```

```
    performance: { profile: tests/perf/cart.k6.js, target: "2k rps/cell" }
  docs: { user: docs/user.md, admin: docs/admin.md, api: generated, runbooks: docs/
  cost:
    drivers: [carts_active, lines_per_cart]
  risks:
    - id: price-drift
      mitigation: "Re-price at ConvertToCheckout; display 'price changed' notice."
  certification:
    gates: [ARCH-*, SEC-*, CMP-*, TEST-*, PERF-*, A11Y-*, DOC-*, OBS-*]
  evolution:
    deprecates: []
    migrations: [migrations/2.2-to-2.3.yaml]
```

8.3 Pack anatomy and packaging

```
packs/commerce.cart/
├── capability.yaml            # the genome (above)
├── schemas/                   # object types, value types (YAML → JSON Schema)
├── api/                       # OpenAPI fragments, GraphQL SDL, protobuf
├── events/                    # AsyncAPI + CloudEvents schemas
├── workflows/                 # workflow DSL
├── decisions/                 # DMN / decision tables with defaults
├── policies/                  # Cedar policies (+ OPA for infra)
├── ui/                        # view definitions (UI meta-model), widgets
├── src/                       # pack behaviour (only place code lives)
│   ├── handlers/              # command handlers, event handlers
│   ├── ports/                 # outbound interfaces (e.g. PaymentProviderPort)
│   └── adapters/              # vendor adapters (only here can vendor SDKs be imported)
├── migrations/
├── tests/
├── docs/
└── i18n/                      # message bundles (no hard-coded strings in src/ or ui/)
```

- Packs are built into an **OCI artifact**, then signed with Sigstore/cosign. Each carries an SBOM (CycloneDX) and SLSA level-3 provenance.
- Pack runtime: a containerised service implementing the **Pack Runtime Contract** (§13.4), or a WASM module for lightweight extension points (validators, computed fields). WASM is strongly sandboxed.

- Publishing a pack requires certification (§28) and registers it in the Capability Registry and the Knowledge Graph.

8.4 Dependency semantics

Keyword	Meaning	Resolver behaviour
`requires`	Hard dependency with a SemVer range	Pull into the closure. If unsatisfiable → `MANDATORY` gap.
`optional`	Unlocks listed features when present	Not pulled automatically. Suggested if the PD intent mentions an enabled feature.
`conflicts`	Cannot coexist	Conflict gap; the builder must choose.
`provides`	Object types/commands/events this pack is the **sole** owner of	Collision with another pack → publish rejected (L4).
`extensionPoints`	Typed slots others may fill	Contributions are validated against the contract.
`implements` (interface)	Pack implements an abstract capability, e.g. `payment.provider`	Several implementations may coexist (adapters). Context selects one at runtime (e.g. country → PSP).

Abstract vs concrete capabilities: `payment.provider` is abstract; `payment.provider.stripe`, `payment.provider.razorpay` and `payment.provider.adyen` implement it. The Composition Substrate chooses implementations from context rules. For example, `jurisdiction=IN ∧ method=UPI → razorpay|payu`. That is how "no payment gateway is hard-coded" is achieved.

8.5 Capability Registry & Marketplace

The Capability Marketplace (Layer 005 of the source) is the Capability Registry plus a storefront:

- **Discovery:** search by outcome, industry, object type, compliance coverage, maturity, rating, cost.
- **Packaging:** packs, plus **Capability Bundles**, which are curated sets such as "Marketplace Starter" or "Subscription Media".
- **Licensing & billing:** each pack can declare a price model (free, per-tenant, usage-based, revenue-share). This is metered by the Entitlement and Ledger universes, so third-party pack vendors can be paid.
- **Trust:** publisher verification, certification badge, security score, runtime health and adoption.

- **Governance:** pack-level policies, allowed-publisher lists per tenant, and kill-switch (revocation propagates to all cells within a defined SLA).

---

9. Universal Temporal Universe

The chat elevated Temporal to a **top-tier constitutional universe** and did not leave it as a Calendar feature. This section is its build specification. **Every** universe that deals with time consumes it.

9.1 Canonical temporal objects

Object	Definition	Representation
TimePoint	An instant	RFC 3339 UTC + original TZ: `{ "utc": "2026-10-01T06:30:00Z", "tz": "Asia/Kolkata" }`
LocalDateTime	Wall-clock time without an instant (e.g. "store opens 10:00")	`{ "local": "10:00", "tz": "Asia/Kolkata" }` resolved per date
TimeRange	Interval `[start, end)`	`{ start: TimePoint, end: TimePoint? }` (open-ended allowed)
Duration	Length of time	ISO 8601 `PT2H` , `P3D` , `P2M` , plus business-time variants `{ value: "P2D", calendar: "ucos:cal:in-business" }`
Recurrence	Repeating pattern	RFC 5545 RRULE + EXDATE + calendar binding: `FREQ=WEEKLY;BYDAY=SA,SU`
Calendar	Named set of working/non-working periods	Types: Gregorian, Fiscal (4-4-5, custom FY), Holiday (country/region/tenant), Business-hours, Resource, Settlement, Trading session
AvailabilityWindow	When a resource/offer/store can be used	`Recurrence × TimeRange × Calendar × Capacity`
Slot	A bookable unit derived from availability	`{ resource, range, capacity, remaining }`
Reservation	A claim on slot capacity	`{ slot, holder, status: held
SLAClock	Elapsed business time against a target, with pauses	`{ target: Duration, calendar, pausedOn: [states], breachAt }`
Trigger	A temporal or hybrid condition	§9.3

	that fires an event	
Timeline	Ordered events of an object (past, present, scheduled future)	Projection over events and schedules
TemporalPolicy	**Reusable** bundle of validity + recurrence + windows + timezone + calendar	§9.2
Forecast	Predicted temporal series	Produced by the Forecasting engine, stored as time-series

9.2 TemporalPolicy (Knowledge Once in practice)

Never model `offer.start_date/end_date`, `coupon.start_date/end_date`, `subscription.start_date/end_date` or `booking.start_date/end_date` independently.

```yaml
kind: TemporalPolicy
id: ucos:tmp:TMP-001
name: "Weekend Evening Window — India"
tenant: ucos:tenant:acme
validity:   { start: "2026-01-01T00:00:00+05:30", end: "2026-12-31T23:59:59+05:30" }
recurrence: "FREQ=WEEKLY;BYDAY=SA,SU"
window:     { local: "10:00-22:00" }
timezone:   Asia/Kolkata
calendar:   { include: [], exclude: [ucos:cal:in-national-holidays-2026] }
blackouts:  [ { start: "2026-11-01", end: "2026-11-03" } ]   # Diwali stock freeze
precedence: 100
```

The same policy is referenced by 100 offers, 50 campaigns, 20 coupons, loyalty multipliers, seller incentives and marketplace promotions:

```yaml
kind: Offer
id: ucos:offer:01J...
type: percentage-discount
temporalPolicy: ucos:tmp:TMP-001          # ← reference, not dates
eligibilityPolicy: ucos:pol:new-buyers
pricingPolicy: { percent: 15, cap: { amount: "500.00", currency: INR } }
governancePolicy: ucos:pol:marketing-approval-required
```

9.3 Trigger Engine

Trigger kind	Example	Implementation
Time	"At 10:00 Asia/Kolkata daily"	Durable timer (scheduler shards)
Recurrence	"Every Monday", "Every month end", "Every festival (holiday calendar tag)"	RRULE expansion, materialised in a 90-day rolling horizon
Relative	"2 hours after payment", "3 days before expiry"	Event-anchored timer: `anchor: payment.payment.captured, offset: PT2H`
Inactivity	"Cart idle 2h"	Timer reset on matching events
Event	"After shipment"	Event subscription
Condition	"If inventory < 10", "If risk score > 90"	Continuous query on a stream (stateful stream processor)
Hybrid	"If SLA breached during business hours"	Composition of the above via Decision engine
Geo-temporal	"When driver enters geofence between 18:00–22:00"	Geo stream + window
Policy/Compliance	"Retention period elapsed", "Certificate expires in 30 days"	Temporal facts on objects
AI	"When forecast demand exceeds capacity"	Forecast series + condition trigger

Triggers are **data**: `Trigger { kind, spec, target: (event | command | workflowSignal), idempotencyKey, misfirePolicy: fire-once|skip|catch-up, owner }`.

Guarantees: at-least-once delivery with idempotency keys; misfire handling defined per trigger; clock source is cell-synchronised NTP/PTP with monotonic guards; DST transitions follow the IANA tz database rules (a nonexistent local time shifts forward, an ambiguous one uses the first occurrence unless specified).

9.4 Temporal consumers (binding map)

Domain	Uses
Commerce/Offer	offer validity, flash sales (10:00–11:00 every Friday), coupon expiry, campaign schedules, price-change effective dates, inventory allocation windows

Subscription/Billing	renewal recurrence, billing cycles, proration (Duration math), grace periods, dunning schedules
Logistics	pickup windows, delivery slots, transit durations, cut-off times, driver shifts, route schedules
Marketplace	seller onboarding windows, auction/bid periods, settlement periods (T+N on settlement calendar)
Finance	interest accrual (day-count conventions), EMI schedules, trading sessions, fiscal periods, period close
ERP/HRMS	payroll cycles, attendance, leave, shifts, production schedules
CRM/Support	follow-ups, reminders, SLA clocks, escalation timelines
AI	training windows, retraining cadence, feature freshness
Security	credential, session, token and certificate expiry; key rotation schedules
Compliance	retention periods, filing deadlines, audit periods

9.5 Universal Calendar Product (generated from Temporal)

A Calendar capability (`temporal.calendar`) renders any calendar-typed object: personal (with Google/Outlook sync connectors), resource (rooms, vehicles, machines), commerce (sales, offers, campaigns), logistics (delivery/pickup), workforce (shifts, attendance), manufacturing (production), financial (fiscal, settlement), AI (training). It provides the views `calendar.month|week|day|agenda|timeline|resource-grid` (see wireframe W10).

9.6 Temporal governance, simulation and twin

- **Governance:** timezone database version pinned per release; holiday calendars have stewards per country and are versioned; fiscal calendars are approved by Finance authority.
- **Simulation:** `ContextEnvelope.time.asOf` lets any decision, price or availability be evaluated "as of" a past or future time. Scenarios such as "what if a 20% discount runs every weekend in Q4" run against the Simulation universe (§29.6).
- **Temporal twin questions:** what happened (event store), what is happening (live state), what will happen (schedules), what could happen (simulation), what should happen (optimisation).

9.7 Temporal API (excerpt)

```
POST /v1/temporal/availability:query
{ "resources": ["ucos:resource:room-12"], "range": {"start":"2026-10-05T00:00:00+05
  "duration": "PT30M", "calendar": "ucos:cal:clinic-hours" }
→ 200 { "slots": [ {"start":"...","end":"...","capacity":1,"remaining":1}, ... ] }

POST /v1/temporal/reservations          (Idempotency-Key required)
{ "slot": "ucos:slot:01J...", "holder": "ucos:person:01J...", "hold": "PT10M" }

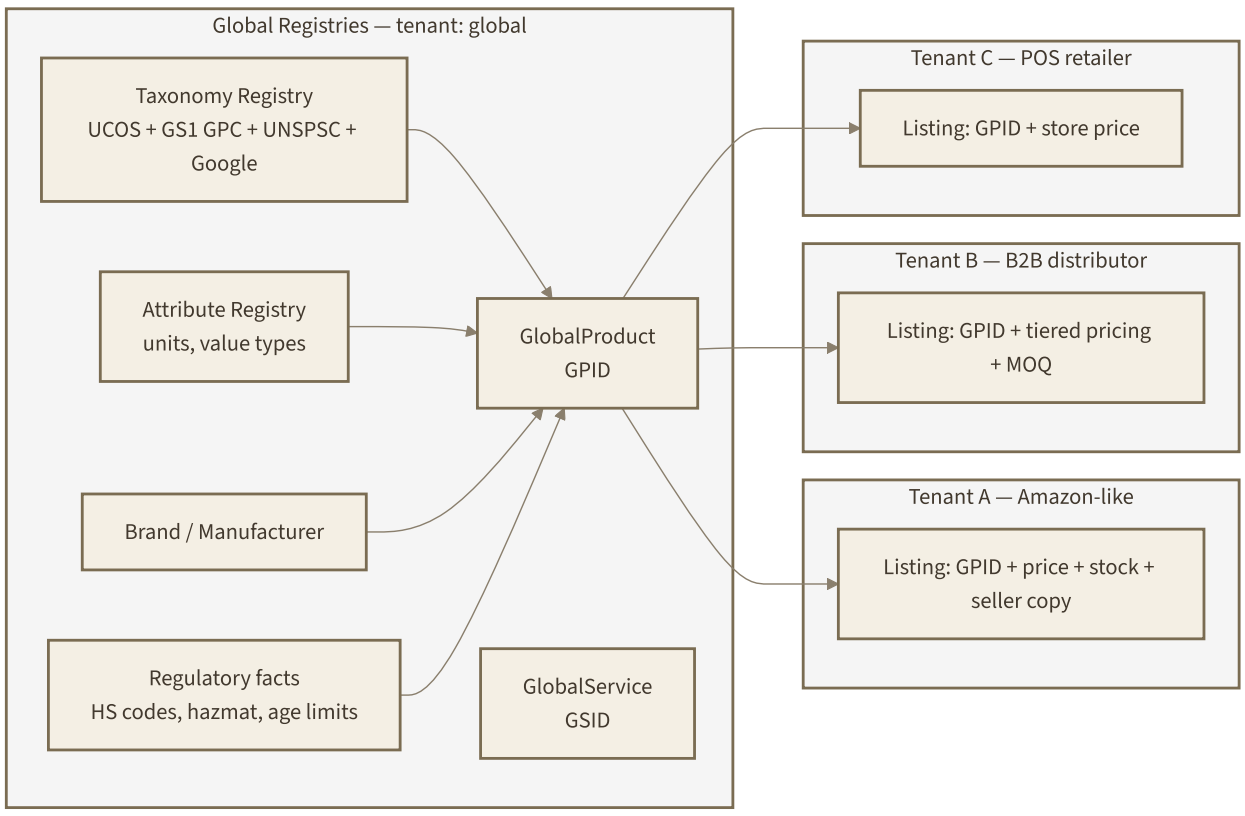
POST /v1/temporal/policies:evaluate
{ "policy": "ucos:tmp:TMP-001", "at": "2026-10-03T19:00:00+05:30" }
→ 200 { "active": true, "nextChange": "2026-10-03T22:00:00+05:30" }
```

---

10. Global Product & Service Intelligence Network

The chat asked for "every product and service in existence available for reuse". It is built as a **knowledge network** with provenance, not as a static catalog.

10.1 Master vs tenant separation



Example: **"Apple iPhone 17, 256 GB, Black"** exists **once** as `ucos:gproduct:01J…` with GTINs, specs, compliance facts and media rights. An Amazon-like store, a Flipkart-like store, a B2B marketplace, an ERP, a POS, a WMS, analytics, the recommendation engine and AI agents all reference that GPID. Tenant-specific facts (price, stock, seller description, warranty terms) live in tenant **Listings**.

10.2 Global product model

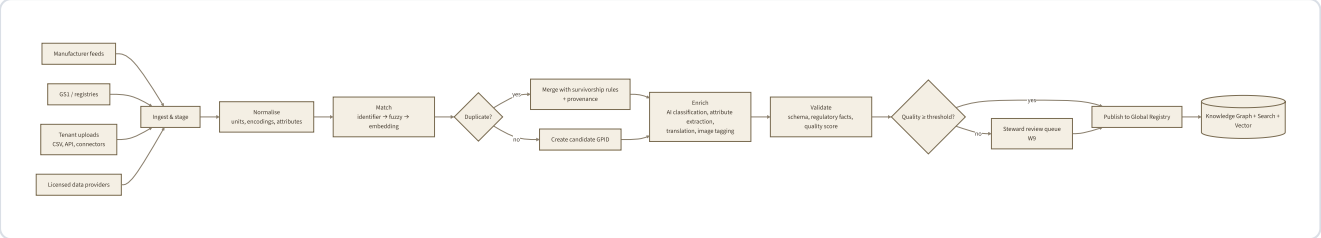
```
GlobalProduct:
  id: ucos:gproduct:01J...
  identifiers: [ {scheme: GTIN-13, value: "0195949xxxxxx"}, {scheme: MPN, value: "M
  brand: ucos:brand:apple
  manufacturer: ucos:org:apple-inc
  productModel: ucos:gproductmodel:iphone-17        # family/model level
  variantAxes: [storage, color]
  variantValues: { storage: {value: 256, unit: GB}, color: ucos:color:black }
  classification:
    ucos: ucos:tax:electronics/phones/smartphones
    gs1-gpc: "10008047"
    unspsc: "43191501"
    hs: "8517.13"
  attributes:                               # each key is a registered Attribute
    display.diagonal: { value: 6.3, unit: in }
    battery.capacity: { value: 3600, unit: mAh }
    connectivity.5g: true
  regulatory: { batteryHazmat: UN3481, eWasteCategory: "...", minAge: null }
  media: [ucos:asset:...]                   # with rights metadata
  provenance: { sources: [manufacturer-feed, gs1-registry, tenant-contribution], co
  lifecycle: { status: Published, launchedOn: TemporalRef }
```

Services follow the same pattern (`GlobalService` with `GSID`): consulting, transport, healthcare, education, streaming, banking, telecom, legal, repair, AI services, robot services. Each has a delivery model (on-site, remote, hybrid, async), SLA templates, scheduling requirements (Temporal) and capacity semantics (Resource).

Attribute Registry: every attribute has an id, value type, unit dimension (length, mass, power, data size, currency, rate…), allowed values (taxonomy-backed enumerations), localisation and validation. Physical, digital, financial, behavioural

operational and regulatory attribute families exist. New attributes are added as data.

10.3 Ingestion & enrichment pipeline



- **Survivorship rules** (data, per attribute): `manufacturer > gs1 > licensed > tenant > ai` , overridable by a steward.
- **Quality score:** completeness, consistency, freshness, source trust, conflict count.
- **AI enrichment** is always flagged `provenance.method=ai` with a confidence value. Facts below the confidence threshold are never auto-published.
- **Legal:** a crawled or licensed source carries a licence and usage rights. The registry enforces rights on redistribution to tenants. "Auto crawl" is allowed only for sources whose terms permit it.

10.4 Commercial intelligence (attached, not merged)

Pricing intelligence (market price ranges), demand signals and competitor observations are separate `KnowledgeObject` s linked to GPIDs. They carry their own provenance and trust score, and are aggregated across tenants **only** with consent and k-anonymity thresholds (§20.5).

10.5 Industry & template registry

Industry templates bundle a taxonomy subset, attribute sets, typical capability bundles, workflows, compliance packs and UI templates. Starter templates: Retail, Grocery, Fashion, Electronics, Pharmacy, Restaurants/Food delivery, Mobility, Hospitality, Travel, Healthcare clinic, Education/LMS, Media/OTT, Fintech-brokerage, B2B distribution, Manufacturing, Government services. Using a template pre-fills a Platform Definition (§11.2).

---

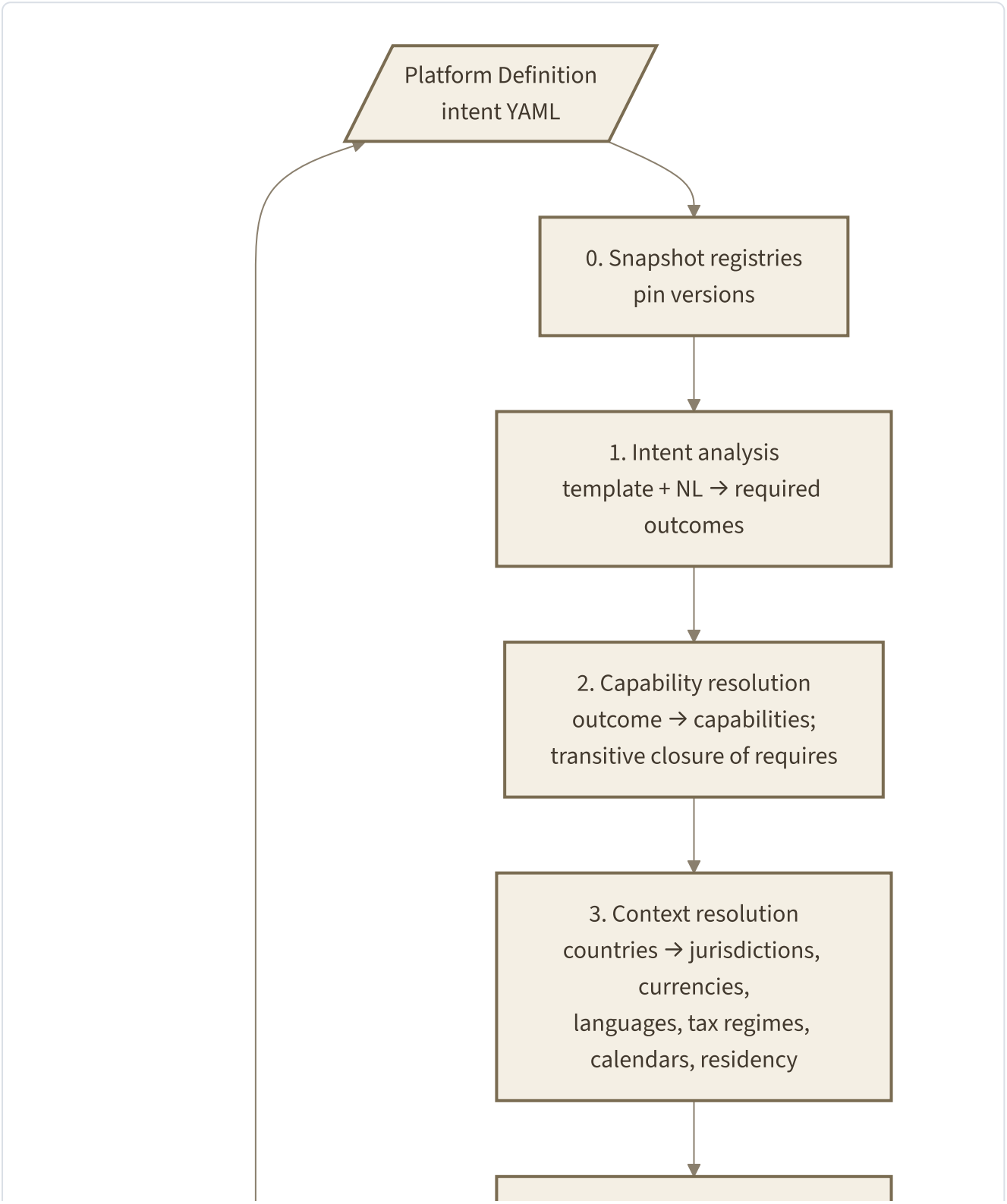
11. Universal Composition Substrate

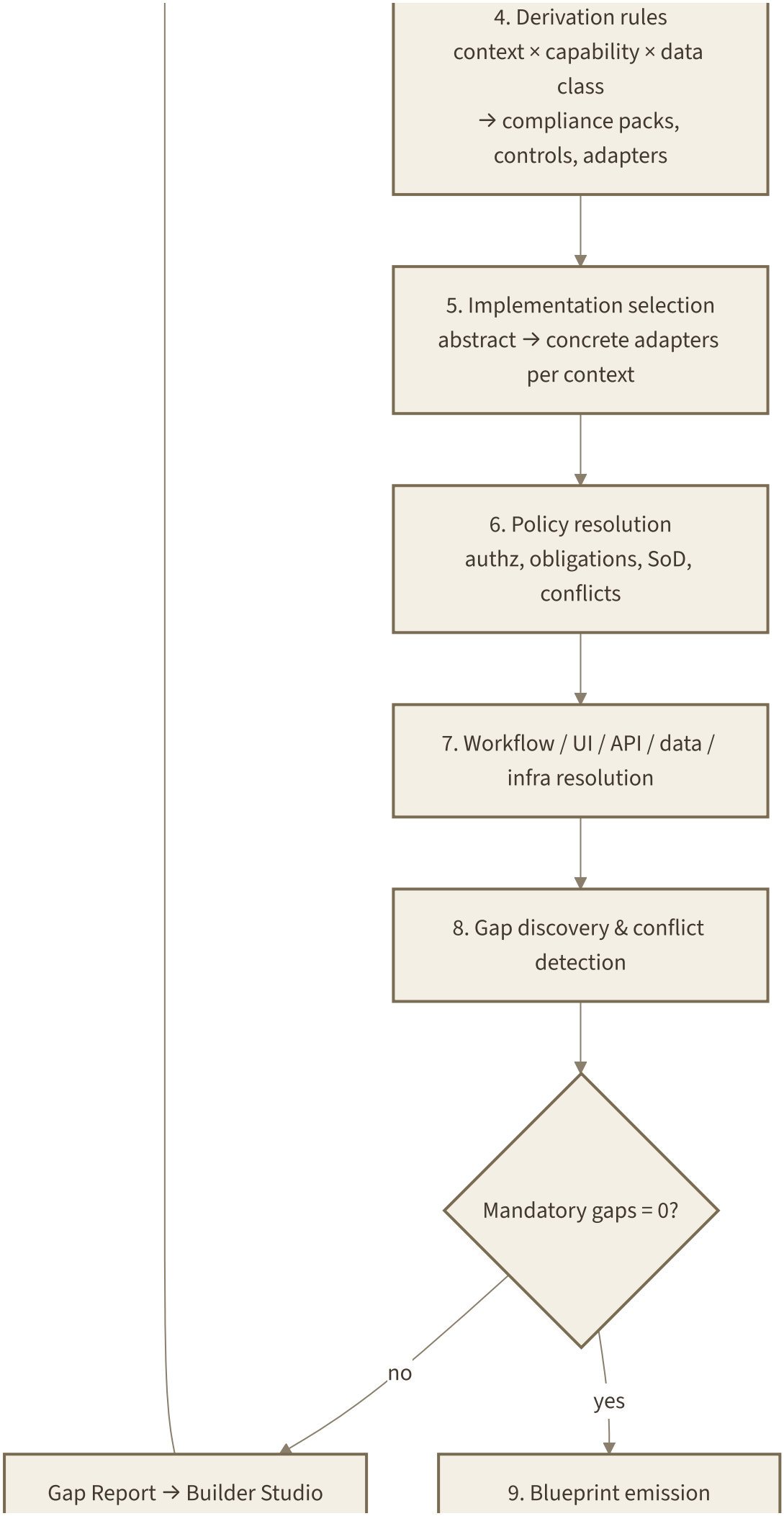
The chat identified this as **"the constitutional core"** and "the most critical missing

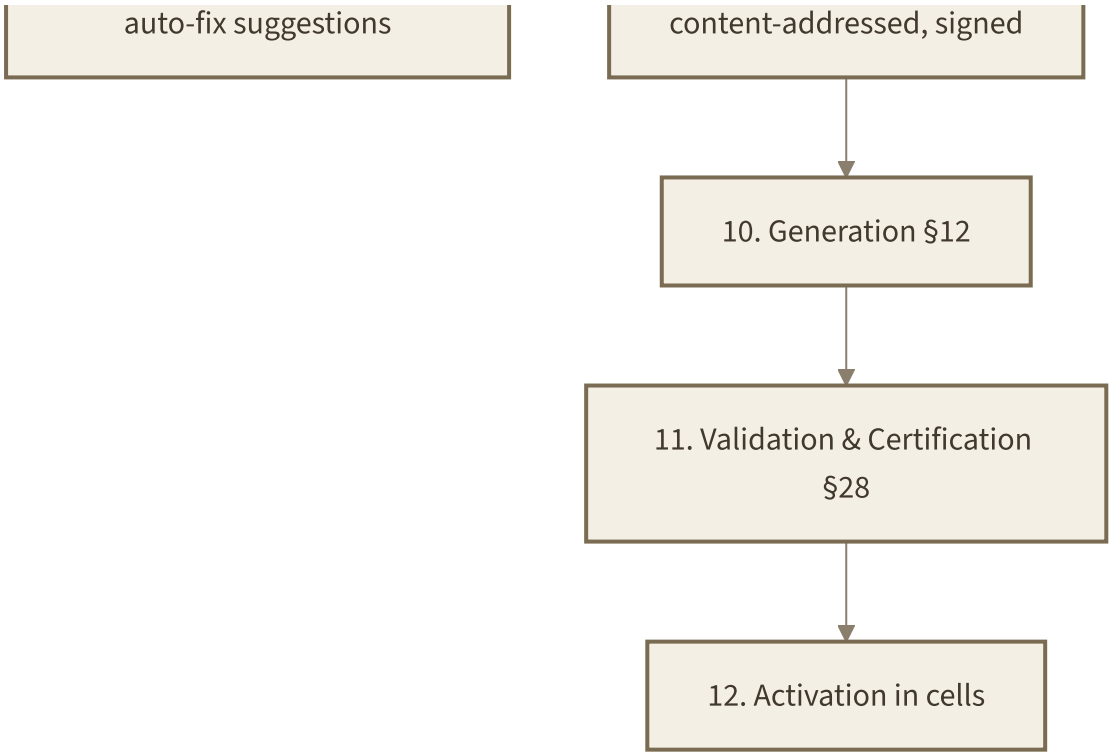
document". It is the engine that answers, *without hard-coding*:

> Given Industry = Commerce, Countries = 40, Channels = Web + Mobile, Capabilities = Marketplace + Logistics + Payments, which entities, workflows, APIs, databases, UI, tests, security controls, compliance controls and AI agents must exist, and which are missing?

11.1 Pipeline overview




4. Derivation rules
context × capability × data class
→ compliance packs, controls, adapters
5. Implementation selection
abstract → concrete adapters per context
6. Policy resolution
authz, obligations, SoD, conflicts
7. Workflow / UI / API / data / infra resolution
8. Gap discovery & conflict detection
Mandatory gaps = 0?
no
yes
Gap Report → Builder Studio
9. Blueprint emission



11.2 Platform Definition (input)

```
apiVersion: ucos.io/v1
kind: PlatformDefinition
metadata:
  id: ucos:pd:bazaar-marketplace
  tenant: ucos:tenant:bazaar
  version: 1.0.0
spec:
  template: ucos:industry-template:multi-vendor-marketplace@2
  industry: [ucos:industry:retail/general-merchandise, ucos:industry:retail/electro
  businessModel: [B2C, marketplace, C2C?]       # '?' = exploring, generates recomm
  geography:
    countries: [IN, AE]
    primaryCurrency: { IN: INR, AE: AED }
    languages: [en, hi, ar]
    dataResidency: strict                       # data stays in-country where law req
  channels: [web, ios, android, whatsapp, agent-api]
  capabilities:                                 # explicit asks (resolver adds the re
    include: [marketplace.core, logistics.multi-carrier, payment.core, offer.engine
    exclude: [loyalty.*]
  portals: [storefront, seller, admin, support, finance, governance]
  compliance:
```

```
    frameworks: [PCI-DSS, SOC2, ISO27001]     # declared; jurisdictional ones are d
  security:
    profile: ucos:secprofile:high             # e.g. MFA for sellers, step-up for p
  ai:
    profile: ucos:aiprofile:assistive         # agents may propose, humans approve
    providers: { default: anthropic, allowed: [anthropic, openai, self-hosted] }
  scale:
    expected: { monthlyActiveBuyers: 5_000_000, peakOrdersPerMinute: 3000, sellers:
  operatingModel:
    cells: auto                               # resolver picks cells honoring resid
    availabilityTier: T1                      # 99.95%
  brand: { theme: ucos:theme:bazaar, domains: [bazaar.in, bazaar.ae] }
```

11.3 Derivation rules ("how the platform knows")

Required items are derived by **rules stored as data** in the Rule Registry. These are Datalog-style facts and rules evaluated over the Knowledge Graph. Adding a new country, regulation or capability means adding facts or rules. **The composer itself never changes.**

```
% Facts (registry data)
jurisdiction_of(country:IN, jur:IN).
regulation_applies(reg:DPDP-2023, jur:IN).
regulation_applies(reg:RBI-PA-Guidelines, jur:IN) :- capability(payment.core).
regulation_applies(reg:UAE-PDPL, jur:AE).
data_class(payment.core, PCI).
data_class(identity.core, PII).
requires_control(reg:DPDP-2023, ctrl:consent-notice).
requires_control(reg:DPDP-2023, ctrl:dsr-access-erasure).
control_implemented_by(ctrl:consent-notice, privacy.consent).
control_implemented_by(ctrl:dsr-access-erasure, privacy.dsr).
adapter_for(payment.provider, jur:IN, method:UPI, payment.provider.razorpay).
adapter_for(tax.determination, jur:IN, tax.gst-india).
adapter_for(tax.determination, jur:AE, tax.vat-uae).
residency_required(jur:IN, data_class:PCI, payment-data-localisation).

% Rules (generic — never change for new countries)
required_capability(C) :- requested(C).
required_capability(D) :- required_capability(C), requires(C, D).
```

```
required_capability(Impl) :- required_capability(Abs), abstract(Abs), in_scope(J),
required_capability(Impl) :- in_scope(J), regulation_applies(R, J), requires_contro
                             control_implemented_by(Ctrl, Impl).
gap(missing_capability, C) :- required_capability(C), not available(C).
gap(uncertified, C)       :- required_capability(C), available(C), not certified(C)
gap(conflict, C1, C2)     :- required_capability(C1), required_capability(C2), conf
gap(residency, Cell, J)   :- assigned_cell(Cell, J), residency_required(J, _, _), r
```

Reference engine: a Datalog evaluator (e.g. Soufflé or a Rust Datalog crate embedded in the composer) over a snapshot of the graph. The **ResolutionTrace** records which rule produced each requirement. This powers the "**Why is this here?**" explanation in the UI (wireframe W5).

11.4 Blueprint (output)

```
apiVersion: ucos.io/v1
kind: Blueprint
metadata:
  id: ucos:blueprint:sha256:7f3c...            # content-addressed
  platformDefinition: ucos:pd:bazaar-marketplace@1.0.0
  registrySnapshot: ucos:snapshot:2026-10-02T09:00Z
  composerVersion: 3.4.1
spec:
  capabilities:                                  # fully pinned
    - { id: identity.core, version: 1.9.2 }
    - { id: temporal.core, version: 1.4.0 }
    - { id: catalog.listing, version: 1.6.1 }
    - { id: commerce.cart, version: 2.3.0 }
    - { id: marketplace.core, version: 2.0.3 }
    - { id: payment.core, version: 3.1.0 }
    - { id: payment.provider.razorpay, version: 1.8.0, scope: { jurisdiction: IN } }
    - { id: payment.provider.network-intl, version: 1.2.0, scope: { jurisdiction: A
    - { id: tax.gst-india, version: 4.0.1, scope: { jurisdiction: IN } }
    - { id: tax.vat-uae, version: 2.2.0, scope: { jurisdiction: AE } }
    - { id: privacy.consent, version: 2.0.0 }
    - { id: privacy.dsr, version: 1.3.0 }
    # ... ~60 more
  objectTypes: 214
  commands: 486
```

```
  events: 371
  workflows: [order.lifecycle@3, seller.onboarding@2, payout.cycle@2, return.flow@2
  policies: { authz: 912, obligations: 140, sod: 22 }
  controls: { total: 318, byFramework: { PCI-DSS: 112, DPDP: 41, UAE-PDPL: 38, SOC2
  ui: { portals: 6, views: 402, journeys: 37, locales: [en-IN, hi-IN, en-AE, ar-AE]
  integrations: [razorpay, network-intl, delhivery, aramex, gupshup-whatsapp, ...]
  ai: { agents: [catalog-enrichment, support-triage, seller-assistant], autonomy: L
  infrastructure:
    cells: [ { id: cell-in-mum-1, jurisdiction: IN }, { id: cell-ae-dxb-1, jurisdic
    tier: T1
  certification: { required: [ARCH, SEC, CMP, TEST, PERF, A11Y, DOC, OBS, DR] }
  resolutionTrace: ucos:trace:sha256:...
```

11.5 Conflict & overlap detection

Check	Example	Resolution
Ownership collision	Two packs both `provide Coupon`	Publish-time rejection (L4)
Capability conflict	`inventory.single-location` vs `inventory.multi-node`	Builder chooses; resolver suggests based on scale
Policy conflict	Allow and deny on the same action/resource with equal precedence	Policy analyser (Cedar validator / SMT) flags it; precedence must be set
Temporal conflict	Two TemporalPolicies on one offer with overlapping windows and different discounts	Decision table hit-policy must be declared
Compliance conflict	Data-retention minimum (tax law: 8 years) vs erasure request (privacy law)	Rule: legal hold overrides erasure; the record is minimised, not deleted (encoded as a control precedence)
Residency conflict	Cell placement violates residency	Resolver places the tenant shard in a compliant cell, or emits a gap
SoD violation	Same role can create a payout and approve it	Gap `SOD` with suggested role split

11.6 Gap discovery: taxonomy, severity and report

Gap kinds (consolidating every "Missing X Detection" list in the source):

`CAPABILITY, DOMAIN, OBJECT, RELATIONSHIP, WORKFLOW, DECISION, POLICY, SECURITY_CONTROL, COMPLIANCE_CONTROL, DATA`

(contract/retention/residency), INTEGRATION, API, UI (view/journey/locale/a11y), AI (guardrail/eval), TEST, DOCUMENTATION, OBSERVABILITY (signal/SLO/alert/runbook), RESILIENCE (backup/DR), INFRASTRUCTURE, GOVERNANCE (steward/approver), CERTIFICATION, TRACEABILITY, FAILURE_SCENARIO, USER_JOURNEY, PERFORMANCE, SCALABILITY, COST .

Severity: `MANDATORY` (blocks activation), `RECOMMENDED` (blocks certification unless waived), `OPTIONAL` (advisory).


```
{
  "kind": "GapReport",
  "blueprintCandidate": "ucos:pd:bazaar-marketplace@1.0.0",
  "summary": { "mandatory": 3, "recommended": 7, "optional": 12 },
  "gaps": [
    {
      "id": "GAP-0001",
      "kind": "COMPLIANCE_CONTROL",
      "severity": "MANDATORY",
      "title": "Card data localisation required for India",
      "because": ["regulation_applies(RBI-PA-Guidelines, IN)", "data_class(payment.
      "fixes": [
        { "action": "add-capability", "target": "payment.tokenization.network", "ef
        { "action": "assign-cell", "target": "cell-in-mum-1", "effort": "config" }
      ],
      "authorityToWaive": "none"
    },
    {
      "id": "GAP-0004",
      "kind": "UI",
      "severity": "MANDATORY",
      "title": "Arabic (RTL) translations missing for 41 checkout strings",
      "fixes": [ { "action": "machine-translate-and-review", "effort": "ai+human" }
    },
    {
      "id": "GAP-0009",
      "kind": "OBSERVABILITY",
      "severity": "RECOMMENDED",
      "title": "No SLO alert route for payout.cycle workflow",
      "fixes": [ { "action": "apply-default-alert-policy", "effort": "config" } ]
```

```
    }
  ]
}
```

Activation rule (L10): `activate(blueprint) ⇔ mandatory = 0 ∧ ∀ recommended: (resolved ∨ waived_by_authority ∧ waiver.expiry > now) ∧ certification.status = PASSED .`

11.7 Composer properties

- **Pure and deterministic** (L18): `compose(PD, snapshot, composerVersion) → blueprint` is a pure function, hence cacheable and reproducible.
- **Explainable:** every element of the blueprint links to the rule(s) and input(s) that caused it.
- **Incremental:** a PD diff recomposes only the affected subgraph. The Studio shows a blueprint diff before apply (W5).
- **Simulatable:** a blueprint can be activated in a **simulation cell**, where external adapters are replaced by contract-faithful simulators.

---

12. Generation Engine & Platform Factory

12.1 Key architectural decision: interpret first, generate second (ADR-001)

The source asked for UI, API, DB, workflow, infra and test generation. There are two strategies:

Strategy	Pros	Cons
Generate code per tenant	Familiar output; tenants can "eject"	N tenants × M versions of code to patch and secure; drift; slow rollout; violates L6 and L2 in practice
Interpret metadata at runtime (metadata-driven multi-tenant runtime, like Salesforce/ServiceNow)	One codebase to secure and scale; instant config changes; tenant changes are data	Runtime engines must be excellent; performance work goes into the engines

Decision: UCOS **interprets** blueprints at runtime for data, API, workflow, policy, decision and UI. It **generates** only artifacts that are consumed *outside* the runtime or

that benefit from static compilation:

Generated artifact	Generator	Consumer
Client SDKs (TS, Kotlin, Swift, Python, Go)	OpenAPI/GraphQL → SDK generator	tenant developers, mobile apps
API reference & developer portal	spec → docs	developers
User/admin manuals, runbooks	capability docs + blueprint → doc generator (with AI drafting + human review)	users, SRE
Physical DB migrations for core object types	schema diff → SQL migration	data plane
Mobile app shells (white-label)	blueprint + theme → React Native/Expo build config	app stores
Infrastructure-as-code for cells	blueprint infra section → OpenTofu/Crossplane claims	GitOps
Test suites (contract, scenario, load, a11y)	blueprint → test generator	CI / certification
Dashboards & alert rules	signals → Grafana/alert manifests	ops
Compliance evidence packs, SBOM	evidence + SBOM → reports	auditors
Diagrams (C4, ERD, flows)	graph → Mermaid/Structurizr	everyone

12.2 Generator contract

```
kind: Generator
id: gen.sdk.typescript
version: 2.1.0
input:  [Blueprint.api, Blueprint.objectTypes]
output: { kind: npm-package, path: generated/sdk/ts }
deterministic: true                # same input → byte-identical output (hash reco
templates: templates/sdk-ts/       # Handlebars/Jinja; no AI in deterministic gene
validation: [tsc --noEmit, contract-tests]
```

AI-assisted generators (documentation drafting, translations, test-idea generation) are marked `deterministic: false`. Their output is always a **proposal** that goes through review and certification. It is never activated directly.

12.3 Platform Factory flow (end to end)

```
sequenceDiagram
  autonumber
  actor B as Tenant Builder
  participant ST as Builder Studio
  participant CO as Composer
  participant GE as Generators
  participant CE as Certification
  participant GV as Governance
  participant DP as Data Plane (cells)
  B-->>ST: Choose template, countries, channels, capabilities
  ST-->>CO: compose(PD draft) [simulate=true]
  CO-->>ST: Blueprint candidate + Gap Report + Cost estimate
  B-->>ST: Apply auto-fixes / configure / request waivers
  ST-->>CO: compose(PD v1.0.0)
  CO-->>ST: Blueprint (0 mandatory gaps)
  ST-->>GE: generate(blueprint)
  GE-->>CE: SDKs, docs, tests, IaC, dashboards
  CE-->>CE: Run gates (ARCH, SEC, CMP, TEST, PERF, A11Y, DOC, OBS, DR)
  CE-->>GV: Certification report + evidence
  GV-->>B: Approval request (release authority)
  B-->>GV: Approve (MFA step-up)
  GV-->>DP: Activate blueprint (canary → full)
  DP-->>ST: Live; telemetry streaming to Platform Twin
```

12.4 Builder Studio (no-code builders)

Tenant/Company/Marketplace builder · Product & Service builder · Commerce-model builder · Object Designer · Workflow Designer · Decision Table editor · Temporal Policy editor · Policy editor · UI/Journey designer · Theme/brand designer · Integration/mapping builder · AI agent builder · Analytics/report builder · Compliance profile builder · Operations builder (SLOs, alerts) · Governance builder (authorities, approval chains).

Every builder writes **definitions** (canonical objects in Draft). It never writes code.

12.5 Specification repository generation (the "Genesis" process)

The Master Genesis Directive asked for 50 artifacts per universe across ~72 universes. That corpus is produced **incrementally, universe by universe**, from this document, using the repository layout in §31.3:

```
Constitution (this doc) → Universe constitution → Ontology → Taxonomy → Object type
```

```
→ Relationships → Events → Commands/APIs → Workflows → Decisions → Policies → Contr
→ UI models → Tests → Docs → Certification criteria
```

The per-universe template (the 50 sections from the Genesis Directive) is consolidated into **12 required sections**, with no loss of content:

#	Section	Covers Genesis items
1	Constitution, authority, principles, laws	1–4
2	Ontology & taxonomy	5–6
3	Registry model & canonical objects/schemas/relationships	7–10
4	Events, APIs, service & runtime contracts	11–12, 21–22
5	Workflows, state machines, rules	13–14, 16
6	Policies, security, encryption, audit, compliance	15, 17–20
7	Validation, certification, versioning, dependency, reuse rules	23–24, 26–28
8	Discovery, gap detection, self-healing, evolution rules	25, 29–31
9	AI & agent governance	32–33
10	Twin, analytics, reporting, monitoring, operations	34–38
11	UI, UX, accessibility, localisation	39–42
12	Scalability, performance, DR, HA, multi-tenant/region/cloud	43–50

PART III

RUNTIME & DATA

13. Runtime Fabric

13.1 Engines (one execution fabric, generic engines)

Engine	Responsibility	Reference implementation (§31.1)
API Gateway +	TLS, authn, rate limit, context	Envoy-based gateway + custom filters

API Gateway + PEP	TLS, authn, rate limit, context resolution, policy enforcement	Envoy-based gateway + custom filters
Command Bus	Routes commands to the owning capability; idempotency; audit	In-house (Go), backed by Postgres outbox
Object Service	Generic CRUD/query for any registered object type; schema validation; RLS	In-house (Go)
Policy Decision Point	Authorisation and obligations	Cedar (app authz), OPA (infra/admission)
Durable Workflow Engine	Long-running orchestrations, sagas, human tasks, timers	Temporal.io-class engine (name collision note: the *engine* vs the *Temporal Universe*)
Decision Engine	DMN decision tables, scoring, rule sets	DMN engine (e.g. GoRules ZEN)
Temporal Scheduler	Recurrences, timers, triggers, SLA clocks	In-house on top of the workflow engine + stream processor
Event Bus	Domain events, CDC, streams	Kafka-compatible (Kafka / Redpanda)
Stream Processor	Condition triggers, aggregates, fraud features	Flink
Query Engine	Read models, GraphQL federation	GraphQL federation router + read-model projections
Search Engine	Full-text, faceting, hybrid	OpenSearch
Vector Engine	Semantic retrieval	pgvector (small), Qdrant (scale)
Graph Engine	Knowledge Graph, traceability, dependency	Neo4j (control plane); Postgres + recursive CTE for tenant graphs
AI Gateway & Agent Runtime	Model routing, guardrails, tool calling, budgets	In-house gateway; MCP for tools
Simulation Engine	Runs blueprints/scenarios with simulated adapters and `asOf` time	Simulation cells
Twin Engine	Maintains twin state from definitions + telemetry	Stream projections

13.2 Request lifecycle (command)

```
sequenceDiagram
  autonumber
  participant C as Client (web/app/agent)
```

```
  participant C as Client (web/app/agent)
  participant G as Gateway (PEP)
  participant X as Context Resolver
  participant P as PDP (Cedar)
  participant B as Command Bus
  participant K as Capability Pack (e.g. commerce.cart)
  participant D as Tenant DB (RLS)
  participant O as Outbox → Event Bus
  participant A as Audit Log
  C->>G: POST /v1/commerce/carts/{id}:addLine (JWT, Idempotency-Key)
  G->>X: resolve context (tenant, jurisdiction, locale, risk)
  X-->>G: ContextEnvelope (signed)
  G->>P: isAuthorized(principal, action=cart:AddLine, resource=cart, context)
  P-->>G: ALLOW + obligations [mask:none, audit:standard]
  G->>B: Command(AddLine) + context
  B->>B: idempotency check (key, hash)
  B->>K: handle(AddLine)
  K->>D: BEGIN; SELECT ... FOR UPDATE (version); INSERT line; UPDATE cart(version+1); INSI
  K-->>B: Result(cartVersion=8)
  B->>A: append(audit record, hash-chained)
  B-->>G: 200 {cart}
  G-->>C: 200 + ETag: "8"
  O-->>O: relay outbox → commerce.cart.line-added
```

13.3 Transaction, consistency and concurrency model

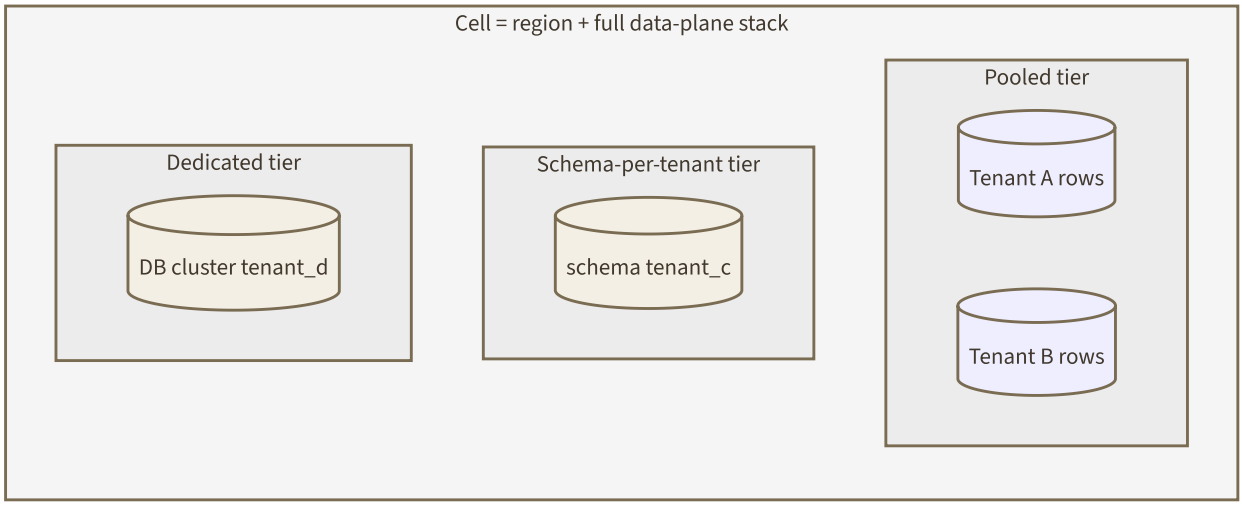
Concern	Rule
Transaction boundary	One **aggregate** per transaction (e.g. Cart, Order, LedgerJournal). ACID inside it (Postgres).
Cross-aggregate / cross-capability	**Sagas** in the workflow engine, with compensating commands (e.g. `ReleaseStockReservation`).
Event publication	**Transactional outbox** → relay → event bus. No dual writes.
Concurrency	Optimistic locking (`version`), `If-Match` ETags on APIs; pessimistic `SELECT … FOR UPDATE` only inside short command handlers.
Idempotency	`Idempotency-Key` required on all non-GET commands; stored 24h–30d per command type; same key + different payload → 409.
Read models	Eventually consistent projections (target lag p99 < 2s); read-your-writes supported via `X-Consistency: strong` for the owning aggregate.
Money	Decimal strings, banker's rounding only where regulation says; ledger is double-entry and immutable; corrections are reversing entries.
Ordering	Per-aggregate ordering via event-bus partition key = aggregate id.
Exactly-once effects	At-least-once delivery + idempotent consumers (consumer offset + processed-event table).

13.4 Pack Runtime Contract

Each pack service MUST:

1. Expose `/.well-known/ucos/pack` (manifest hash, version, health, readiness).
2. Accept commands via gRPC `ucos.pack.v1.CommandHandler/Handle(CommandEnvelope) → ResultEnvelope`.
3. Accept events via subscription on the bus with the standard consumer library (idempotency, retries, DLQ).
4. Use **only** the platform SDK for DB access, so that tenancy/RLS, encryption, auditing and outbox are enforced automatically.
5. Emit OTel traces/metrics/logs with standard attributes (`ucos.tenant` , `ucos.capability` , `ucos.command`).
6. Never call another pack's database. It calls commands/queries or consumes events only.
7. Run with a non-root, read-only filesystem, network policy limited to declared ports and adapters, and secrets via workload identity.

13.5 Multi-tenancy model



Tier	Isolation	Use for
Pooled	Shared tables + Postgres Row-Level Security on `tenant_id` + per-tenant DEKs	SMB tenants, trials
Schema	Dedicated schema, shared cluster	Mid-market, custom indexes

Dedicated	Dedicated DB cluster (and optionally dedicated cell)	Enterprise, regulated (banking, health, government), residency

Tier moves are online migrations orchestrated by the platform (logical replication + cutover). Noisy-neighbour controls: per-tenant rate limits, connection quotas, workload priority classes, and per-tenant Kafka quotas.

---

14. Data Architecture

14.1 Polyglot storage mapping

Data kind	Store	Notes
Transactional canonical objects	PostgreSQL 16+ (Citus or cloud-managed, partitioned)	Core universes: generated physical tables; tenant extensions: `JSONB` + GIN indexes (ADR-003)
Ledger	PostgreSQL, append-only tables, hash-chained journals	Separate cluster in regulated tiers
Events / streams	Kafka-compatible	Retention per topic; compacted topics for state
Search	OpenSearch	Projections from events
Vector	pgvector / Qdrant	Embeddings for products, knowledge, support
Graph	Neo4j (control-plane UKG); Postgres for tenant relationships	
Documents / binaries	S3-compatible object storage	Encrypted, versioned, lifecycle rules
Time-series	Prometheus/Mimir (ops); ClickHouse (business series)	
Analytics / lakehouse	Apache Iceberg on object storage + Trino/Spark; ClickHouse for real-time	CDC from Postgres via Debezium
Cache	Redis (tenant-namespaced keys)	Never the source of truth
Audit	Append-only store (Postgres + WORM object storage replica)	Hash chain anchored daily

14.2 Physical modelling of dynamic object types (ADR-003)

- **Core object types** (owned by certified packs) → generated physical tables with typed columns, constraints and indexes. Migrations are produced by the schema-diff generator and are always backward-compatible (expand → migrate → contract).
- **Tenant custom attributes** (`Extension` fields) → `custom JSONB` column validated against the tenant's extension schema. Tenants may request indexes, which the platform creates as expression indexes within quotas.
- **Registry-only types** (reference data, low volume) → generic `objects` table (`type, id, attributes JSONB`).

14.3 Residency, sovereignty and cells

- Each cell is pinned to a jurisdiction. A tenant's **data shards** are placed in cells that satisfy `residency_required` rules (§11.3).
- Cross-border flows (e.g. global catalog → tenant cell) carry only non-personal data unless a transfer mechanism (SCCs, adequacy, consent) is recorded as a fact in the Compliance universe.
- Global registries (products, taxonomies, reference data) replicate read-only to all cells.

14.4 Data contracts, lineage and quality

```
kind: DataContract
id: dc.order.placed.v3
producer: commerce.order
schema: events/order.placed.v3.json
semantics: "Emitted once per order when payment authorisation succeeded or COD acce
sla: { freshness: "p99 < 5s", completeness: "100%" }
classification: { dataClasses: [PII, FINANCIAL] }
consumers: [fulfillment.core, ledger.core, analytics.lakehouse, recommendation.basi
breakingChangePolicy: "new major version; dual-publish for 90 days"
```

Lineage: OpenLineage events from pipelines plus the event-catalog graph give column-level lineage from source to report. Quality: Great-Expectations-style checks are attached to contracts, and failures create `DATA` gaps and alerts.

14.5 Data lifecycle

Classification → retention rule (resolved per jurisdiction × data class × legal hold) → archival (cold storage) → deletion or crypto-shredding (destroy the per-tenant or per-subject DEK). Legal hold suspends deletion. Every deletion produces evidence.

14.6 Bitemporal model

Canonical objects record **valid time** (`validFrom/validTo` , when it is true in the world) and **transaction time** (`recordedAt/supersededAt` , when the system knew it). This enables `asOf` queries, regulatory reconstruction ("what did we show the customer on 3 Oct?"), and simulation. It is implemented with history tables (temporal tables pattern) for core types and event replay for the rest.

---

15. API & Event Specification

15.1 API styles

Style	Use	Contract
REST (JSON)	Public/tenant APIs; commands & queries	OpenAPI 3.1 generated from capability manifests
GraphQL (federated)	Experience/BFF reads, portals	One subgraph per universe
gRPC	Internal service-to-service, pack runtime contract	Protobuf
Events	Async integration, projections	AsyncAPI 3 + CloudEvents 1.0
Webhooks	Tenant/external notification	Signed (HMAC-SHA256 + timestamp), retries with backoff
MCP	AI agents (internal & external) using platform tools	Tools generated from commands with authority scopes

15.2 REST conventions

```
Base:           https://{tenant-domain}/api/v1
Resources:      /{universe}/{collection}/{id}
                /commerce/carts/{cartId}
Commands:       POST /{universe}/{collection}/{id}:{command}
```

```
                POST /commerce/carts/{cartId}:addLine
Queries:        GET  /{universe}/{collection}?filter=...&sort=...&page[cursor]=...
Generic object: /objects/{type}/{id}   (any registered type, policy-checked)
Headers:        Authorization: Bearer <JWT> | Idempotency-Key | If-Match | Accept-
                X-UCOS-Context-AsOf (simulation/time-travel, privileged) | tracepa
Versioning:     URL major (/v1); additive changes only within a major; deprecation
Errors:         RFC 9457 application/problem+json
Pagination:     cursor-based; total counts opt-in
Rate limits:    RateLimit-* headers (IETF draft), per tenant × API product × princ
```

Error example:

```
{
  "type": "https://errors.ucos.io/policy/denied",
  "title": "Action not permitted",
  "status": 403,
  "detail": "Payout approval requires a different approver than the creator (SoD).'
  "instance": "/api/v1/marketplace/payouts/01J...:approve",
  "ucos": { "code": "POLICY_SOD_VIOLATION", "policy": "marketplace.payout.sod", "tr
}
```

15.3 Generic object API (works for any registered type)

```
POST /api/v1/objects/healthcare.encounter.Encounter
GET  /api/v1/objects/healthcare.encounter.Encounter/{id}?expand=patient,practitione
POST /api/v1/objects/{type}:query     { "where": {...}, "select": [...], "asOf": "
GET  /api/v1/meta/types/{type}        # schema, commands, views — enables generic U
```

15.4 Event envelope (CloudEvents + UCOS extensions)

```
{
  "specversion": "1.0",
  "id": "01J9Z4...",
  "source": "ucos://cell-in-mum-1/commerce.order",
  "type": "order.order.placed",
  "dataschema": "ucos://schemas/order.placed/3.0.0",
  "subject": "ucos:order:01J9Z3...",
```

```
  "time": "2026-10-02T09:15:22.481Z",
  "datacontenttype": "application/json",
  "ucostenant": "ucos:tenant:bazaar",
  "ucosblueprint": "sha256:7f3c...",
  "ucoscontext": "eyJqdXIiOlsiSU4iXSwibG9jYWxlIjoiaGktSU4i...",
  "ucoscausation": "01J9Z4...prev",
  "ucoscorrelation": "checkout-01J9Z3...",
  "ucosclassification": "confidential",
  "traceparent": "00-4bf92f...-01",
  "data": {
    "orderId": "ucos:order:01J9Z3...",
    "orderNumber": "ORD-2026-00042117",
    "buyer": "ucos:party-role:01J...",
    "sellerSplit": [ { "seller": "ucos:org:...", "lines": 2, "subtotal": { "amount"
    "total": { "amount": "3127.40", "currency": "INR" },
    "placedAt": { "utc": "2026-10-02T09:15:22Z", "tz": "Asia/Kolkata" }
  }
}
```

15.5 Event naming & catalog

- `type = <universe>.<object>.<past-tense-verb>` (kebab-case verbs): `order.order.placed`, `payment.payment-intent.captured`, `temporal.trigger.fired`, `governance.blueprint.activated`.
- Topics: `ucos.<cell>.<universe>.<object>.v<major>`, partition key = aggregate id.
- Every event type is registered (schema, owner, classification, retention, consumers). Unregistered events are rejected by the producer SDK.
- Compatibility: schema registry enforces BACKWARD_TRANSITIVE within a major version.

15.6 Canonical event families (seed)

Universe	Events
Identity	identity.identity.registered, .verified, .suspended; identity.session.started, .ended; identity.role.granted, .revoked
Catalog	catalog.listing.created, .published, .unpublished, .updated
Offer	offer.offer.created, .activated, .expired, .redeemed

Pricing	pricing.pricelist.changed, pricing.price.overridden
Cart	commerce.cart.created, .line-added, .line-removed, .coupon-applied, .abandoned, .converted
Order	order.order.placed, .confirmed, .cancelled, .amended, .completed; order.return.requested, .approved
Payment	payment.payment-intent.created, .authorized, .captured, .failed; payment.refund.issued; payment.dispute.opened
Ledger	ledger.journal.posted, ledger.period.closed, ledger.settlement.completed
Inventory	inventory.stock.adjusted, .reserved, .released, .depleted
Fulfillment	fulfillment.shipment.created, .dispatched, .delivered, .failed
Temporal	temporal.trigger.fired, temporal.reservation.held, .confirmed, .expired; temporal.sla.breached
Governance	governance.proposal.submitted, .approved, .rejected; governance.blueprint.certified, .activated, .rolled-back
Compliance	compliance.control.failed, compliance.evidence.collected, privacy.dsr.received, .completed
AI	ai.agent-run.started, .tool-called, .completed, .escalated; ai.guardrail.triggered

16. Workflow, Decision & Rules

16.1 Workflow DSL (state machine + orchestration)

```
apiVersion: ucos.io/v1
kind: WorkflowDefinition
id: order.lifecycle
version: 3.0.0
object: commerce.order.Order
states:
  placed:      { on: { PaymentAuthorized: confirmed, PaymentFailed: payment_failed,
  confirmed:   { entry: [ reserveStock, createFulfillmentOrders, postLedger(ORDER_C
                 on: { AllShipmentsDispatched: shipped, Cancel: cancelling } }
  shipped:     { on: { AllShipmentsDelivered: delivered, DeliveryFailed: exception
  delivered:   { entry: [ startTimer(returnWindow) ],
                 timers: { returnWindow: { temporalPolicy: ref(decision:returns.win
                 on: { ReturnRequested: return_in_progress } }
```

```
                  on: { ReturnRequested: return_in_progress } }
  completed:   { final: true, entry: [ releaseSellerPayoutHold ] }
  cancelling:  { saga: [ { do: releaseStock, compensate: none },
                         { do: voidOrRefundPayment, compensate: alertFinance },
                         { do: notify(template: order-cancelled) } ],
                 then: cancelled }
  cancelled:   { final: true }
  payment_failed: { timers: { retry: { duration: PT15M, then: cancelled } }, on: {
  exception:   { humanTask: { queue: support.logistics-exceptions, sla: ref(tmp:SLA
                 on: { Resolved: shipped, Refund: cancelling } }
  return_in_progress: { subworkflow: return.flow@2, then: completed }
guards:
  Cancel: "state in [placed, confirmed] and not anyShipmentDispatched"
observability: { slo: { "placed→confirmed": "p95 < 60s" } }
```

Human tasks support assignment rules (Decision), approvals (n-of-m, sequential, parallel), delegation, escalation chains bound to SLA clocks, and four-eyes/SoD guards from policy.

16.2 Decision tables (configurable business logic)

YAML

```
kind: DecisionTable
id: returns.window-by-category
hitPolicy: FIRST
inputs:  [ listing.category, buyer.tier, context.jurisdiction ]
outputs: [ returnWindow (Duration), restockingFee (Percentage) ]
rules:
  - [ "electronics/*", "*",       "IN", "P7D",  "0%"  ]
  - [ "fashion/*",     "premium", "*",  "P30D", "0%"  ]
  - [ "fashion/*",     "*",       "*",  "P15D", "0%"  ]
  - [ "grocery/*",     "*",       "*",  "PT0S", "n/a" ]
  - [ "*",             "*",       "AE", "P14D", "10%" ]
  - [ "*",             "*",       "*",  "P10D", "0%"  ]
```

Other seed decision models: pricing waterfall, offer stacking, fraud scoring thresholds, seller risk tiering, payout hold rules, dispatch matching weights, support routing, credit limits, KYC escalation.

16.3 Automation modes

Event-driven · rule-driven · workflow-driven · schedule-driven (Temporal) · goal-driven (agent plans validated against policies) · policy-driven (obligations triggering actions).

---

17. Integration Universe

17.1 Connector framework

```
kind: Connector
id: connector.carrier.delhivery
implements: [logistics.carrier-port@2]
auth: { type: api-key, secretRef: vault://tenants/{tenant}/delhivery }
capabilities: [rateQuote, createShipment, cancelShipment, trackingWebhook, pickupSc
mappings:
  createShipment:
    request:  mappings/create-shipment.jsonata
    response: mappings/shipment-response.jsonata
rateLimits: { rps: 20 }
retries: { policy: exponential, max: 6, retryOn: [429, 5xx] }
webhooks: { path: /hooks/delhivery, verify: hmac-sha256 }
certification: [SEC-CONN-*, CONTRACT-*]
```

- **Ports and adapters:** packs call canonical ports (`PaymentProviderPort` , `CarrierPort` , `TaxEnginePort` , `KYCPort` , `MessagingPort` , `LLMPort` , `MapsPort`); connectors implement them.
- **Patterns:** request/response, webhook ingestion, polling, CDC (Debezium), batch/file (SFTP, EDI X12/EDIFACT), streaming, and federation (GraphQL subgraph of an external system).
- **Mapping:** declarative (JSONata/JOLT) with AI-assisted mapping suggestions that a human approves.
- **Seed connector families:** PSPs (Stripe, Adyen, Razorpay, PayU, Network International, PayPal), carriers (Delhivery, Shiprocket, Aramex, DHL, FedEx), tax (Avalara, Vertex, local GST/VAT engines), messaging (Twilio, Gupshup, Meta WhatsApp, SES, FCM/APNs), ERP/CRM (SAP S/4, Oracle, Salesforce, Dynamics, ServiceNow, Workday), maps (Google, Mapbox, OSM), KYC (Onfido, Digio), identity (Okta, Entra ID, Google), LLM providers (Anthropic, OpenAI, Google, self-hosted), and clouds (AWS. Azure. GCP).

17.2 Integration reliability

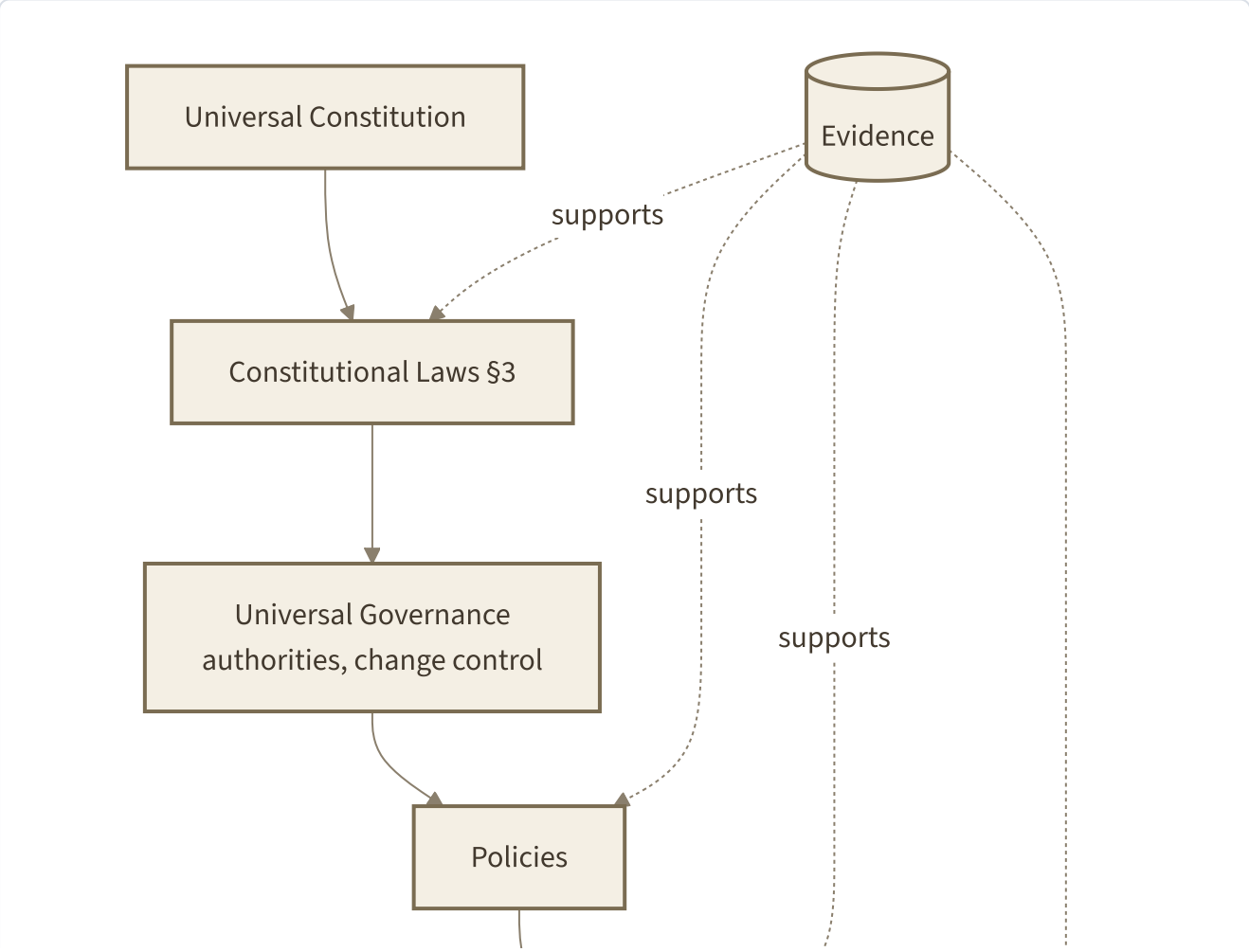
Outbound calls go through a resilience layer (timeouts, circuit breakers, bulkheads, retries with jitter, hedging for idempotent reads). Inbound webhooks are deduplicated by provider event ID. Reconciliation jobs (Temporal recurrences) compare provider state with UCOS state (payments, shipments) and open `Case`s for mismatches.

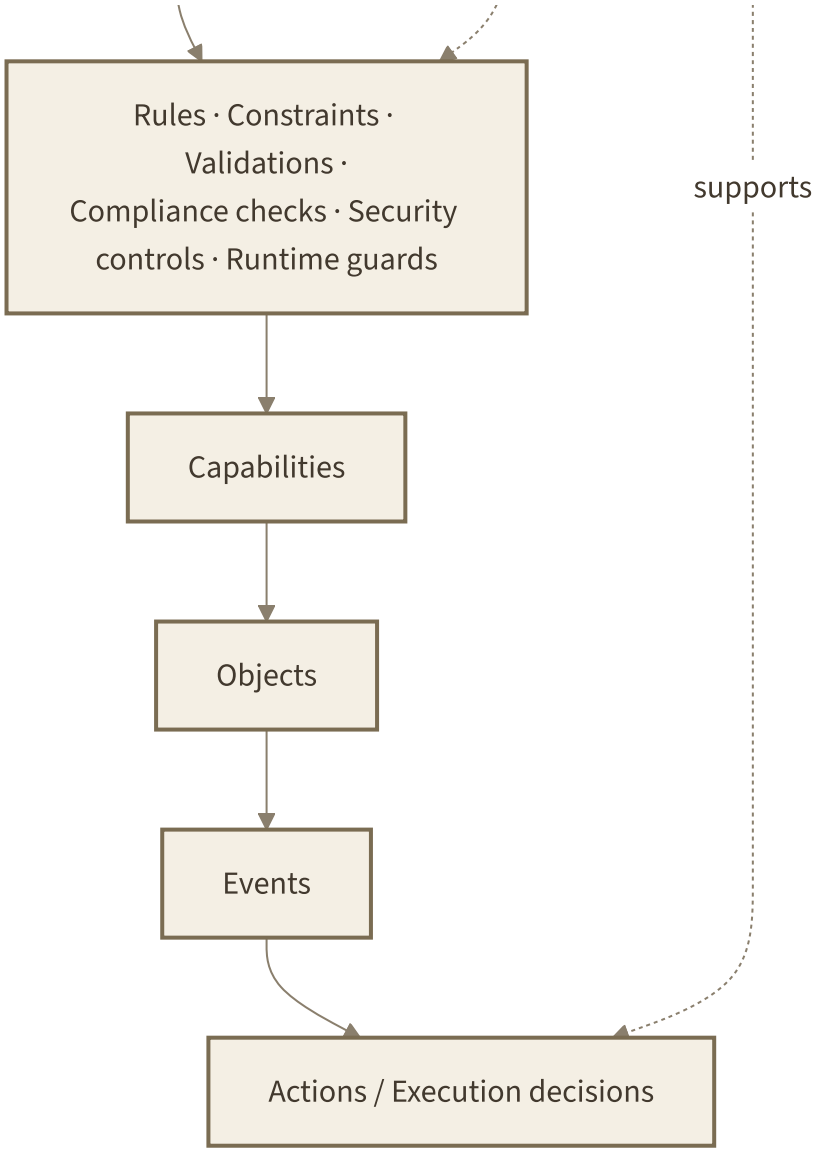
PART IV

TRUST

18. Authority, Governance & Policy

18.1 Authority hierarchy





Nothing executes outside this hierarchy, and no lower level may contradict a higher one. The policy analyser rejects any policy that permits what a Law forbids.

18.2 Authority types

Type	Controls	Examples	Typical holders
Foundational	Constitution, laws	Constitutional amendment, law retirement	Architecture Board (multi-party)
Governance	Policies, approvals, certifications	Capability certification, release approval	Governance council, CISO, DPO
Administrative	Configuration, operations	Tenant setup, environment config	Platform admins, tenant admins
Domain	Universe-specific governance	Product data stewardship, financial controls	Universe stewards
Operational	Runtime operations	Deploy, scale, failover,	SRE on-call

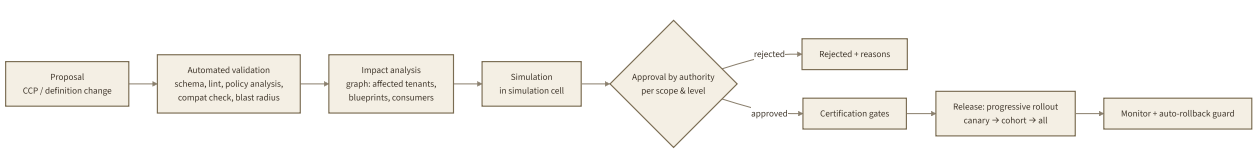
		incident actions	
Autonomous	Bounded automated action	Auto-remediation, agent tool calls	AI agents, automation engines (§21.5)

18.3 Authority & Decision objects

```
Authority:
  id: ucos:authority:01J...
  type: governance
  scope: { universes: [U29-payment, U30-ledger], tenants: [ucos:tenant:bazaar], act
  level: 3                          # 1 (low) … 5 (constitutional)
  holder: ucos:role:bazaar-finance-controller
  constraints: { maxAmount: { amount: "1000000", currency: INR }, requiresMfa: true
  validity: TemporalRef(TimeRange)
  delegation: { allowed: true, maxDepth: 1, mustBeLowerLevel: true }
  revocation: { by: [ucos:role:bazaar-cfo], effect: immediate }
  evidence: [ucos:evidence:board-resolution-2026-07]

DecisionRecord:
  id: ucos:decision:01J...
  type: waiver | approval | certification | policy-change | incident-action | ai-ac
  authority: AuthorityRef
  context: ContextRef
  inputs: [...]        # gap ids, proposals, evaluations
  outputs: [...]
  rationale: Text
  risk: { score, assessment: RiskAssessmentRef }
  compliance: { assessment: ComplianceAssessmentRef }
  approvals: [ { by, at, method: mfa|hardware-key } ]
  audit: AuditRef
  certificationStatus: n/a | pending | certified
```

18.4 Change control (governed change)



Change classes: **Standard** (pre-approved templates, e.g. a new offer), **Normal** (one approval), **Significant** (multi-party, impact > threshold), **Constitutional** (Architecture Board, CCP, minimum review period), **Emergency** (break-glass with retrospective approval within 24h).

18.5 Policy model (authorisation unified)

RBAC, ABAC, ReBAC and PBAC are all expressed as **Cedar policies** evaluated by one PDP. Infrastructure and admission policies use OPA/Rego.

```
// Sellers may update their own listings while their account is in good standing
permit (
  principal in Role::"seller",
  action in [Action::"catalog:UpdateListing", Action::"catalog:PublishListing"],
  resource is Catalog::Listing
)
when {
  resource.seller == principal.organization &&
  principal.organization.standing == "good" &&
  context.risk.score < 70
};

// Separation of duties on payouts
forbid (
  principal,
  action == Action::"marketplace:ApprovePayout",
  resource is Marketplace::PayoutBatch
)
when { resource.createdBy == principal };

// Agents may only act within delegated authority & budget
forbid (
  principal is Agent,
  action,
  resource
)
unless { principal.authority.actions.contains(action) && context.agent.budgetRemain
```

Policy obligations (returned with ALLOW): `mask(fields)` , `audit(level)` ,

`stepUpAuth` , `notify(role)` , `rateLimit(class)` , `watermark` , `requireReason` . Every PDP decision is logged with the policy IDs that matched.

18.6 Governance domains

Policy · Capability · AI · Data · Security · Compliance · Knowledge · Financial · Operational · Evolution · Architecture · Certification. Each has a steward authority and its own registry view in the Governance portal (W11).

---

19. Security Architecture

19.1 Principles

Zero trust · identity-first · policy-first · encryption-first · least privilege · separation of duties · multi-party approval for critical actions · secure by design / by default / by construction · continuous verification, monitoring and audit · crypto-agility.

19.2 Identity types & authentication

Principal	Authentication	Notes
End user (buyer, patient, student)	OIDC; passkeys (WebAuthn) preferred; OTP fallback; social login optional	Risk-based step-up
Business user (seller, admin, agent supervisor)	OIDC + mandatory MFA (passkey/TOTP/hardware key); SSO federation (SAML/OIDC) with tenant IdP	Session binding to device
Workload / pack service	SPIFFE/SPIRE workload identity, mTLS	No static secrets
AI agent	Agent identity (`ucos:agent:*`) + signed delegation token from a human/org authority	Scoped, time-bound, budgeted
Device / IoT / POS	Device certificates (X.509), attestation where available	
External system	OAuth2 client credentials / mTLS; API keys only for low-risk read	Keys hashed, rotatable

Identity Intelligence (KYC/KYB, sanctions/PEP screening, document verification, fraud signals) is a capability family (`identity.kyc` , `identity.kyb`). Jurisdiction rules select providers.

19.3 Authorisation

19.3 Authorisation

PDP (Cedar) at the gateway and command bus. Policies are compiled per blueprint and cached in each cell. Decision latency target p99 < 5 ms. Field-level controls come through obligations (masking/redaction). Row-level isolation is enforced by Postgres RLS using the context's tenant and organisation (defence in depth).

19.4 Threat model (STRIDE by layer, summary)

Layer	Key threats	Primary controls
Edge/Gateway	Credential stuffing, DDoS, bot scraping, injection	WAF, bot management, rate limits, passkeys, input validation from schemas
Composition/Studio	Malicious definition (privilege-escalating policy), supply-chain pack	Policy analyser, multi-party approval, signed packs, SLSA provenance, pack sandboxing
Runtime	Cross-tenant data access, IDOR, replay	RLS, PDP on every command, ETag/idempotency, audience-bound tokens
Data	Exfiltration, insider misuse	Field-level encryption, DLP, PAM with session recording, just-in-time access
Integration	Webhook spoofing, SSRF	HMAC verification, egress allow-lists per connector
AI/Agents	Prompt injection, tool abuse, data leakage, excessive agency	Tool scopes, authority tokens, content isolation (untrusted-content tagging), output filters, HITL for high-risk actions
Supply chain	Dependency compromise	SBOM, vulnerability scanning, pinned digests, reproducible builds
Operations	Misconfiguration, secret leakage	GitOps, OPA admission policies, secret scanning, workload identity

Every capability pack ships a threat model (`docs/threat-model.md`). Gate `SEC-TM-001` checks that it exists and was reviewed.

19.5 Cryptography

- In transit: TLS 1.3 everywhere; mTLS service-to-service.
- At rest: envelope encryption. KMS/HSM-backed KEKs; **per-tenant DEKs**; optional **BYOK/HYOK** for enterprise tiers.
- Field-level: `Encrypted<T>` fields (PAN tokens, national IDs, health data) encrypted application-side with per-tenant/per-purpose keys. Searchable fields

use blind indexes.

- Tokenisation: card data never touches UCOS. PSP tokens are used, keeping UCOS cardholder-data scope at SAQ-A / A-EP where possible. Network tokenisation is supported.
- In use: confidential computing nodes for designated workloads (e.g. dedicated healthcare tenants, model inference on sensitive data).
- **Crypto-agility / post-quantum readiness:** algorithm identifiers are stored with ciphertexts and signatures. Hybrid TLS key exchange (X25519+ML-KEM) where the stack supports it. ML-DSA signatures planned for audit-chain anchoring. Key rotation is automated through Temporal schedules.

19.6 Tenant & capability isolation

Network policies per pack; namespace per cell tier; separate KMS keys per tenant; optional dedicated cells; sandboxed WASM extensions with no network by default; per-tenant resource quotas.

19.7 Data classification

`public · internal · confidential · restricted` , crossed with data classes `PII, SPI (sensitive personal), PCI, PHI, FINANCIAL, CREDENTIAL, CHILD, BIOMETRIC, LOCATION, TRADE_SECRET` . Classification drives encryption, masking, retention, residency, logging redaction and DLP rules automatically.

19.8 Security operations

SIEM (all audit + security signals), SOAR playbooks (auto-contain: revoke tokens, isolate tenant, freeze payouts), EDR/XDR on nodes, CSPM/KSPM, continuous vulnerability scanning, continuous automated penetration testing (DAST + attack-simulation), bug bounty, incident response runbooks with legal/regulatory notification timers (Temporal triggers: GDPR 72h, CERT-In 6h, DPDP as applicable).

19.9 Immutable audit

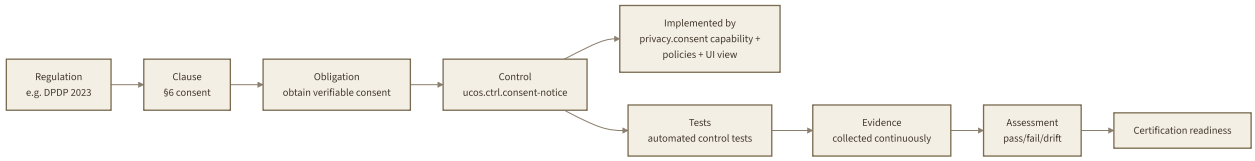
```
AuditRecord:
  id, time (TimePoint), tenant, actor (identity, kind, onBehalfOf), action, resourc
  decision (allow/deny + policy ids + obligations), context hash, request hash, res
  prevHash, hash = SHA-256(prevHash || canonical(record))
```

Records are written synchronously on the command path. Chains are per tenant per day, and daily roots are anchored to WORM storage (optionally to an external timestamping authority, RFC 3161). Audit reads are themselves audited.

---

20. Privacy & Compliance-as-Code

20.1 Compliance model



20.2 Control definition (example)

```
kind: Control
id: ucos.ctrl.dsr-erasure
title: "Erase personal data on verified request within statutory deadline"
objective: "Data subjects can obtain erasure subject to legal holds."
mappings:
  - { regulation: GDPR, clauses: ["Art.17"] }
  - { regulation: DPDP-2023, clauses: ["S.12"] }
  - { regulation: CCPA, clauses: ["1798.105"] }
  - { regulation: UAE-PDPL, clauses: ["Art.15"] }
  - { framework: ISO27701, clauses: ["7.3.6"] }
implementedBy: [privacy.dsr, data.crypto-shredding, workflow:dsr.flow@1]
parameters:
  deadline: { decision: privacy.dsr-deadline-by-jurisdiction }   # e.g. 30 days GDP
tests:
  - id: test.dsr.erasure-e2e         # synthetic subject → request → verify deletio
    schedule: "FREQ=WEEKLY"
evidence:
  collectors: [dsr-completion-log, crypto-shred-receipts, search-index-purge-proof]
owner: ucos:role:dpo
severity: high
```

20.3 Framework library (seed, extensible as data)

Privacy: GDPR, UK GDPR, CCPA/CPRA, DPDP (India), UAE PDPL, PIPL, LGPD, PDPA (SG). Security: ISO 27001, SOC 2, NIST CSF 2.0, NIST 800-53, FedRAMP, CIS. Payments/finance: PCI DSS v4, PSD2/SCA, RBI (payment aggregator, card tokenisation, data localisation), SEBI, FINRA, MiFID II, DORA, SOX. Health: HIPAA, HITECH. AI: ISO/IEC 42001, EU AI Act, NIST AI RMF. Telecom: TRAI. Sector: FDA 21 CFR Part 11, EMA, accessibility (WCAG 2.2, EN 301 549, ADA). Country- and industry-specific packs are added as data, along with future regulations.

20.4 Privacy capabilities

Consent & preference registry (purpose-based, versioned notices, proof of consent) · purpose limitation enforced through policy obligations · data minimisation (schema-level `purpose` tags) · retention engine · DSR portal & workflow (access, rectification, erasure, portability, objection, grievance) · records of processing (auto-generated from the graph) · DPIA templates · cross-border transfer registry · cookie/tracking consent for web channels · children's data rules (age gating, parental consent).

20.5 Cross-tenant learning & aggregation

Cross-tenant learning, the federated intelligence the source mentions, is allowed only through: (1) explicit tenant opt-in recorded as a contract fact; (2) aggregation with k-anonymity ≥ configured threshold and differential-privacy noise for published statistics; (3) federated learning where raw data never leaves the tenant boundary. These are enforced by a dedicated `analytics.cross-tenant` capability with its own controls.

20.6 Continuous compliance

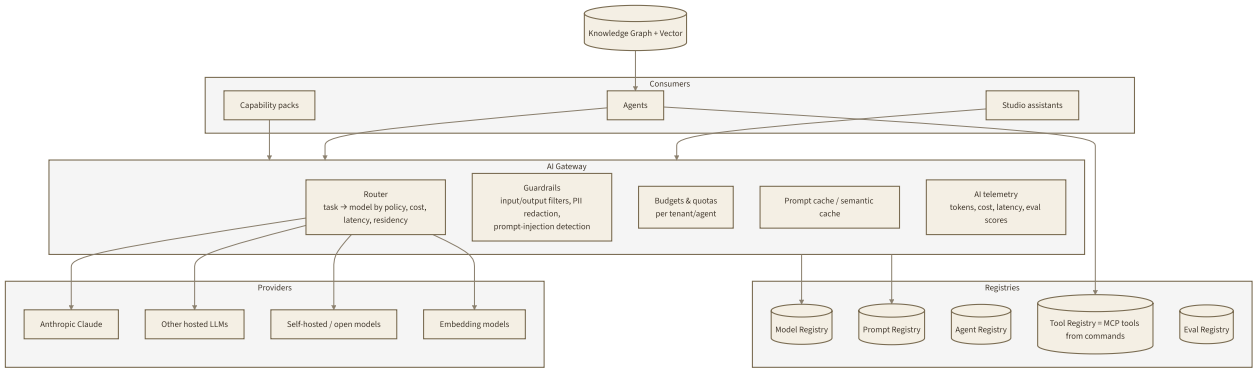
Evidence collectors run on schedules and events. Control drift creates `compliance.control.failed` events, cases and (if mandatory) gaps on affected blueprints. Audit-readiness dashboards per framework sit in the Compliance portal (W11). Auditor read-only access is time-boxed, watermarked and logged.

---

21. AI & Agent Universe

21.1 Architecture

21.1 Architecture



21.2 Model registry & routing

```
kind: Model
id: model.anthropic.claude-opus-5-5
provider: anthropic
modelId: claude-opus-5-5
capabilities: [reasoning, tool-use, vision, long-context, code]
residency: [us, eu]              # where inference may run (per contract)
dataPolicy: { training: none, retention: zero-retention-endpoint }
cost: { inputPerMTok: ..., outputPerMTok: ... }   # from price list registry, not h
evaluations: [eval.support-triage@3: 0.92, eval.catalog-attr-extract@5: 0.95]
status: certified
---
kind: Route
id: route.default
rules:   # evaluated by Decision engine
  - when: { task: complex-reasoning|agentic-planning }                then: model.a
  - when: { task: general|tool-use, latencyBudgetMs: "<3000" }        then: model.a
  - when: { task: classification|extraction|routing, volume: high }   then: model.a
  - when: { context.dataClass: PHI, tenant.policy: on-prem-only }     then: model.s
fallbacks: [model.anthropic.claude-sonnet-5-5, model.selfhosted.general]
```

Default reference routing uses the Claude family: `claude-opus-5-5` for complex reasoning and agent planning, `claude-sonnet-5-5` for general and tool-use tasks, `claude-haiku-4-5-20251001` for high-volume classification and extraction. Providers remain swappable per L15. Routing is policy, not code.

21.3 AI capability families

Recommendation · semantic & hybrid search · conversational commerce · support triage and answer drafting · catalog enrichment (classification, attribute extraction, translation, image tagging) · pricing intelligence · demand forecasting · fraud/risk scoring (ML models, not LLMs) · document understanding (invoices, KYC docs) · code/definition assistance in Studio (proposes definitions, never activates them) · compliance mapping assistance · anomaly explanation for ops · simulation narratives.

21.4 Agent object

```
kind: Agent
id: ucos:agent:catalog-enrichment
version: 1.4.0
purpose: "Enrich seller-submitted listings with taxonomy, attributes and translatio
identity: ucos:agent:catalog-enrichment            # first-class identity
owner: ucos:role:catalog-steward
authority:
  ref: ucos:authority:agent-catalog-enrichment
  actions: [catalog:ProposeAttributeValues, catalog:ProposeClassification, localiza
  forbidden: [catalog:PublishListing, pricing:*, payment:*]
  limits: { maxActionsPerHour: 5000, maxCostPerDay: { amount: "200", currency: USD
autonomy: L2          # propose → steward auto-approves above confidence 0.95, else
model: { route: route.default, task: extraction }
tools: [catalog.get_listing, product.search_global, taxonomy.suggest, attribute.val
memory: { type: episodic+semantic, store: vector://agent-memory/catalog, retention:
knowledge: [kg://taxonomy, kg://attributes, kg://global-products]
guardrails: [no-pii-in-prompts, untrusted-content-isolation, output-schema-validati
evaluation: { suite: eval.catalog-attr-extract@5, minScore: 0.93, cadence: "FREQ=WE
escalation: { queue: catalog.steward-review, when: [confidence<0.8, conflicting-sou
audit: full            # every prompt hash, tool call, output, decision
```

21.5 Agent governance

- **Identity & delegation:** agents act with their own identity. When acting for a user, they carry a delegation token (`onBehalfOf`) whose scope is the intersection of the agent's authority and the user's permissions.
- **Bounded authority:** authority objects (§18.3) with action lists, monetary limits, rate limits and budgets. Financially material, irreversible, or legally significant actions need human approval (L17).

- **Tool safety:** tools are generated from commands, inherit PDP checks, and declare side-effect class (`read` , `reversible-write` , `irreversible-write` , `external-communication`).
- **Prompt-injection defence:** retrieved and user content is tagged untrusted and kept out of the system instructions. Tool-call plans are validated against policy before execution. Sensitive tools require confirmation.
- **Multi-agent orchestration:** a supervisor agent pattern with explicit hand-off contracts. Agent-to-agent messages go through the Communication universe and are audited.
- **Evaluation & drift:** offline eval suites, online quality signals (acceptance rate, edits), drift alerts, and rollback to previous prompt/model versions.
- **AI governance & regulation:** risk classification per use case (EU AI Act classes), model cards, data sheets, human oversight design, and incident reporting. ISO/IEC 42001 control mapping.

21.6 RAG over the Knowledge Graph

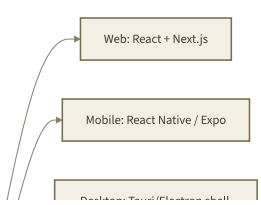
Hybrid retrieval: (1) graph traversal for authoritative facts with provenance, (2) vector search for unstructured content, (3) keyword search for exact identifiers. Answers cite object IDs. Facts below a trust threshold are excluded or flagged. Tenant boundaries are enforced in retrieval filters, never only in prompts.

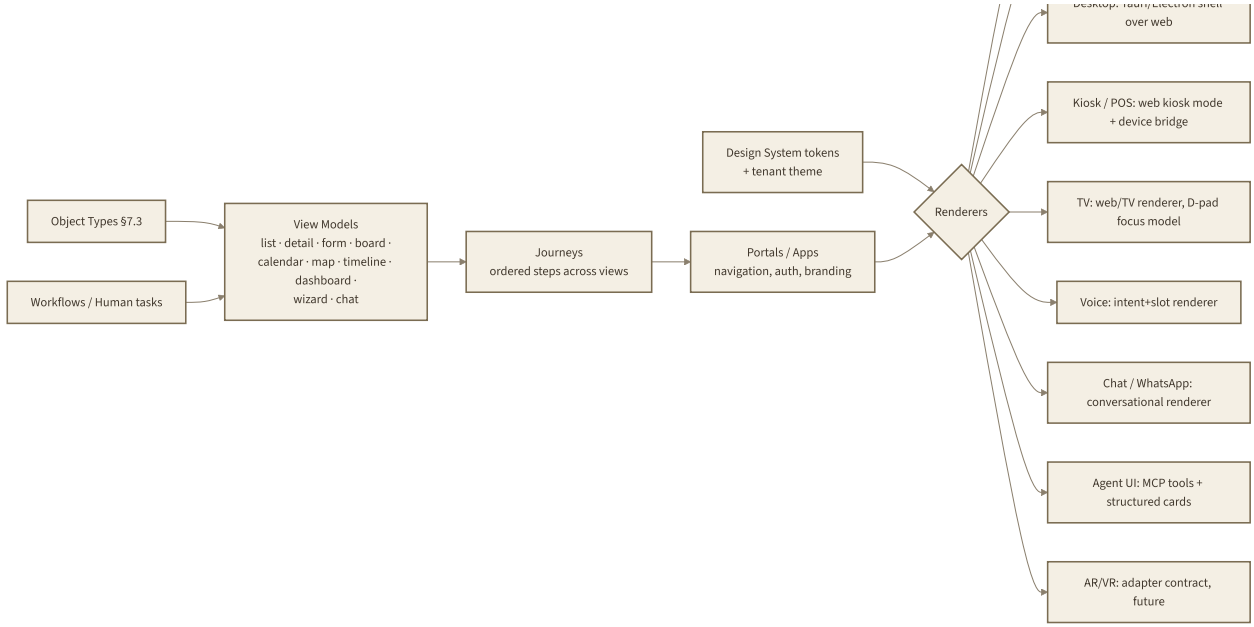
PART V

EXPERIENCE

22. Experience Architecture

22.1 UI is metadata, renderers are code





The same `ViewDefinition` renders on every channel. Each renderer maps abstract components (`Field` , `Collection` , `Action` , `Media` , `Map` , `Calendar` , `Chart`) to native widgets. Channels that cannot render a component fall back through declared fallbacks (for example, a map falls back to an address list on voice).

22.2 View definition (UI meta-model)

```
apiVersion: ucos.io/v1
kind: ViewDefinition
id: catalog.listing.edit
version: 2.1.0
object: catalog.listing.Listing
type: form
audience: [seller, catalog-admin]
layout:
  - section: { title: t("listing.basics"), columns: { sm: 1, md: 2 } }
    fields:
      - gproduct: { widget: product-match-picker, help: t("listing.gproduct.help")
      - title:    { widget: text, maxLength: 150, ai: { suggest: catalog-enrichment
      - condition: { widget: segmented, source: taxonomy:item-condition }
  - section: { title: t("listing.pricing") }
    fields:
      - price:    { widget: money, currencyFrom: context.currency }
      - priceList: { widget: reference-select, target: pricing.PriceList, visibleWh
  - section: { title: t("listing.availability") }
    fields:
```

```
      - stock:    { widget: stock-by-location, source: inventory.StockLevel }
      - availabilityPolicy: { widget: temporal-policy-picker }
  - section: { title: t("listing.media") }
    fields:
      - media:    { widget: asset-gallery, accept: [image/*, video/*], min: 1 }
actions:
  primary:   { command: catalog:PublishListing, label: t("publish"), confirm: false
  secondary: [ { command: catalog:SaveDraft }, { command: catalog:PreviewListing }
validation: schema + decision:catalog.listing-quality-gate
permissions: policy            # fields hidden/masked per PDP obligations
analytics: { track: [view, field-error, publish] }
a11y: { landmarks: auto, errorSummary: top, focusOnError: true }
variants:
  mobile: { layout: single-column, stickyPrimaryAction: true }
  voice:  { mode: slot-filling, required: [gproduct, price, stock] }
```

22.3 Portals & apps (generated per blueprint)

Portal	Audience	Purpose
Platform Console	Platform owner/operators	Cells, tenants, packs, health, governance across all tenants
Builder Studio	Tenant builders, solution architects	Compose platforms: definitions, workflows, UI, policies, integrations
Tenant Admin	Tenant administrators	Users, roles, settings, plans, feature flags, environments
Storefront / Customer apps	End customers	Web + iOS + Android + WhatsApp + voice
Seller / Vendor / Partner portal	Marketplace participants	Listings, orders, payouts, performance, support
Operations portal	Ops staff	Fulfillment, logistics exceptions, inventory, dispatch
Support portal / Agent desktop	Support agents	Cases, conversations, knowledge, customer 360
Finance portal	Finance	Ledger, settlements, reconciliation, invoices, tax
Governance & Compliance portal	Governance, DPO, CISO, auditors	Approvals, policies, controls, evidence, certifications, audit

Developer portal	Developers	API docs, SDKs, sandbox, webhooks, pack publishing
AI Operations portal	AI owners	Models, prompts, agents, evals, guardrail incidents, costs
Observability / Twin portal	SRE, leadership	SLOs, incidents, cost, platform & business twin

22.4 Channel capabilities matrix

Capability	Web	iOS/Android	WhatsApp/Chat	Voice	POS/Kiosk	TV	A
Browse/search	✓	✓	✓ (carousel)	✓ (top-3)	✓	✓	✓
Cart/checkout	✓	✓ (native pay)	✓ (pay link)	✓ (confirm by voice + OTP)	✓ (device pay)	✓ (QR handoff)	✓ c o
Account/DSR	✓	✓	link-out	link-out	—	link-out	✓
Offline	PWA cache	✓ local store + sync	—	—	✓ store-and-forward	—	—
Push/notifications	web push	APNs/FCM	native	—	—	—	e

23. UI/UX Design System & Principles

23.1 UX principles for a composition platform

1. **Intent first, configuration second:** builders describe outcomes ("multi-vendor marketplace in India and UAE"). The system proposes, and the builder refines.
2. **Explain every automatic decision:** every auto-added capability, control or gap has a "Why?" link to its resolution trace (§11.3).
3. **Preview, simulate, diff, then apply:** no change goes live without a visible diff and an optional simulation run.
4. **Safe by default:** destructive or high-risk actions require confirmation, reason capture and step-up auth, with undo where reversible.
5. **Progressive disclosure:** sensible defaults hide advanced settings until needed

Expert mode exposes raw YAML with schema-aware editing.

6. **One mental model:** every object has the same anatomy (header, status, key facts, tabs: Details · Relationships · Timeline · Policies · Audit · Evidence).
7. **Status is always visible:** lifecycle state, certification badge, gap count and environment (sandbox/simulation/production) are pinned in the header.
8. **Accessible and global by construction:** WCAG 2.2 AA minimum, RTL, locale-aware formats, plain language.
9. **Human-in-the-loop clarity:** AI suggestions are visually distinct, show confidence and source, and are never auto-applied unless authority allows it.
10. **Performance as UX:** interactive in < 2.5s (LCP) on mid-range mobile over 4G; optimistic UI where safe.

23.2 Design tokens (theming & white-label)

```
{
  "color": {
    "brand":   { "primary": "{tenant.brand.primary|#4F46E5}", "onPrimary": "#FFFFF
    "neutral": { "0": "#FFFFFF", "50": "#F8FAFC", "100": "#F1F5F9", "700": "#33415
    "status":  { "success": "#15803D", "warning": "#B45309", "danger": "#B91C1C", '
    "gap":     { "mandatory": "#B91C1C", "recommended": "#B45309", "optional": "#64
  },
  "font":   { "family": { "base": "{tenant.font|Inter}, system-ui", "mono": "JetBra
             "size": { "xs": 12, "sm": 14, "md": 16, "lg": 20, "xl": 24, "2xl": 32
  "space":  { "1": 4, "2": 8, "3": 12, "4": 16, "6": 24, "8": 32 },
  "radius": { "sm": 4, "md": 8, "lg": 12, "full": 9999 },
  "motion": { "fast": "120ms", "base": "200ms", "reduce": "respect prefers-reduced-
  "breakpoints": { "sm": 0, "md": 768, "lg": 1024, "xl": 1440 },
  "modes": ["light", "dark", "high-contrast"]
}
```

Tenant themes override only tokens, never components. Contrast is validated automatically (≥ 4.5:1 for text). A theme that fails contrast creates a `UI` gap.

23.3 Component library (abstract → per-renderer)

App shell · Navigation (sidebar, top bar, tab bar, breadcrumb) · Object header · Data table (virtualised, column config, saved views, bulk actions) · Card · Form fields (text, number, money, quantity+unit, temporal pickers, reference picker, taxonomy tree

picker, address/geo picker, asset uploader, rich text, code/YAML editor) · Status badge · Gap badge · Certification badge · Timeline · Kanban board · Calendar (month/week/day/resource grid) · Map · Graph canvas (composition/dependency) · Diff viewer · Chart set (via the dataviz standard) · Wizard/stepper · Command palette (⌘K) · AI suggestion chip · Approval panel · Empty states · Skeletons · Toasts · Modal/drawer.

23.4 Accessibility, localisation, personalisation

- **A11y:** semantic landmarks, keyboard paths for every action, focus management on route change, ARIA only where semantics are insufficient, captions/transcripts for media, motion reduction. Automated axe checks in CI plus manual audits on top journeys (gate `A11Y-*`).
- **Localisation:** ICU MessageFormat; no string literals in views (lint). Pseudo-localisation in CI. RTL mirroring. Locale-aware number, currency, date, address and name formats from the Localization universe. Translation workflow: machine translation, then human review for regulated or customer-facing strings.
- **Personalisation:** recommendation placements, saved views, role-based home dashboards, adaptive navigation (frequently used first), all consent-aware.

---

24. Information Architecture & User Journeys

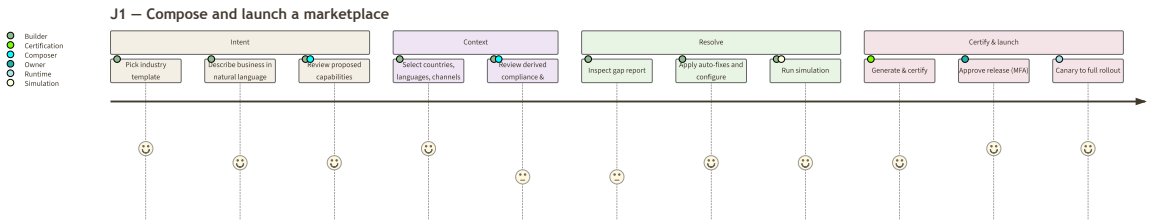
24.1 Platform Console navigation (single canonical IA)

```
UCOS Ω∞
├── 01 Foundation      Identity · Organizations & Tenants · Geography · Temporal · Currenc
├── 02 Knowledge       Products · Services · Industries · Categories · Brands · Manufactur
│                      Assets · Documents · Media · Knowledge Graph · Ontology · Taxonomy
├── 03 Business        Customers · Sellers · Buyers · Vendors · Partners · Distributors ·
│                      (Leads, Opportunities, Accounts, Contacts, Activities, Pipelines)
├── 04 Commerce        Catalogs · Pricing · Price Lists · Offers · Promotions · Coupons ·
│                      Affiliates · Marketplace · Advertising · Merchandising · Cart · Che
│                      Subscriptions · Memberships · Billing · Payments · Settlements · In
├── 05 Operations      Inventory · Warehousing · Procurement · Fulfillment · Logistics · T
│                      Tracking · Returns · Exchanges · Service Delivery · Field Ops · Sup
│                      Cases · Work Orders · Maintenance
├── 06 Experience      Websites · Mobile Apps · Portals · Kiosks · POS · Themes · Brandinc
```

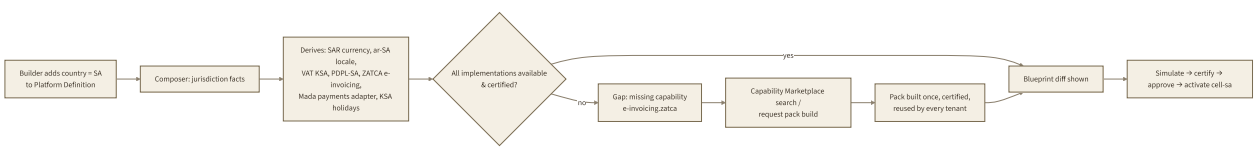
```
│                    UI Components · Channels · Personalisation · Accessibility · Locali
├── 07 Intelligence  Models · Agents · Agent Teams · Prompts · Recommendation · Search
│                    Risk · Optimisation · Simulation · Digital Twins · Learning
├── 08 Governance    Policies · Standards · Regulations · Compliance · Security · Privac
│                    Evidence · Certifications · Audits · Approvals · Change Control · T
├── 09 Platform      Capabilities · Marketplace · Builder Studio · Generators · APIs · I
│                    Decisions · Integrations · Connectors · Runtime · Deployments · Ext
│                    Capability Discovery · Gap Detection · Dependency & Conflict Analys
├── 10 Observability Monitoring · Metrics · Logs · Traces · Dashboards · Alerts · SLO/SL
│                    Health · RCA · Diagnostics
├── 11 Administration Tenants · Companies · Stores · Workspaces · Users · Roles · Permiss
│                    Plans · Billing · Provisioning · Environments · Secrets · Settings
└── 12 Architecture  Constitution · Universes · Ontologies · Taxonomies · Registries · C
                     Schemas · Events · APIs · Workflows · Graphs (Capability, Knowledge
                     Traceability) · Reference Architectures · Build Specs · Evolution
```

Every menu item resolves to one canonical home (Law L3). The same object never appears in two menus as two different things. Cross-links open the canonical page.

24.2 Journey J1: Create a platform (Builder)



24.3 Journey J2: Add a new country (zero code)

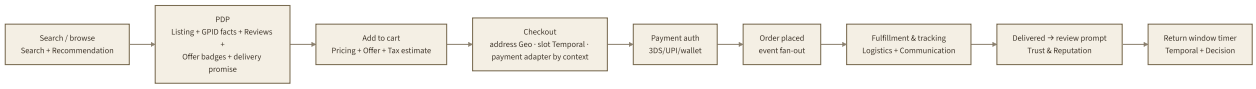


24.4 Journey J3: Seller onboarding (marketplace)

```
sequenceDiagram
```

```
actor S as Seller
participant SP as Seller Portal
participant ID as Identity/KYB
participant WF as Workflow seller.onboarding
participant CM as Compliance
participant LG as Ledger
S-->>SP: Sign up (business email, passkey)
SP-->>ID: Create org identity, start KYB (tax ID, bank, documents)
ID-->>WF: KYB result (auto-pass / review / fail)
WF-->>CM: Screen sanctions/PEP, category restrictions (rules by jurisdiction)
CM-->>WF: Cleared
WF-->>SP: Agreement e-sign (Document universe)
S-->>SP: Sign seller agreement
WF-->>LG: Open seller ledger accounts, payout schedule (Temporal: T+7 settlement calendar
WF-->>S: Welcome; listing tools unlocked (Entitlement)
```

24.5 Journey J4: Buyer purchase (storefront, cross-capability)



24.6 Journey J5: Governed change & approval

Proposal → automated validation → impact analysis (affected tenants, consumers) → simulation → approval (authority by scope; MFA) → certification → progressive rollout → monitoring with auto-rollback. See §18.4 and wireframe W11.

24.7 Journey J6: Data subject request

Request via portal/app/email → identity verification → scope discovery (graph: all objects linked to subject) → legal-hold check → fulfil (export package or erase/crypto-shred) → evidence → response within jurisdictional deadline (Temporal SLA clock) → audit.

---

25. Wireframes

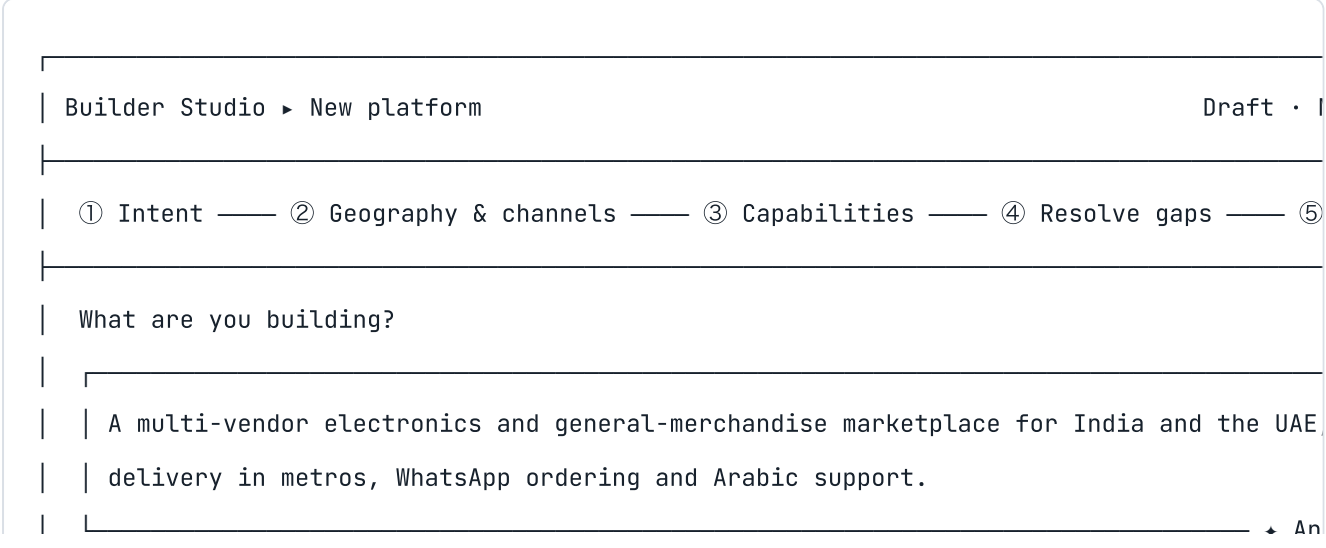
Low-fidelity wireframes for the key screens. They define **layout, hierarchy and behaviour**, not visual styling (§23 covers that). Desktop width is ~120 columns. Mobile wireframes are marked 📱.

W1 — Platform Console · Home (operator)



Behaviour: KPI tiles drill into filtered lists. Gap and approval counts are always in the header. The ⌘K palette searches objects, commands and docs across the graph.

W2 — Builder Studio · New Platform · Step 1: Intent



```
│
│  Or start from a template
│  ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐ ┌
│  │ 🛒 Multi-vendor  │ │ 🏬 Single brand  │ │ 🚗 Mobility /    │ │ 🎬 Subscription
│  │   marketplace    │ │   D2C store      │ │   ride-hailing   │ │   media (OTT)    │ │
│  │ ● selected       │ │                  │ │                  │ │                  │ │
│  └──────────────────┘ └──────────────────┘ └──────────────────┘ └──────────────────┘ └
│
│  ✦ Detected from your description                                    (AI suggestion
│   Industry: Retail › Electronics, Retail › General merchandise      Model: B2C · Market
│   Outcomes: same-day delivery · conversational ordering · RTL localisation        [Accep
│
│                                                                          [Save draf
└──────────────────────────────────────────────────────────────────────────────────────
```

W3 — Builder Studio · Step 2–3: Geography, channels & capabilities

```
┌──────────────────────────────────────────────────────────────────────────────────────
│  ① Intent ✓ ─── ② Geography & channels ●─── ③ Capabilities ─── ④ Resolve gaps ──
├──────────────────────────────────────────┬──────────────────────────────────────────
│ Countries                                │ Derived automatically  (why? ⓘ)
│  [🇮🇳 India ×] [🇦🇪 UAE ×] [+ Add country]   │ Currencies   INR, AED
│ Languages   [English ×][हिन्दी ×][العربية ×]   │ Locales      en-IN, hi-IN, en-AE, a
│ Data residency  (●) Strict  ( ) Flexible │ Tax          GST India 4.0 · VAT UAE 2.
│                                          │ Privacy      DPDP 2023 · UAE PDPL
│ Channels                                 │ Payments     UPI, cards, netbanking (IN
│  [✓] Web  [✓] iOS  [✓] Android           │              Pay (AE)  → adapters: raz
│  [✓] WhatsApp  [ ] Voice  [ ] POS  [✓] Agent API │ Calendars    IN national + state hol
│                                          │ Cells        cell-in-mum-1, cell-ae-dxb
├──────────────────────────────────────────┴──────────────────────────────────────────
│ Capabilities                                  Search capability marketplace… [
│  REQUESTED              AUTO-ADDED (dependencies & obligations)                  OPT
│  ☑ marketplace.core     ☑ identity.core        why? ⓘ  ☑ privacy.consent  why? ⓘ   □ l
│  ☑ logistics.multi-     ☑ catalog.listing      why? ⓘ  ☑ privacy.dsr      why? ⓘ   □ a
│    carrier              ☑ pricing.engine       why? ⓘ  ☑ tax.gst-india    why? ⓘ   □ s
│  ☑ payment.core         ☑ temporal.core        why? ⓘ  ☑ tax.vat-uae      why? ⓘ   □ r
│  ☑ offer.engine         ☑ inventory.multi-node why? ⓘ  ☑ payment.tokenization     □ sel
│  ☑ reviews.core         ☑ ledger.core          why? ⓘ  ☑ communication.whatsapp
```

```
│  ☑ support.case          … 38 more  [show]                                  │
│                                                  Est. cost: $18.4k/mo at expected s
└──────────────────────────────────────────────────────────────────────────────┘

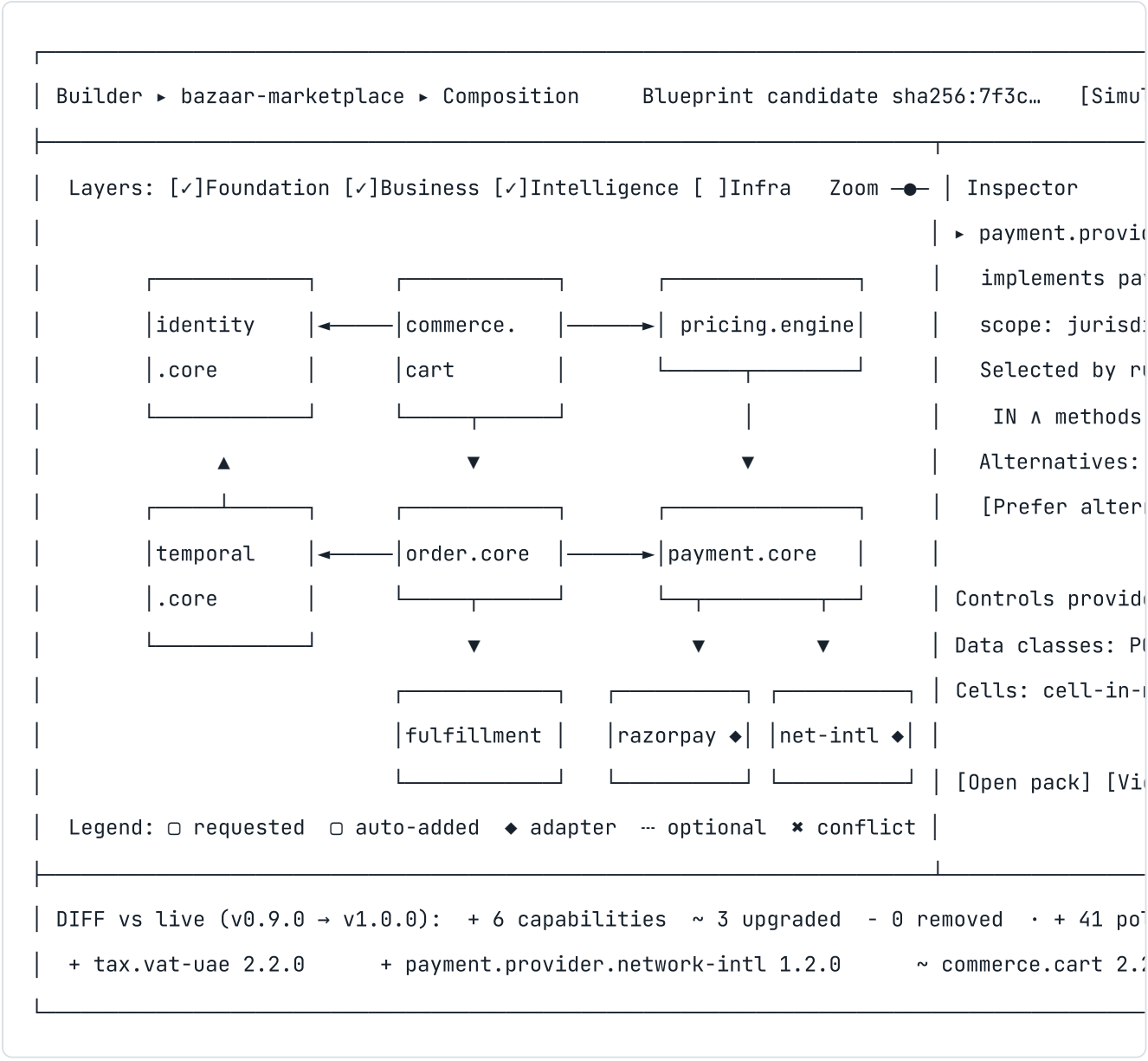
  "why? ⓘ" popover example:

  ┌──────────────────────────────────────────────────────────┐
  │ privacy.dsr is required because:                         │
  │  • DPDP 2023 applies to jurisdiction IN (you selected India)  │
  │  • DPDP §12 → control ucos.ctrl.dsr-erasure              │
  │  • control implemented by privacy.dsr@1.3                │
  │ Rule: required_capability/1 (rule R-CMP-004)   [View trace]  │
  └──────────────────────────────────────────────────────────┘
```

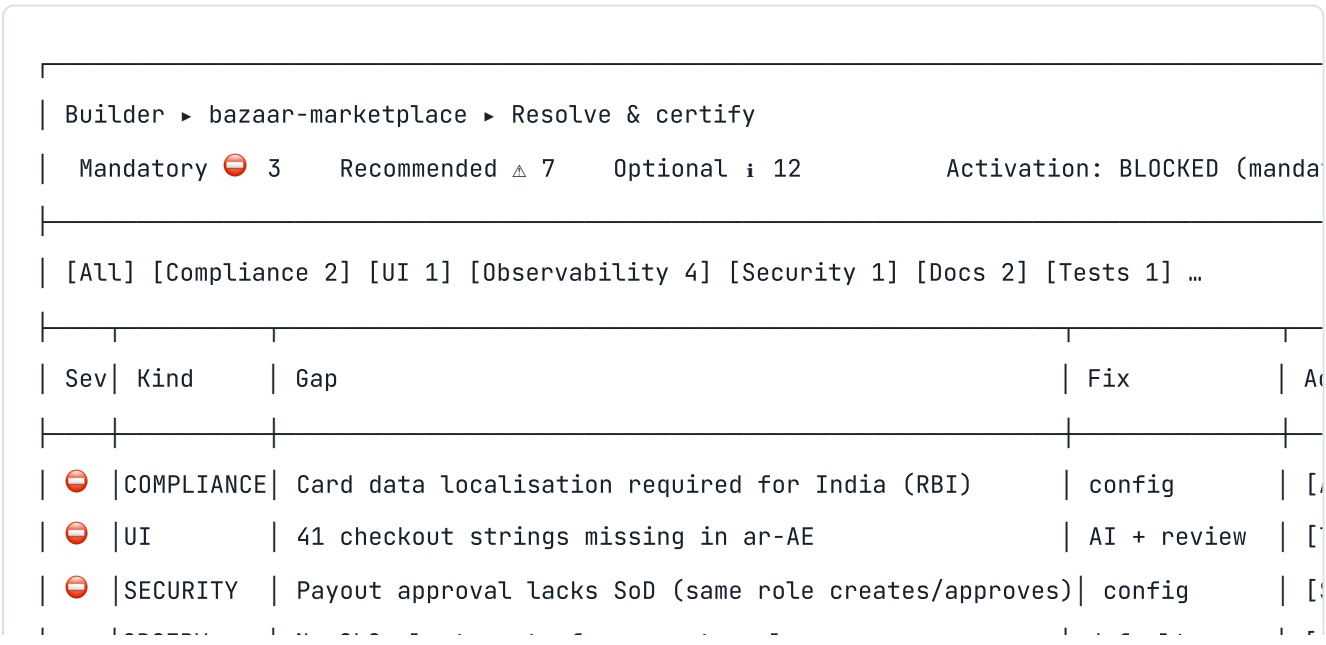
W4 — Capability Marketplace · Browse & detail

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ Platform ▸ Capability Marketplace          [Search: "subscription billing"          ]
├──────────────────────┬───────────────────────────────────────────────────────┤
│ Filters              │  48 results                                       Sor
│ Universe             │ ┌─────────────────────────────────────────────────────┐
│  ☐ Commerce (21)     │ │ subscription.core  v3.2.0     ✔ Certified   ★ 4.8 (312 tenants)
│  ☑ Subscription (9)  │ │ Plans, renewals, proration, dunning, usage records. Requires t
│  ☐ Finance (12)      │ │ pricing.engine, payment.core, ledger.core.    Controls: PCI-DS
│ Maturity             │ │                                          [Details] [A
│  ☑ Certified         │ ├─────────────────────────────────────────────────────┤
│  ☐ Beta              │ │ billing.usage-metering v1.4.1  ✔ Certified  ★ 4.6   Usage-base
│ Compliance           │ ├─────────────────────────────────────────────────────┤
│  ☐ PCI-DSS ☐ HIPAA   │ │ entitlement.core  v2.0.3      ✔ Certified   ★ 4.7   Free
│ Publisher            │ └─────────────────────────────────────────────────────┘
│  ☑ UCOS  ☐ Verified  │
├──────────────────────┴───────────────────────────────────────────────────────┤
│ DETAIL ▸ subscription.core v3.2.0
│ [Overview] [Objects (7)] [Commands (22)] [Events (15)] [Workflows (4)] [UI (11)] [Contr
│ [Dependencies] [SLOs] [Changelog] [Certification report]
│  Dependency graph:  subscription.core —requires→ temporal.core, pricing.engine, payment
│                                      —optional→ entitlement.core (feature gating), tax
│  Security score A · Last pen-test 2026-09-14 · SBOM ✓ · Signed ✓ (cosign) · SLSA L3
└──────────────────────────────────────────────────────────────────────────────┘
```

W5 — Composition Canvas · Blueprint graph, "Why?" & diff



W6 — Gap Report & Certification Gate



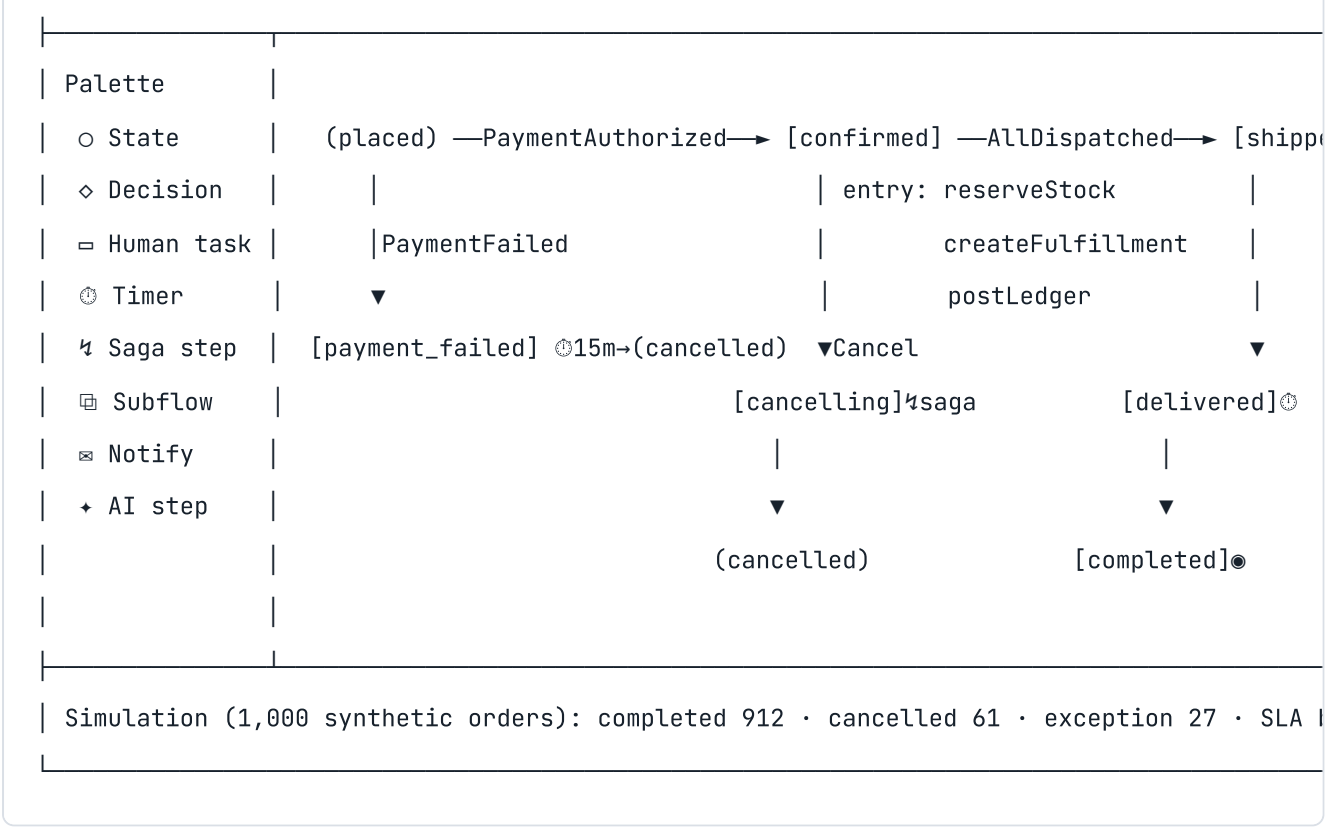
```
| ⚠  |OBSERV.   | No SLO alert route for payout.cycle                    | default     | [A
| ⚠  |DOCS      | Seller runbook missing for COD reconciliation          | AI draft    | [D
├────┴──────────┴────────────────────────────────────────────────────────┴─────────────┴──
| Certification gates
|  ARCH ✔   SEC ⏳ running (14/22)   CMP ⛔ blocked   TEST ✔ 4,812 passed   PERF ⏳   A11
|
| Waive "No SLO alert route…" ──────────────────────────────────────────────────────────
|  Reason* [Alert routing will be configured by NOC on onboarding day                  ]
|  Expires* [2026-10-30 ▾]   Authority: bazaar-ops-lead (level 2)   □ Notify governance
└──────────────────────────────────────────────────────────────────────────────────────
```

W7 — Object Designer (no-code data modelling)

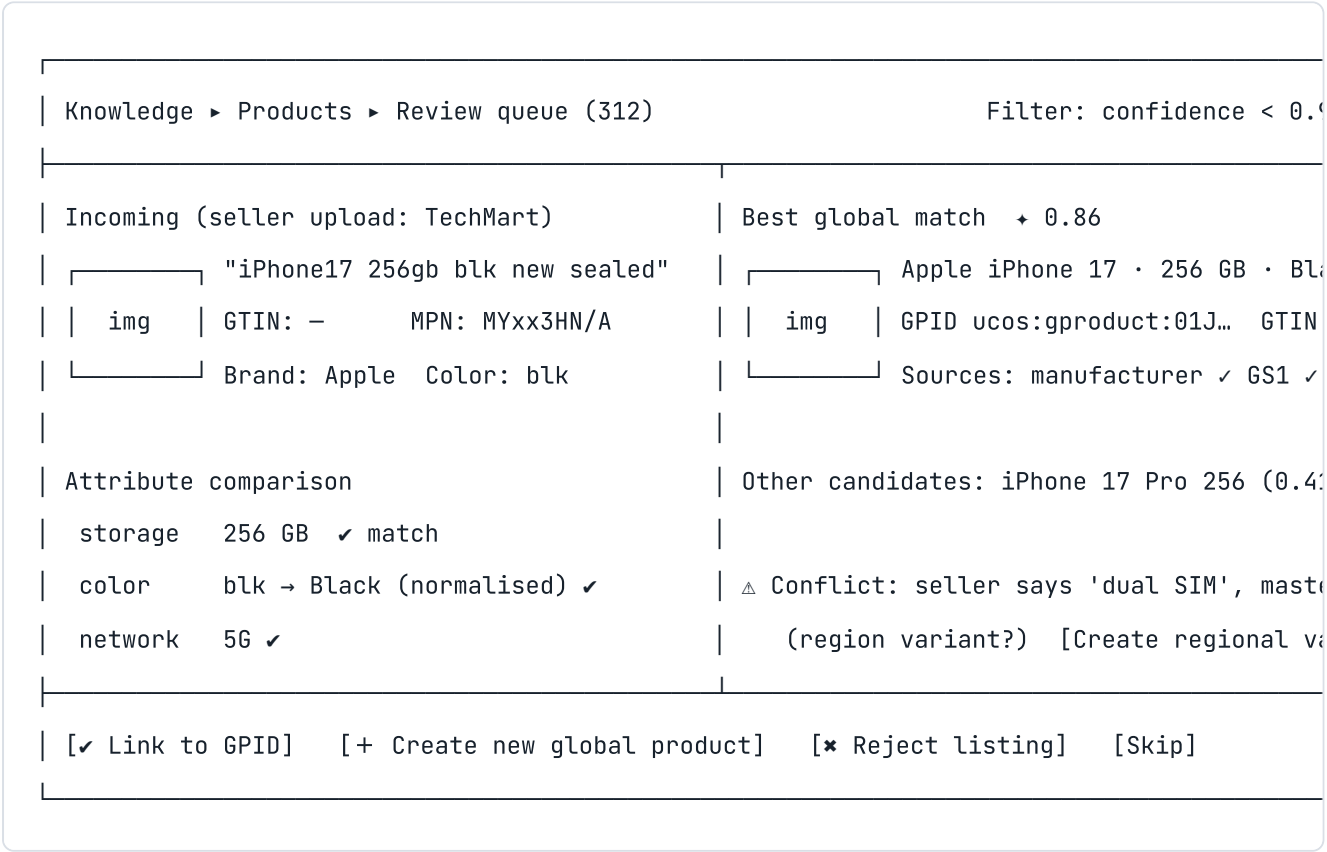
```
┌──────────────────────────────────────────────────────────────────────────────────────
| Architecture ▸ Object types ▸ healthcare.encounter.Encounter  v0.3.0 (Draft)        [Va
├──────────────────────────┬──────────────────────────────────────────────────────────
| Types                    | Fields                                        [+ Add field]
| ▾ healthcare             | ┌──────────────┬──────────────────┬──────┬─────────┬─────────┐
|   ▸ Patient (ref Person) | | Name         | Type             | Card.| Class   | Index   |
|   ● Encounter            | ├──────────────┼──────────────────┼──────┼─────────┼─────────┤
|   ▸ Practitioner         | | patient      | Reference→Pers.  | 1    | PHI     | ✓       |
|   ▸ Observation          | | practitioner | Reference→Prac.  | 1..N | PII     |         |
| ▾ commerce               | | period       | TemporalRef      | 1    | PHI     | ✓       |
|   ▸ Order                | | location     | Reference→Place  | 0..1 | intern. |         |
|   ▸ Cart                 | | reasonCode   | Enum(tax: ICD)   | 0..N | PHI     | facet   |
|                          | | notes        | Encrypted<Text>  | 0..1 | PHI     |         |
| Relationships            | | custom.*     | Extension        |      | tenant  |         |
|  patient —subjectOf→ Enc.| └──────────────┴──────────────────┴──────┴─────────┴─────────┘
|  Enc. —partOf→ EpisodeOfC| Lifecycle: planned → arrived → in-progress → finished
|                          | Policies: 4  Events: 5  Views: 3  Controls: HIPAA 9
├──────────────────────────┴──────────────────────────────────────────────────────────
| Validation:  ✔ schema   ✔ naming   ⚠ 'notes' PHI requires purpose tag   ✔ no duplicate c
└──────────────────────────────────────────────────────────────────────────────────────
```

W8 — Workflow Designer

```
┌──────────────────────────────────────────────────────────────────────────────────────
| Platform ▸ Workflows ▸ order.lifecycle v3.0.0                  [Test with sample] [Simula
```

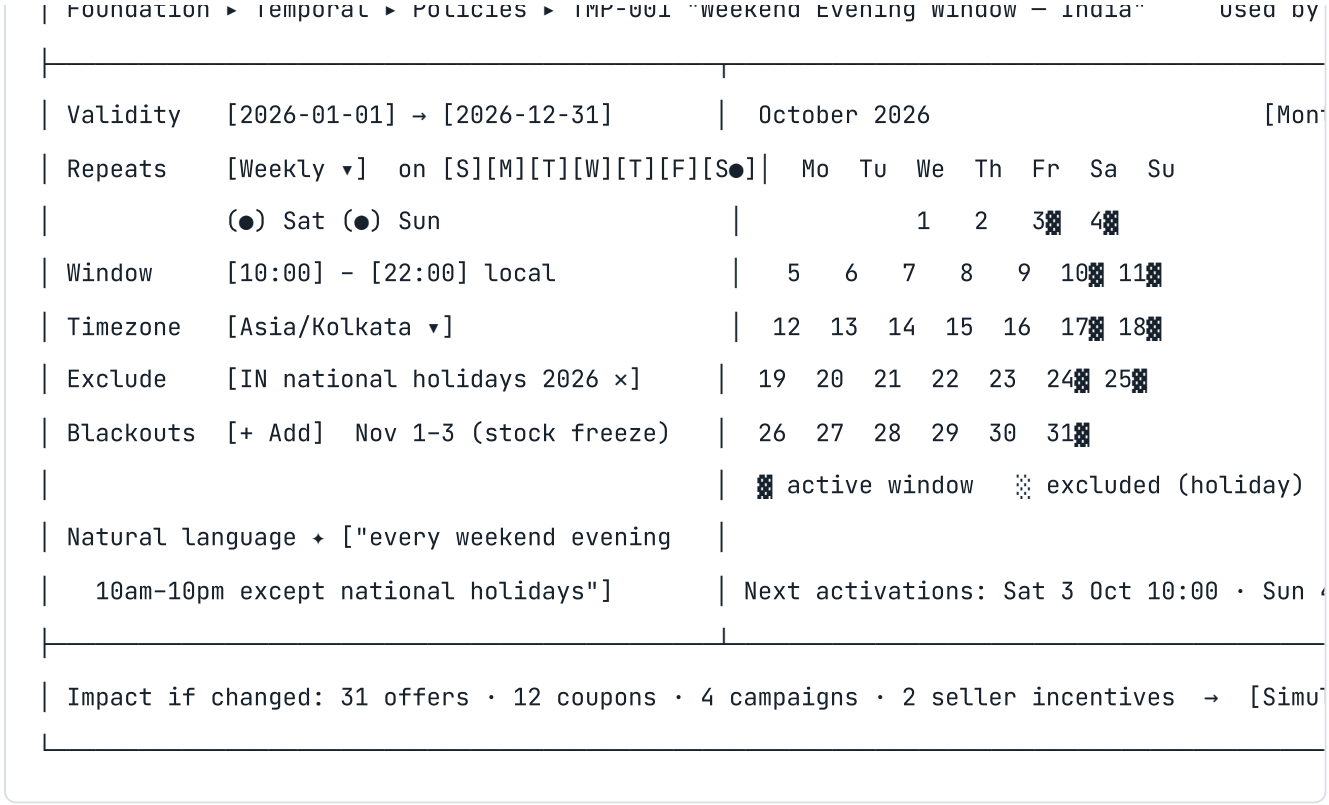


W9 — Product Intelligence · Match & steward review

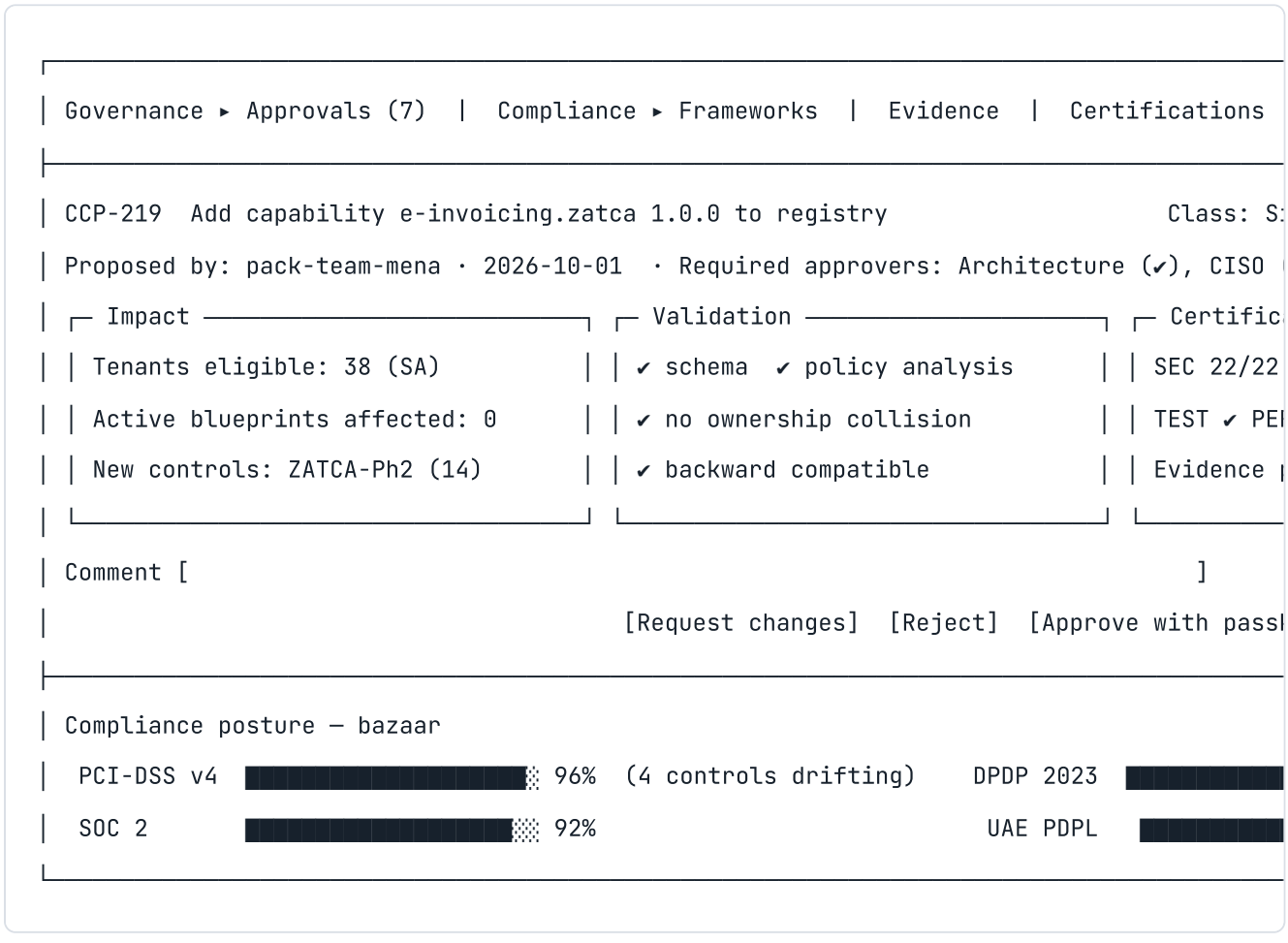


W10 — Temporal Policy editor & calendar

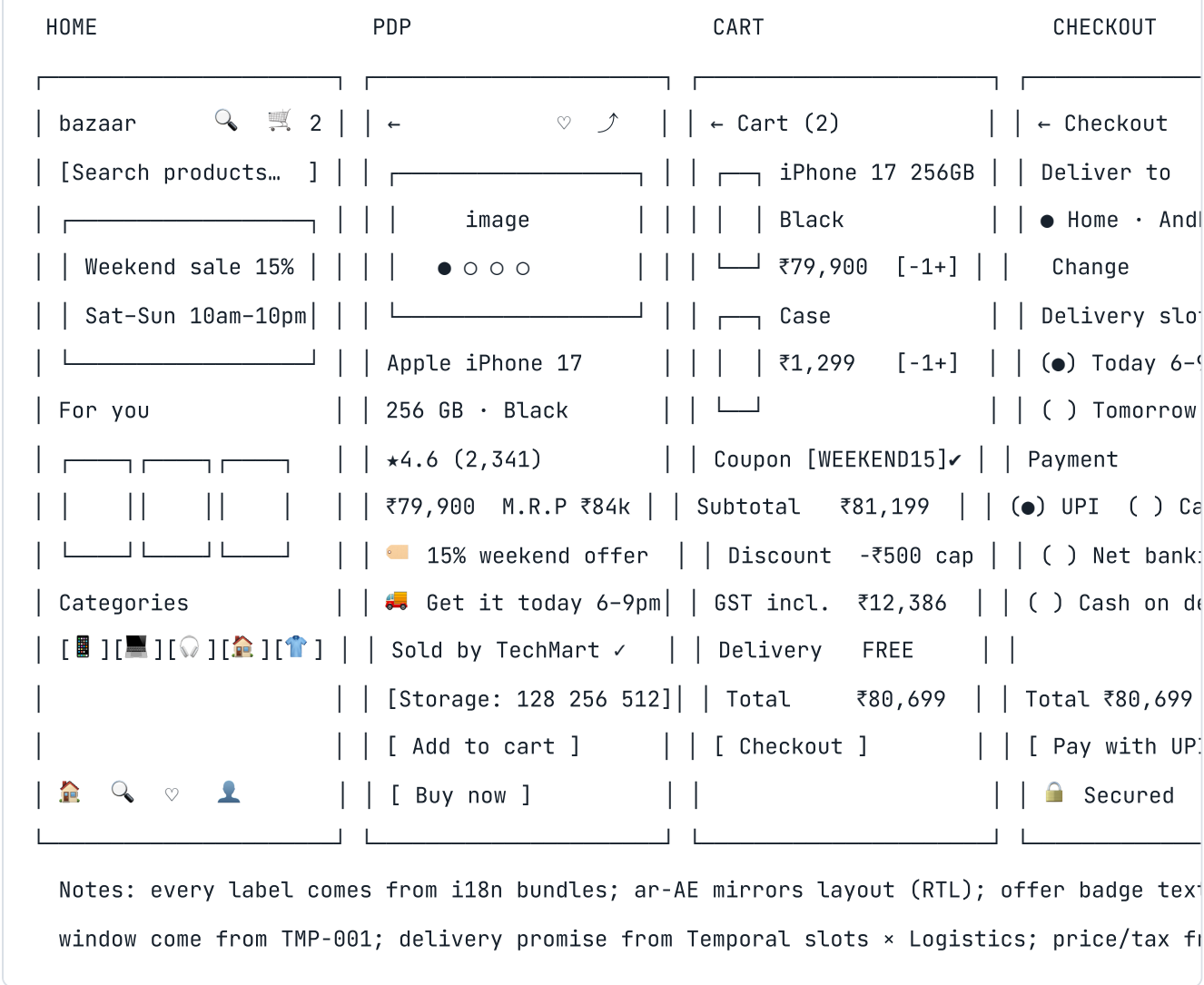




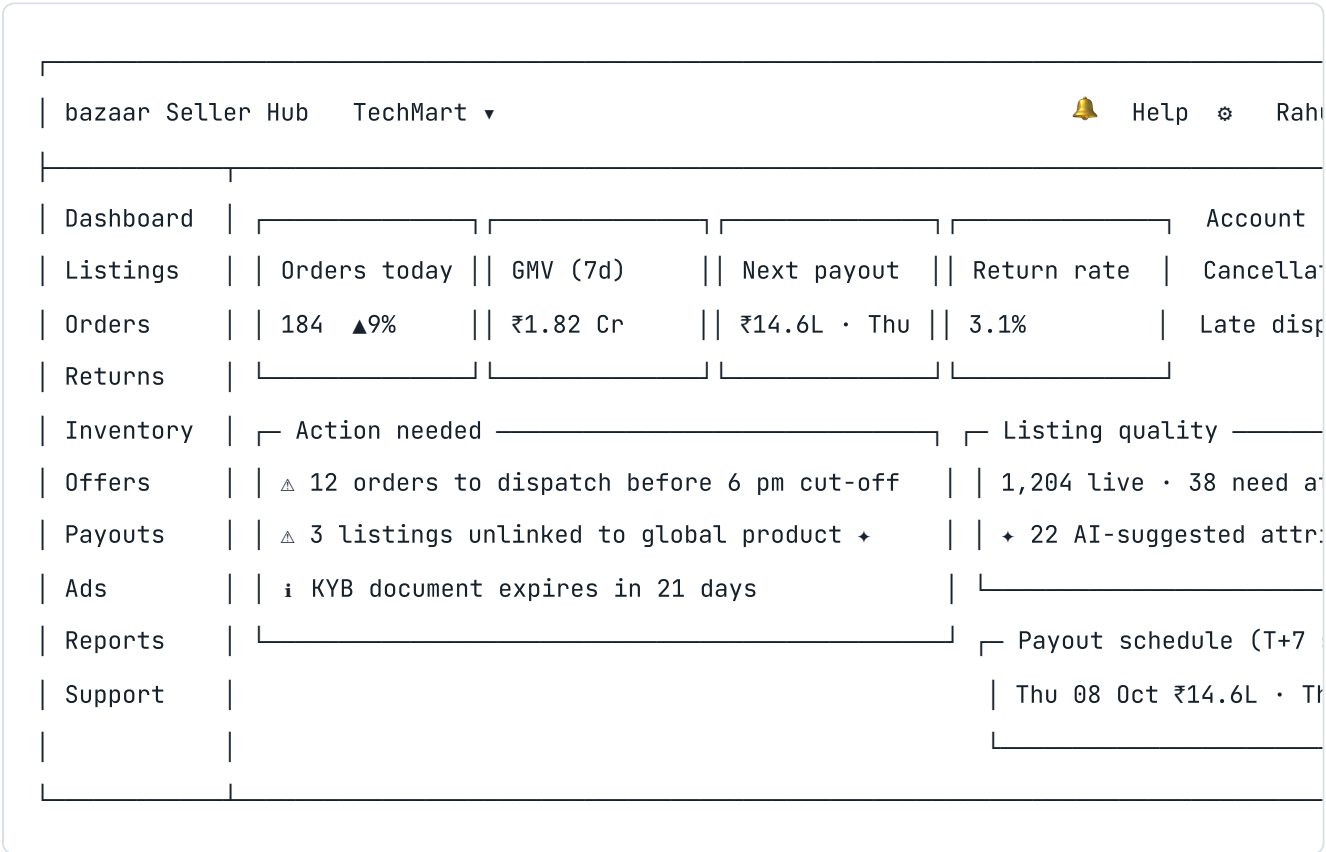
W11 — Governance & Compliance portal



W12 — Generated storefront 📱 (mobile: Home → PDP → Cart → Checkout)



W13 — Seller portal · Dashboard



W14 — Observability · Platform Twin

```
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│ Observability ▸ Platform Twin ▸ bazaar (blueprint sha256:7f3c…)          Live ●  | Time ▾ │
├──────────────────────────────────────────────────────┬───────────────────────────────────┤
│ Capability health map                                │ SLOs (28d)                        │
│  identity.core ●  catalog ●  cart ●  order ●  payment ◐ │  Checkout success     99.71%  │
│  ledger ●  tax ●  fulfillment ●  logistics ◐  search ● │  AddLine p99          132ms   │
│  ◐ payment: razorpay adapter error rate 2.4% (↑)     │  Payment auth p95     2.8s    │
│  ◐ logistics: carrier X tracking webhook lag 14m     │  Error budget left    41%     │
├──────────────────────────────────────────────────────┼───────────────────────────────────┤
│ Business signals      orders/min ▁▂▅▇▆▃  GMV ▂▃▅▇    │ Suggested actions ✦ (AIOps)       │
│ Security signals      blocked logins 1.2k  anomalies 3 │ 1. Shift 30% UPI traffic to …  │
│ Compliance signals    controls drifting 4            │    [Simulate] [Apply w/ approval] │
│ AI signals            agent actions 18k  escalations 92 │ 2. Open case with carrier X   │
│ Cost                  $612/day (▲4%)                 │    [Create case]                  │
└──────────────────────────────────────────────────────┴───────────────────────────────────┘
```

W15 — AI Operations · Agent console

```
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│ Intelligence ▸ Agents ▸ catalog-enrichment v1.4.0   Autonomy L2  ● Running   Budget $1… ▾ │
├──────────────────────────────────────────────────────────────────────────────────────────┤
│ [Runs] [Authority] [Tools] [Prompts] [Evals] [Guardrail events] [Costs] [Audit]          │
│ ┌─ Last 24h ──────────────────────────┐ ┌─ Eval suite catalog-attr-extract@5 ──────────┐ │
│ │ Runs 18,204 · Proposals 61,390      │ │ Score 0.951 (min 0.93) ✔   Trend ▃▅▆▇▇▇      │ │
│ │ Auto-approved 48,117 (≥0.95 conf.)  │ │ Model: claude-haiku-4-5 · Route: extraction   │ │
│ │ Human review 13,273 · Rejected 2.1% │ └──────────────────────────────────────────────┘ │
│ └─────────────────────────────────────┘                                                  │
│ Run 01J9Z… ▸ listing ucos:listing:01J…                                                   │
│  1 tool catalog.get_listing ✔   2 tool product.search_global ✔ (3 candidates)            │
│  3 model → proposal {classification: phones/smartphones, attrs: 7, confidence: 0.97}     │
│  4 policy check: catalog:ProposeAttributeValues ALLOW (authority ucos:authority:agent-ca… │
│  5 auto-approved (confidence ≥ 0.95)   [View prompt hash] [Replay in sandbox] [Report is… │
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

W16 — Audit Explorer

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ Governance ▸ Audit     Tenant [bazaar ▾]  Actor [any ▾]  Action [marketplace:* ▾]  Rang      │
├──────────────────────┬──────────────────────┬──────────────────────────┬─────────┬──────────┤
│ Time (IST)           │ Actor                │ Action                   │ Result  │ Policy   │
├──────────────────────┼──────────────────────┼──────────────────────────┼─────────┼──────────┤
│ 02 Oct 14:02:11.481  │ anita@bazaar (MFA)   │ marketplace:ApprovePayout│ ALLOW   │ payout.a │
│ 02 Oct 14:01:58.120  │ ravi@bazaar          │ marketplace:ApprovePayout│ DENY    │ payout.s │
│ 02 Oct 13:55:03.007  │ agent:support-triage │ support:ProposeReply     │ ALLOW   │ agent.s  │
├──────────────────────┴──────────────────────┴──────────────────────────┴─────────┴──────────┤
│ Record detail: hash 9c1f…  prev 77ab…  context {jur: IN, risk: 12, device: trusted}  ob     │
│ [Export evidence (signed)]  [Show related decisions]  [Open trace 4bf9…]                     │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

PART VI

OPERATE & EVOLVE

26. Observability, Operations & Resilience

26.1 Signals

All components emit **OpenTelemetry** traces, metrics and logs with standard resource attributes (`ucos.cell` , `ucos.tenant` (hashed in shared backends), `ucos.capability` , `ucos.blueprint` , `ucos.command` , `ucos.agent`).

Signal family	Examples	Consumers
Runtime	latency, errors, saturation, queue lag, DB locks	SRE, AIOps
Business	orders/min, GMV, conversion, payout volume, cart abandonment	Tenants, leadership, anomaly detection
Security	auth failures, policy denies, anomaly scores, WAF blocks	SOC/SIEM
Compliance	control pass/fail/drift, evidence freshness, DSR SLA	DPO, compliance

AI	tokens, cost, latency, eval scores, guardrail triggers, escalation rate	AI Ops
Governance	pending approvals, waivers expiring, certification status	Governance
Knowledge	registry quality scores, conflicts, stale facts	Stewards
Cost	per tenant, per capability, per cell (FinOps)	Finance, platform

Correlation: every signal carries `traceId + ucoscorrelation`, so business events, policy decisions, AI outputs and infra metrics can be joined for a single customer interaction.

26.2 SLOs & availability tiers

Tier	Availability	RPO	RTO	Topology
T0 (control plane, identity, payments path)	99.99%	≤ 1 min	≤ 15 min	Multi-AZ active-active; cross-region warm standby
T1 (core commerce, orders, ledger)	99.95%	≤ 5 min	≤ 30 min	Multi-AZ; cross-region DR
T2 (search, recommendations, analytics, reporting)	99.9%	≤ 1 h	≤ 4 h	Multi-AZ; rebuildable from events
T3 (batch, simulation, sandbox)	99.5%	≤ 24 h	≤ 24 h	Single region

Each capability declares SLOs in its manifest. Error-budget policies gate releases: when the budget is exhausted, only reliability fixes ship.

26.3 Operations model

- **Incident management:** severity matrix, on-call rotations per universe team, incident commander role, status page per tenant, post-incident reviews feeding Evolution proposals.
- **Runbooks:** generated per capability (from manifest + failure scenarios), maintained as code, executable steps via SOAR/automation, with authority checks.
- **AIOps:** anomaly detection on signals; probable-cause ranking using the dependency graph; suggested remediations with simulation (W14). Auto-remediation only for pre-approved runbook steps (autonomy L3, §29.2).
- **Capacity & FinOps:** per-tenant cost attribution, forecasted capacity from business

signals (Temporal forecasts), rightsizing recommendations, budgets and anomaly alerts.

- **Sustainability:** carbon intensity per cell and per workload, available to tenants as a signal.

26.4 Resilience engineering

Cell-based architecture limits blast radius · bulkheads per tenant tier · backpressure and load shedding (priority classes: checkout > browse > analytics) · graceful degradation (e.g. search falls back to category browse; recommendations to popular items; tax estimate to "calculated at checkout") · backups (PITR for Postgres, Kafka tiered storage, object versioning) with **automated restore tests** weekly · DR drills quarterly (gate `DR-*`) · chaos experiments in pre-prod continuously and in prod with guardrails (game days).

26.5 Digital twins (scoped, practical)

Twin	Built from	Questions it answers
Platform Twin	blueprint + live telemetry + dependency graph	What runs where, what's healthy, what breaks if X fails?
Business Twin	events + ledger + forecasts	Revenue, margin, inventory position; what-if pricing/offers
Supply-chain Twin	inventory + fulfillment + logistics events	Stock-outs, delivery promise accuracy, capacity
Customer Twin (consented, aggregate-first)	CRM + behaviour	Segments, churn risk, journeys
Security/Compliance Twin	controls + evidence + findings	Posture, drift, audit readiness
Organization Twin	org, roles, authority graph	Who can approve what; SoD coverage

Twins are projections maintained by the Twin engine. The Simulation universe runs scenarios against twin snapshots (`asOf` , synthetic load, policy changes).

---

27. Infrastructure & Deployment

27.1 Topology

```
flowchart TB
  subgraph Global
    DNS[Global DNS / Anycast + CDN + WAF]
    CP[Control Plane cluster(s)<br/>multi-region active-passive]
    REGG[(Global registries<br/>read replicas to cells)]
    PKGR[(OCI registry<br/>signed packs & images)]
  end
  subgraph Region_IN[Region: India]
    C1[Cell in-mum-1<br/>K8s + Postgres + Kafka + OpenSearch + ...]
    C2[Cell in-hyd-1 (DR)]
  end
  subgraph Region_AE[Region: UAE]
    C3[Cell ae-dxb-1]
    C4[Cell ae-auh-1 (DR)]
  end
  subgraph Region_EU[Region: EU]
    C5[Cell eu-fra-1]
  end
  DNS ⟶ C1 & C3 & C5
  CP ⟶ C1 & C3 & C5
  C1 -. async replication .→ C2
  C3 -. async replication .→ C4
  REGG ⟶ C1 & C3 & C5
  PKGR ⟶ C1 & C3 & C5
```

- **Cloud-agnostic by design:** Kubernetes for compute; OpenTofu + Crossplane for provisioning; only managed services with portable equivalents (Postgres, Kafka API, S3 API, Redis API). Each cloud-specific service is wrapped in an infrastructure port.
- **Deployment targets:** public cloud (AWS/Azure/GCP), sovereign cloud, on-prem (for government/regulated tenants, using the same Helm/Crossplane bundles), edge nodes (stores, warehouses, factories) running a reduced data-plane profile with store-and-forward sync.
- **GPU pools:** for self-hosted models and embeddings; scheduled by priority; per-tenant quotas.

27.2 Environments

`dev` → `ci-ephemeral` (per PR) → `integration` → `staging` (production-like, synthetic tenants) → `production` cells. Plus **sandbox** (per tenant, for builders and developers) and **simulation** cells (no external side effects).

27.3 Delivery pipeline

```
flowchart LR
  PR[PR: pack / platform / definition] ⟶ CI[CI: build, unit, lint (L2/L1 rules),<br/>SAS
  CI ⟶ SIGN[Sign image & pack<br/>cosign + SLSA provenance]
  SIGN ⟶ EPH[Ephemeral env:<br/>contract + scenario + a11y tests]
  EPH ⟶ CERT[Certification gates §28]
  CERT ⟶ STG[Staging: perf, chaos, DR smoke]
  STG ⟶ APPROVE{Release authority<br/>approval}
  APPROVE ⟶ GITOPS[GitOps (Argo CD):<br/>canary cell → wave 1 → wave 2 → all]
  GITOPS ⟶ GUARD[Automated rollback on SLO burn /<br/>error-rate guard]
```

Definitions (tenant blueprints) follow the same flow, minus compilation: validate → compose → certify → approve → progressive activation (tenant cohorts) → guard.

27.4 Platform kernel vs packs: release cadence

Artifact	Cadence	Compatibility
Platform kernel & engines	Bi-weekly minor; quarterly major (with 12-month support for N-1)	Engine APIs SemVer; pack runtime contract versioned
Capability packs	Continuous (per pack)	SemVer; compatibility checked against blueprints using them
Global registries (reference data, taxonomies, holiday calendars, regulations)	Continuous, governed by stewards	Additive by default; breaking changes via Evolution
Tenant blueprints	On demand	Pinned versions; auto-upgrade policy per tenant (patch auto, minor opt-in, major planned)

---

28. Testing, Certification & Release

28.1 Test strategy (generated + authored)

Level	What	Generated?
Unit	Pack handlers, decision logic	Authored
Schema & contract	API/event schemas vs consumers (Pact-style); data contracts	Generated from manifests
Policy	Cedar policy tests (allow/deny matrices), SoD checks, policy analyser proofs	Generated + authored

Scenario (BDD)	Business flows per capability and per blueprint (Gherkin)	Generated skeletons + authored
Integration	Pack ↔ engines ↔ adapters (with provider sandboxes/simulators)	Generated
End-to-end journeys	Top journeys per portal/channel (Playwright, Detox)	Generated from Journey definitions
Performance	Load profiles per capability (k6), soak, spike	Generated from SLOs + scale in PD
Resilience	Chaos experiments, failover, restore tests	Templates
Security	SAST, DAST, dependency, container, IaC scans; fuzzing of APIs from schemas; pen tests	Generated + external
Compliance	Control tests (§20.2), evidence presence & freshness	Generated from controls
Accessibility	axe-core on all views; screen-reader journeys on top flows	Generated + manual
Localisation	Pseudo-locale, RTL snapshots, missing-key checks	Generated
AI	Eval suites, red-teaming prompts, guardrail tests, regression on golden sets	Authored + generated
Composition	Determinism (same input → same hash), gap rule regression	Generated

28.2 Certification gates

Gate	Criteria (summary)	Applies to
`ARCH-*`	Manifest complete; layer and dependency rules respected; no ownership collision; traceability chain complete; naming conventions	packs, blueprints
`SEC-*`	Threat model reviewed; no critical/high vulns open; secrets scan clean; SBOM + signature + provenance; policy tests pass; pen-test (for T0/T1 packs) within 12 months	packs, blueprints
`CMP-*`	All required controls mapped and implemented; evidence collectors active; residency satisfied	blueprints, packs
`TEST-*`	Coverage thresholds (unit ≥ 80% of handlers, 100% of decision tables), all contract/scenario tests green	packs, blueprints
`PERF-*`	Meets SLO targets at declared scale + 30% headroom	packs, blueprints
`A11Y-*`	Zero critical axe violations; top journeys pass manual audit	UI views, portals

`DOC-*`	User/admin/API docs and runbooks exist and are current	packs, blueprints
`OBS-*`	Signals, dashboards, alerts and SLOs defined; alert routes configured	packs, blueprints
`DR-*`	Backup policy defined; last restore test passed within 30 days (T0/T1)	blueprints, cells
`AI-*`	Eval ≥ threshold; guardrails configured; authority bounded; model card present	agents, AI features
`GAP-000`	`mandatory_gap_count = 0` ; recommended gaps resolved or waived	blueprints

28.3 Certification record

```
kind: Certification
id: ucos:cert:01J...
subject: ucos:blueprint:sha256:7f3c...
gates:
  - { id: SEC-007, result: pass, evidence: [ucos:evidence:sast-run-8812], at: 2026-
  - { id: CMP-021, result: pass, evidence: [ucos:evidence:control-test-dsr-erasure-
  - { id: GAP-000, result: pass, details: { mandatory: 0, recommended: 2, waived: 2
waivers: [ucos:waiver:W-88, ucos:waiver:W-91]
validUntil: 2027-01-02            # re-certification cadence (or on change)
signedBy: ucos:svc:certification   # Sigstore-signed attestation (in-toto format)
approvedBy: ucos:decision:01J...   # release authority decision
```

Certification is **continuous**. Material changes (pack upgrade, policy change, control drift, expiring waiver) trigger re-evaluation, and expired certifications raise gaps.

---

29. Evolution & Autonomy

29.1 Evolution engine

Inputs: telemetry, gap scans, incidents, AI insights, new regulations (regulatory-change feeds), new providers, usage analytics, feedback.

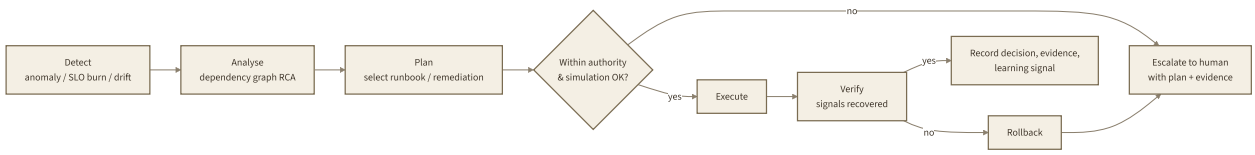
Outputs: **EvolutionProposals** (new capability, new universe, schema evolution, policy change, deprecation, optimisation, new registry entries, constitutional

policy change, deprecation, optimisation, new registry entries, constitutional amendment proposals). Every proposal goes through governed change (§18.4). Nothing self-applies above its autonomy level.

29.2 Autonomy levels

Level	Name	The system may…	Examples
L0	Observe	detect and report	Gap scans, drift detection
L1	Suggest	propose changes with rationale and simulation	"Add SLO alert", "Split role for SoD"
L2	Draft & auto-approve low risk	create definitions in Draft; auto-approve within explicit confidence/risk thresholds	catalog enrichment ≥ 0.95, translation of non-legal strings
L3	Act within runbooks	execute pre-approved remediation steps	restart, scale out, failover to DR, shift PSP traffic per pre-approved rule, revoke compromised token
L4	Bounded autonomy	optimise parameters within guardrails continuously	dynamic pricing within min/max bands; bid optimisation within budget
—	**Never autonomous**	—	constitutional changes, authority grants, irreversible financial movements above thresholds, data deletion outside DSR/retention rules, disabling security controls

29.3 Self-healing loop



29.4 Schema & capability evolution rules

- **Additive changes** (new optional field, new event, new command) are minor versions and auto-compatible.
- **Breaking changes** need a major version. The **expand → migrate → contract** pattern is mandatory. Events are dual-published for ≥ 90 days. APIs announce deprecation with `Sunset` headers ≥ 6 months ahead (T0/T1 APIs: ≥ 12 months).
- **Migrations** are definitions (`Migration` objects) with forward and backward

scripts, dry-run on a snapshot, and progress tracking per tenant.

- **Deprecation** requires a published successor or explicit retirement rationale. Consumers are notified, and the graph tracks remaining consumers until zero.

29.5 Continuous discovery questions (Completeness Engine schedule)

Question	Detector	Cadence
What is missing?	Gap rules over blueprints & registries	On change + nightly
What is duplicated / overlapping?	Ontology similarity + `provides` collisions + embedding similarity on definitions	Weekly
What conflicts?	Policy analyser, temporal overlap, capability conflicts	On change
What violates law/architecture/security/compliance?	Lint rules, policy analysis, control tests	On change + continuous
What is obsolete/unused?	Usage telemetry + graph consumer counts	Monthly
What under-performs?	SLO/error-budget + cost per outcome	Daily
What should be created?	Unresolved intents from Studio, failed searches in marketplace, support cases, regulatory feeds	Weekly

29.6 Simulation

Simulation cells run blueprints with simulated adapters, synthetic tenants/users (agent-driven load), and `asOf` time control. Scenario types: business (pricing, offers, demand), security (attack simulation), compliance (new regulation impact), infrastructure (failover, capacity), AI (model swap), and market (competitor price moves). Outputs are compared to baselines and attached to proposals as evidence.

29.7 Future horizon (extension points, not current scope)

Quantum computing, robotics fleets, autonomous vehicles, space/interplanetary commerce, synthetic economies, digital humans, brain-computer interfaces and AGI/ASI-class agents appear in the source vision. They are **not built now**. The architecture absorbs them through:

- `Future` `Object` → new ObjectTypes via the Meta universe;

- "Future" object → new ObjectTypes via the Meta universe;
- new **channel renderers** (Experience) and **device identities** (Identity);
- new **compute pools** (Infrastructure ports, e.g. quantum job port);
- new **agent authority classes** (Authority);
- new **regulations** (Compliance as data);
- the Evolution engine's proposal process.

Under L20 (Discover the Unknown), the measure of success is that none of these require re-architecting, only new registry entries and new packs.

PART VII

BUILD

30. Reference Compositions (Worked Examples)

Each reference composition is a **Platform Definition + capability bundle** maintained in the Industry Template registry. They prove the "no code change" claim: certification of the second, third and fourth vertical MUST NOT require kernel changes (Phase 6 exit criterion, §32).

30.1 Composition matrix

Capability ↓ / Platform →	Amazon-like marketplace	Uber-like mobility	Netflix-like OTT	Airbnb-
identity.core / org	●	●	●	●
identity.kyc / kyb	sellers	drivers	—	hosts
temporal.core	●	● shifts, surge windows	● renewals	● availa booking
geography	●	● geofence, routing	● licensing regions	● maps
catalog / product intel	●	—	● titles	● listing
offer / pricing	●	● dynamic pricing	● plans	● night

cart / order	●	● trip = order	—	● reser
marketplace	●	● two-sided	—	● two-s
subscription / entitlement	○ prime	○ pass	●	—
payment	●	●	●	●
ledger / finance	●	●	●	●
tax	● GST/VAT	●	● digital-services tax	● occu
inventory / fulfillment / logistics	●	—	—	—
mobility.dispatch / matching	—	●	—	—
media / streaming / DRM	—	—	●	—
crm / support	●	●	●	●
workflow / decision	●	●	●	●
trading (orders, risk, market data)	—	—	—	—
recommendation / search AI	●	○	● **core**	●
compliance packs (examples)	PCI, DPDP, consumer protection	transport licensing, PCI	content licensing, PCI, age rating	short-te rules

● required · ○ optional · — not used

30.2 Deep example: Uber-like composition (key definitions only)

```
# PlatformDefinition excerpt
spec:
  template: ucos:industry-template:ride-hailing@1
  geography: { countries: [IN], cities: [BLR, HYD, MUM] }
  capabilities:
    include: [mobility.dispatch, mobility.matching, pricing.dynamic, trust.ratings,
---
# Dynamic pricing = Decision + Temporal + Geo (no code)
```

```
kind: DecisionTable
id: mobility.surge-multiplier
hitPolicy: FIRST
inputs:  [ zone.demandSupplyRatio, temporal.window, context.weather ]
outputs: [ multiplier ]
rules:
  - [ ">2.5", "*",                 "*",     "2.0" ]
  - [ ">1.8", "tmp:peak-hours-blr","*",     "1.6" ]
  - [ ">1.3", "*",                 "rain",  "1.4" ]
  - [ "*",    "*",                 "*",     "1.0" ]
guardrails: { max: 2.0, regulatoryCap: ref(compliance:in-ka-aggregator-fare-cap) }
---
# Driver shift availability = Temporal AvailabilityWindow on Resource(vehicle+drive
kind: AvailabilityWindow
resource: ucos:resource:driver-01J...
recurrence: "FREQ=DAILY"
window: { local: "07:00-15:00" }
calendar: ucos:cal:in-ka-holidays
```

30.3 Deep example: Netflix-like OTT

Subscription plans (Subscription) → Entitlements (resolution: HD/4K, screens: 1/2/4, downloads) → content rights windows (Temporal policies per territory from content licences, Compliance + Geography) → playback authorisation (`media:Play` policy checks entitlement × territory × device count × parental rating) → recommendations (AI) → dunning workflow (Temporal recurrences + Communication). New plans, territories or rating systems are added as data.

30.4 Deep example: SAP-like ERP slice (Procure-to-Pay)

PurchaseRequisition → approval workflow (Decision: amount × cost centre → approver chain; SoD) → PurchaseOrder → GoodsReceipt (Inventory) → Supplier Invoice (Document + AI extraction) → three-way match (Decision) → Ledger posting (double-entry, fiscal calendar from Temporal) → Payment run (Temporal recurrence, Payment) → period close.

---

31. Technology Stack, Repository & NFRs

31.1 Reference technology stack

These choices are the recommended defaults. Each sits behind a port, so alternatives can be swapped per L15 without changing packs.

Concern	Default	Alternatives
Kernel/engines language	**Go** (gateway filters, command bus, object service, scheduler, connectors runtime)	Rust for hot paths
Packs language	**Go** or **Kotlin/JVM** (SDKs for both); **TypeScript** for UI-adjacent and lightweight packs	Any language implementing the gRPC pack contract
Extensions sandbox	**WASM** (Wasmtime) via Extism-style host functions	—
Studio, Console, portals	**TypeScript, React, Next.js**; design system in Storybook	—
Mobile	**React Native + Expo** (meta-model renderer)	Flutter renderer
API gateway	**Envoy** + custom filters (authn, context, PEP)	Kong, APISIX
AuthN / IdP	**Keycloak** or **Ory** (OIDC, passkeys) + tenant IdP federation	Auth0, Entra ID
AuthZ	**Cedar** (app), **OPA** (infra/admission)	OpenFGA for ReBAC-heavy cases
Workflow	**Temporal.io** (durable execution)	Camunda 8 (BPMN-native)
Decisions	**DMN** engine (GoRules ZEN)	Drools
Composer reasoning	**Datalog** (Soufflé or embedded Rust Datalog)	Answer-set programming
OLTP	**PostgreSQL 16+** (managed, Citus for scale-out)	CockroachDB/YugabyteDB for global-write tenants
Events	**Kafka** API (Apache Kafka / Redpanda / MSK), Schema Registry	Pulsar
Stream processing	**Apache Flink**	Kafka Streams
CDC	**Debezium**	—
Search	**OpenSearch**	Elasticsearch, Vespa
Vector	**pgvector** → **Qdrant** at scale	Weaviate, Milvus
Graph (UKG)	**Neo4j**	Amazon Neptune, Apache AGE
Analytics	**ClickHouse** (real-time) + **Iceberg** lakehouse	Snowflake/BigQuery via connectors

	+ **Trino**	
Cache	**Redis** / Valkey	—
Object storage	**S3 API** (cloud or MinIO)	—
Secrets / keys	**HashiCorp Vault** / OpenBao + cloud KMS/HSM	—
Observability	**OpenTelemetry** → Grafana stack (Mimir, Loki, Tempo)	Datadog, Honeycomb
Infra	**Kubernetes**, **OpenTofu**, **Crossplane**, **Helm**, **Argo CD**	—
Supply chain	**Sigstore/cosign**, **SLSA**, **CycloneDX**, **Trivy/Grype**	—
Developer portal	**Backstage** (catalog fed from Capability Registry)	—
AI	AI Gateway (in-house) → Anthropic Claude (default), others via adapters; **MCP** for tools	—

31.2 Non-functional requirements (targets)

Area	Target
API latency (gateway → response), simple command	p50 < 40 ms, p99 < 150 ms (in-cell)
Policy decision	p99 < 5 ms
Read-model freshness	p99 < 2 s
Search query	p95 < 120 ms
Storefront LCP (4G, mid-range mobile)	< 2.5 s; INP < 200 ms; CLS < 0.1
Checkout success (excluding user abandonment)	≥ 99.5%
Scale per cell (reference)	5k orders/min sustained, 50k RPS reads, 100k events/s
Scale overall	horizontal by cells; 10k+ tenants; billions of objects
Composition time	< 60 s for a 100-capability blueprint (cold); < 5 s incremental
Activation rollout	canary within 10 min; full within 2 h (configurable)
Availability	per tier (§26.2)
Security	0 critical vulns > 7 days; high < 30 days; MFA 100% for business

	users; keys rotated ≤ 90 days
Audit	100% of commands and policy decisions recorded; chain verification daily
Accessibility	WCAG 2.2 AA
Localisation	100% of UI strings externalised; RTL supported
Data	RPO/RTO per tier; restore tests weekly (T0/T1)
Cost	per-tenant cost attribution ≥ 95% of spend

31.3 Specification repository + code monorepo

This structure merges the "constitutional repository" and the "generated outputs" layouts from the source into one monorepo:

```
ucos/
├── README.md
├── UCOS-MASTER-ARCHITECTURE.md          # ← this document (constitution-level)
├── constitutions/                       # per-universe constitutions (12-section templat
│   ├── U13-temporal/constitution.md
│   └── ...
├── spec/                                # machine-readable canonical definitions (source
│   ├── meta/                            # CanonicalObject, field types, relationship typ
│   ├── ontology/                        # concepts & mappings (RDF/OWL or YAML)
│   ├── taxonomy/
│   ├── reference-data/                  # ISO codes, units, tz pins, holiday calendars
│   ├── object-types/
│   ├── events/                          # AsyncAPI/CloudEvents schemas
│   ├── apis/                            # OpenAPI/GraphQL/protobuf (generated + curated)
│   ├── workflows/
│   ├── decisions/
│   ├── policies/                        # Cedar / Rego
│   ├── controls/                        # compliance controls & mappings
│   ├── rules/                           # composition derivation rules (Datalog)
│   ├── ui/                              # view definitions, journeys, design tokens
│   ├── ai/                              # models, routes, prompts, agents, evals
│   └── industry-templates/
├── kernel/                              # platform engines (Go)
│   ├── gateway/  commandbus/  objectsvc/  scheduler/  composer/
│   └── certification/  governance/  registry/  audit/  aigateway/
```

```
│   │   certification/  governance/  registry/  audit/  gateway/
│   └── sdk/  (go, kotlin, ts)          # pack SDKs (tenancy, outbox, audit, OTel enforce
├── packs/                              # capability packs (only place business code live
│   ├── identity.core/  temporal.core/  catalog.listing/  commerce.cart/ ...
│   └── adapters/ payment.provider.razorpay/ carrier.delhivery/ ...
├── experience/
│   ├── design-system/                  # tokens, components, Storybook
│   ├── renderers/ web/ mobile/ voice/ chat/ agent/
│   └── portals/ console/ studio/ seller/ governance/ ...   # portal shells (navigation fr
├── generators/                         # SDK, docs, IaC, tests, dashboards, diagrams
├── infrastructure/                     # OpenTofu modules, Crossplane compositions, Helr
├── tests/                              # cross-cutting suites (journeys, chaos, compliar
├── tools/ucos-cli/                     # `ucos lint|compose|trace|certify|simulate|gen`
├── docs/                               # ADRs, runbooks, guides (generated + authored)
│   └── adr/
├── generated/                          # build outputs (never edited by hand; .gitignore
├── gaps/                               # latest gap reports (CI artefacts)
└── evolution/                          # evolution proposals & decisions
```

Forbidden: `apps/<tenant-name>/`, tenant-specific branches, and vendor SDK imports outside `packs/adapters/`. CI enforces this (§34).

31.4 CLI (developer & CI interface)

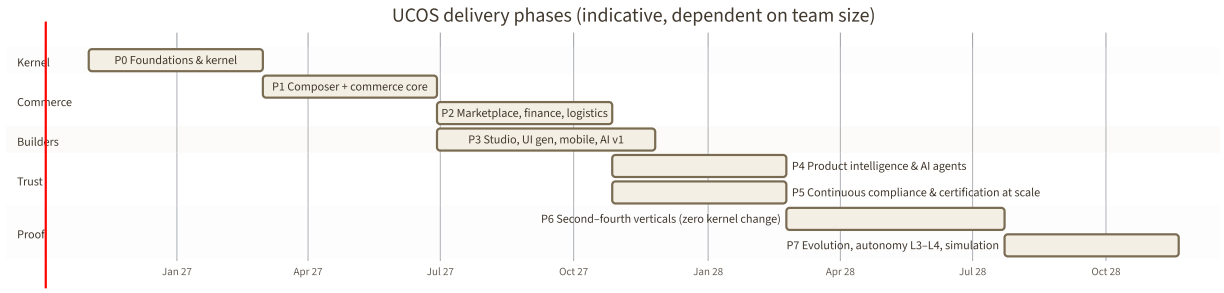
BASH

```
ucos lint spec/ packs/            # L1/L2/L3/L4 rules, naming, i18n, banned literal
ucos compose pd.yaml --snapshot latest --out blueprint.yaml --explain
ucos gaps blueprint.yaml          # gap report (exit≠0 if mandatory gaps)
ucos trace ucos:capability:commerce.cart
ucos simulate blueprint.yaml --scenario scenarios/diwali-peak.yaml
ucos gen sdk|docs|tests|iac|dashboards blueprint.yaml
ucos certify blueprint.yaml --gates all
ucos pack build|sign|publish packs/commerce.cart
```

---

32. Delivery Roadmap, Team & ADRs

32.1 Phased delivery (each phase ends with a certified, running increment)



Phase	Scope	Exit criteria
P0 Foundations	Meta-model, registries, reference data, Identity, Org/Tenant, Geography, Temporal core, Currency/Unit, Localization, Policy (Cedar PDP), Audit, Object service, Command bus, Event bus, pack SDK, CLI lint, cells (1 region), CI/CD + supply chain	A sample pack defines an object type; tenant isolation proven by tests; audit chain verified; Temporal triggers working
P1 Composer + Commerce core	Composition Substrate (Datalog rules, gap engine), Blueprint store, certification v1; Catalog, Product (tenant-level), Offer, Pricing, Cart, Order, Payment (1 PSP per country), Inventory (single-node), Tax (1 country), Communication (email/SMS), Storefront web renderer, Tenant admin	A single-brand D2C store composed **from a PD** and certified with `GAP-000`; end-to-end purchase works
P2 Marketplace, finance, logistics	Marketplace (sellers, commission, payouts), Ledger (double-entry), Settlement, Fulfillment, Logistics (multi-carrier), Returns, Seller portal, Support/Case, second country (residency + cell)	Amazon-like reference composition live in 2 countries; adding the 2nd country needed **no kernel change**
P3 Builders & experience	Builder Studio (W2–W8, W10), UI meta-model renderers (web, mobile), design system, journeys, i18n/RTL, developer portal + SDK generation, AI gateway v1 + Studio assistant	A non-engineer builds a new storefront variant end-to-end in Studio
P4 Intelligence	Global Product & Service Intelligence Network (ingestion pipeline, GPID, W9), search/vector, recommendation, agents (catalog enrichment, support triage) with authority & evals	≥ 1M global products with quality ≥ threshold; agents at L2 with eval ≥ target
P5 Continuous compliance	Control library (PCI, DPDP, GDPR, SOC2, ISO27001), evidence collectors, DSR, consent, Governance portal (W11, W16), certification at scale, DR automation	External audit (SOC 2 Type I) passes using UCOS-generated evidence
P6 Vertical proofs	Subscription/OTT, Mobility, CRM/ITSM templates; Resource universe; Dispatch	**Three new verticals certified with zero kernel code changes** (only packs + definitions)
P7 Evolution &	Evolution engine, simulation cells, twins, AIOps auto-	Measured MTTR reduction;

autonomy	remediation (L3), bounded optimisation (L4)	≥ N evolution proposals accepted via governance

32.2 Team topology (indicative)

- **Platform kernel team(s)**: gateway/command bus/object service; scheduler/temporal; composer/certification; registry/graph.
- **Universe stream-aligned teams**: Foundation (Identity/Geo/Temporal/Currency/Localization), Commerce core, Marketplace & Finance, Operations (Inventory/Fulfillment/Logistics), Experience (design system + renderers), Intelligence (AI gateway, agents, search/reco), Knowledge (product/service intelligence).
- **Enabling teams**: Security, Compliance engineering, SRE/Platform ops, Developer experience.
- **Governance roles**: Architecture Board, universe stewards, DPO, CISO, release authority.

32.3 Architecture Decision Records (initial)

ADR	Decision	Rationale
ADR-001	Interpret metadata at runtime; generate only external artifacts	One codebase to secure/scale; instant config; avoids per-tenant code drift (§12.1)
ADR-002	Control plane / data plane split with cells	Blast radius, residency, scale-out (§5.4, §27)
ADR-003	Hybrid physical model: generated tables for core types, JSONB for tenant extensions	Performance + flexibility (§14.2)
ADR-004	Temporal is a foundation universe; no raw start/end dates in other universes	Knowledge Once (§9)
ADR-005	Single PDP (Cedar) for RBAC/ABAC/ReBAC/PBAC	One policy language, analysable (§18.5)
ADR-006	Composition rules in Datalog over the Knowledge Graph	Deterministic, explainable, data-extensible (§11.3)
ADR-007	Transactional outbox + sagas; no distributed 2PC	Reliability at scale (§13.3)
ADR-008	Global Product master separated from tenant Listings	Reuse without leakage of tenant data (§10.1)
ADR-	Packs as signed OCI artifacts with SBOM + SLSA	Supply-chain security (§8.3)

009	provenance; WASM for in-process extensions	
ADR-010	AI via gateway with policy-based routing; agents are identities with bounded authority	Vendor agnosticism + safety (§21)
ADR-011	Bitemporal canonical objects	Audit, regulation, simulation (§14.6)
ADR-012	Kafka-API event backbone with CloudEvents envelope	Ecosystem, portability (§15.4)

---

33. Consolidation Traceability Matrix

This matrix shows where every distinct concept from `Chat.txt` now lives. Repeated lists in the source have been merged.

Source concept (all variants)	Canonical home in this document
"Application = Capability Composition", composition examples (Amazon/Uber/Netflix/Salesforce/SAP…)	§1.1, §30
Universal Capability Stack (Application compositions → Execution substrate)	§5.2 (L5–L11)
Layer 000 Constitutional Foundation; Constitution Engine; Constitutional Laws; Authority model	§3, §18 (L0)
Layer 001 Existence Foundation; Level-0 Existence (Identity, Entity, Agent, Asset, Ownership, Reputation…); "Everything that exists"	§7 (Canonical Object, object hierarchy), §6 (U09–U17)
Layer 002 Knowledge Foundation; Universal Knowledge Graph; Ontology/Taxonomy/Truth/Evidence/Provenance	§6 (U05–U08), §10, §21.6 (L3)
Layer 003 Context Foundation; all context dimensions (18+ variants)	§7.6
Layer 004 Composition Foundation; Platform Composer; Composition Substrate; Resolvers	§11
Layer 005 Capability Marketplace; Capability Store; installable capabilities	§8.5, W4
Layer 006 Commerce; Commerce Engine; Commerce Model Universe (B2B…P2P, freemium, usage-based…)	§6 (U21–U33), §30
Layer 007 Financial System; Fintech Engine; Economic Universe	U29–U30, §13.3 (money rules), §30
Layer 008 Logistics; Mobility Engine	U34–U37, §30.2

Layer 009 Experience Platform; Experience Universe (web…brain interface)	§22–§25
Layer 010 Communication	U40, §17
Layer 011 Workflow Engine (BPM, rules, decisions, approvals, case mgmt)	§16, U53, U60
Layer 012 AI Universe; AI Fabric; AI engines; Agent Universe; multi-model router	§21
Layer 013 Trust; Reputation & Trust Universe	U70, §19
Layer 014 Governance; Governance Machine/Runtime	§18
Layer 015 Security (zero trust, IAM, RBAC/ABAC/PBAC/ReBAC/ZBAC, KMS, HSM, SIEM/SOAR, PQC…)	§19
Privacy & Consent Universe	§20.4
Compliance Universe (GDPR…DORA, compliance-as-code/metadata/policy)	§20
Audit Universe; Auto Audit; immutable cryptographic trails	§19.9, U68
Layer 016 Observability (business/AI/security/compliance telemetry)	§26.1
Layer 017 Data Platform; Data Universe/Constitution (mesh, fabric, lineage, contracts, sovereignty)	§14
Layer 018 Platform Services (gateway, mesh, SDK, CLI, Studio, sandbox, federation)	§13, §31
Layer 019 Operations (DevOps…TrustOps)	§26, §27
Layer 020 Evolution Engine; Self-Evolution; Auto Learning; Learning Universe	§29
Layer 021 Future Existence (quantum, robotics, space, BCI, AGI…)	§29.7
Universal Configuration Model / Configuration Engine / Configuration Flow	§11.2, §12.3, J1
Universal Automation Layer ("Auto *" lists)	§29.2 autonomy levels, §28 (auto testing/certification), §12 (generation)
Meta-Intelligence ("what exists / missing / duplicates / conflicts…")	§3.1, §29.5
Gap Discovery Universe / Completeness Engine / "Mandatory Gap Count = 0"	§11.6, §28.2 (GAP-000)
Digital Twin Universe (product…civilization twin)	§26.5

Digital Twin Universe (productive/realization twin)	§29.5
Simulation Universe; What-if	§29.6
Certification Universe/Engine (architecture…release)	§28
Platform Factory (all ~10 variants of the pipeline)	§11.1, §12.3
Generation Engine (API, UI, DB, workflows, tests, docs, IaC, SDKs…)	§12
Universal Runtime Universe/Fabric (execution, policy, workflow, event, AI, agent…)	§13
Integration Universe (API/connector/adapter/protocol registries, EDI, MCP)	§17, §15
Resource Universe	U17
Decision Universe	U53, §16.2
Search Universe	U44
Document / Asset / Catalog / Subscription / Entitlement / Risk / Dispute universes	U42, U43, U21, U27, U28, U69, U71
Identity Intelligence (KYC/KYB)	§19.2
AI Governance Universe	§21.5
Capability Genome / Universal Capability Ontology	§8.1–§8.2
Constitutional Meta-Model, canonical objects, field types, cardinality, relationship types, lifecycle, traceability, "Graph of Everything"	§7
Authority types, Authority object, Decision model	§18.2–§18.3
Global identifiers (UGID, UID, GID, PID, OID, CID, BID, SID, DID, AID, RID, EID, TID, KID, GPID, GSID)	§7.2
Universal Product & Service Database / Intelligence Network	§10